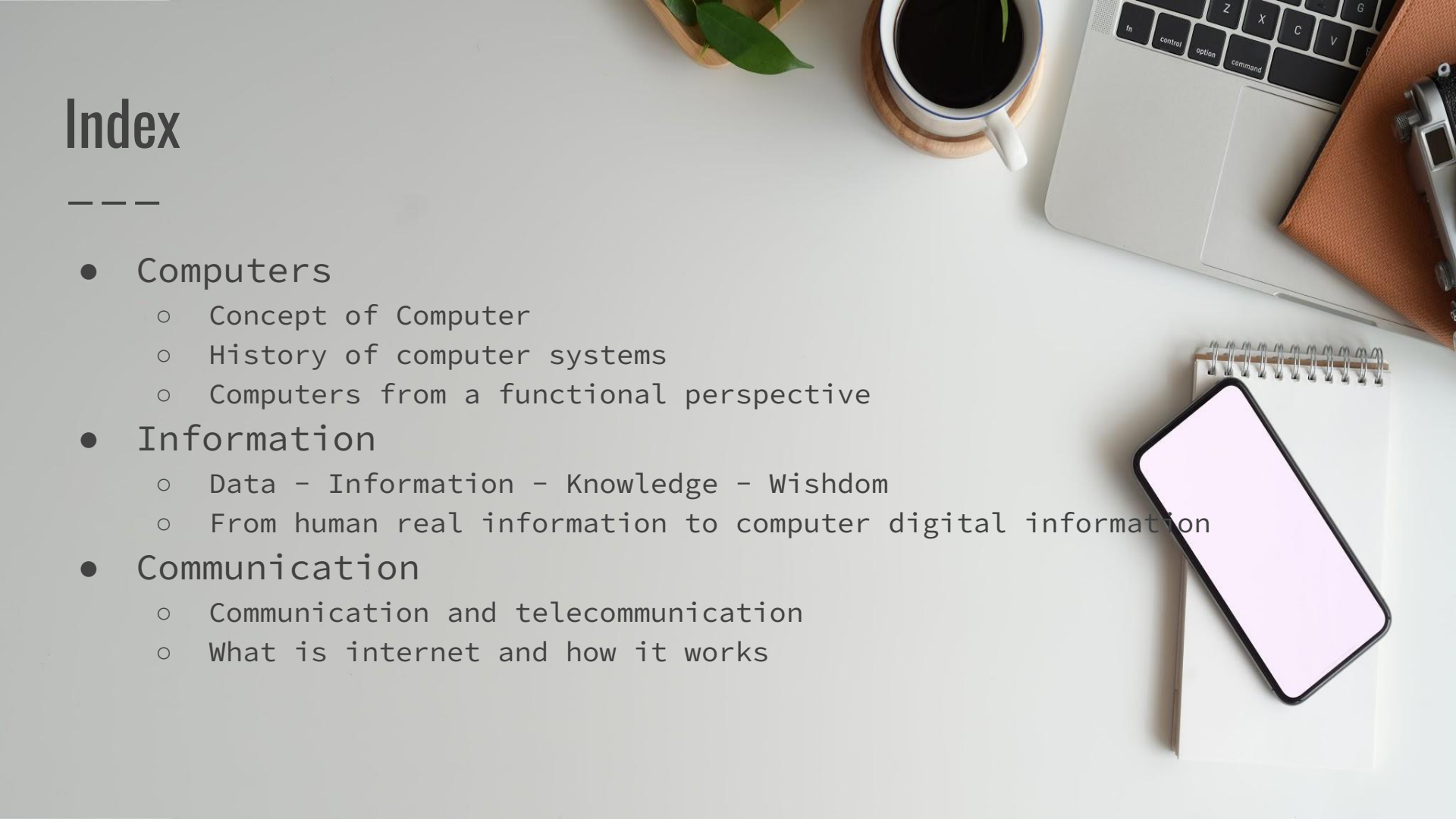


Information and communication technologies

2025
Isaac González



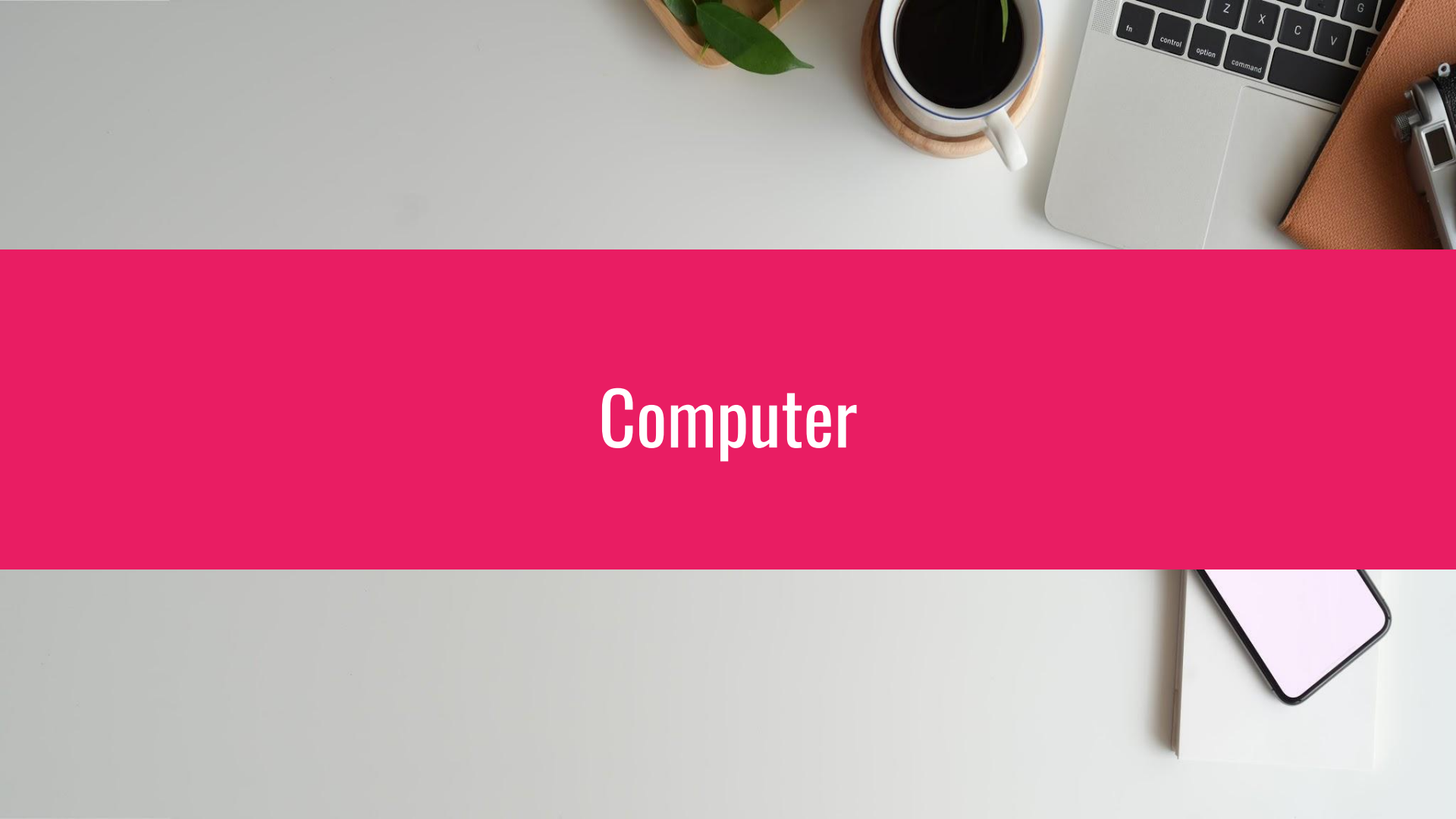
Index

A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' showing. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter is in the top center. In the bottom right, a spiral-bound notebook is open, and a smartphone with a pink screen is resting on it.

- Computers
 - Concept of Computer
 - History of computer systems
 - Computers from a functional perspective
- Information
 - Data - Information - Knowledge - Wisdome
 - From human real information to computer digital information
- Communication
 - Communication and telecommunication
 - What is internet and how it works

Link the actual
knowledge you have
with the general
abstract theory



A top-down view of a workspace. In the upper right, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white cup of black coffee on a wooden coaster. To the left of the cup is a small wooden planter with green leaves. In the lower right, a smartphone with a pink screen lies on a white surface.

Computer

A top-down view of a workspace. In the top right corner, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter is in the top center. A brown leather folder or notebook is on the right. In the bottom right, a spiral-bound notebook is open, and a smartphone with a pink screen is resting on it.

Computer

A big summary

Computer, your knowledge

Do you have one?

What is it?

What is for? What can you do with?

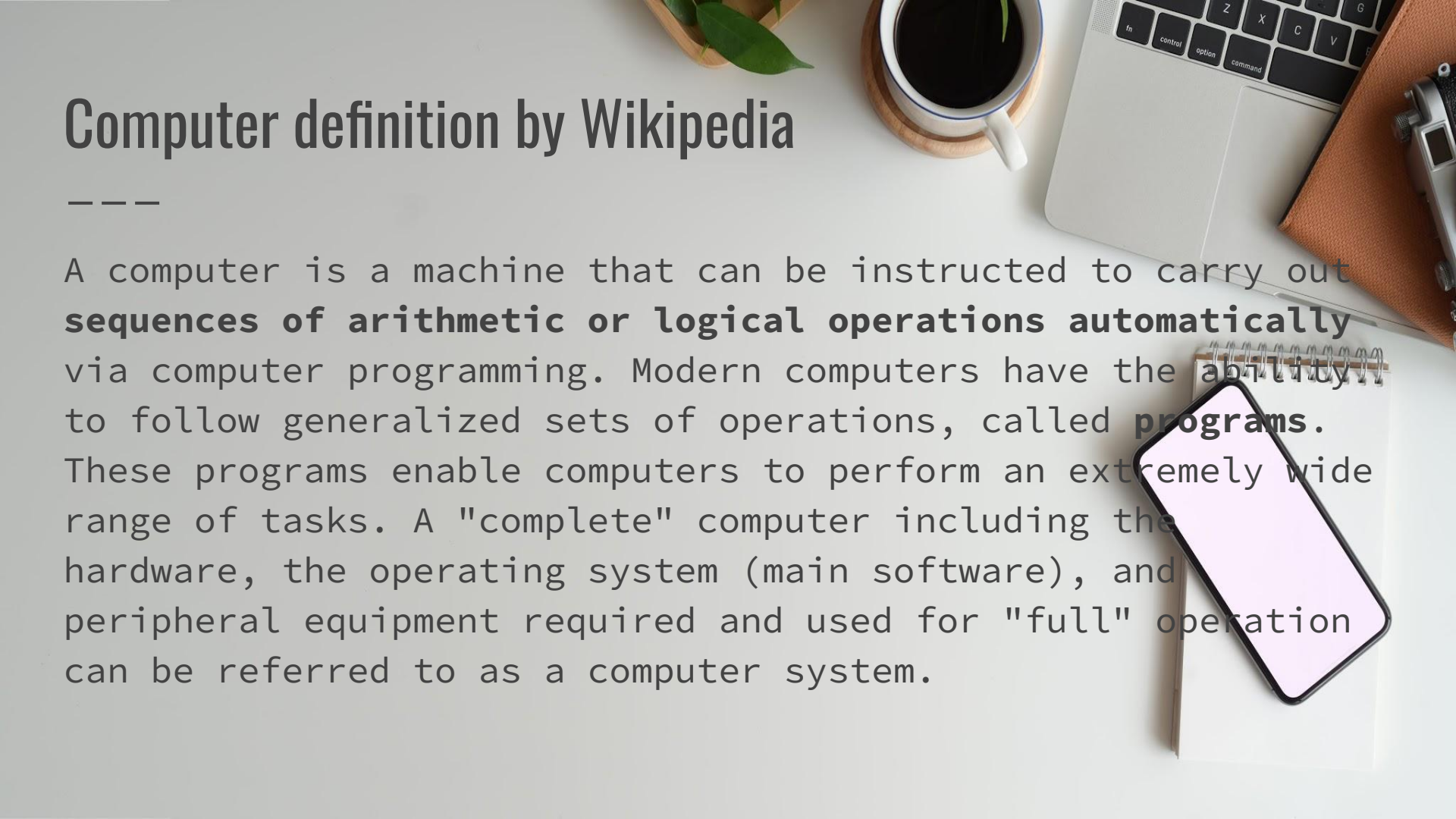
What's the relation with information?

Is it relevant for your daily life or profesion?

How do we arrive here? (history)



Computer definition by Wikipedia

A top-down view of a desk with a laptop, a cup of coffee, a small potted plant, and a smartphone. The laptop is silver and open, showing the keyboard. The coffee cup is white with a blue rim and sits on a wooden coaster. The plant is in a wooden pot. The smartphone is black and lies on a spiral-bound notebook.

A computer is a machine that can be instructed to carry out **sequences of arithmetic or logical operations automatically** via computer programming. Modern computers have the ability to follow generalized sets of operations, called **programs**. These programs enable computers to perform an extremely wide range of tasks. A "complete" computer including the hardware, the operating system (main software), and peripheral equipment required and used for "full" operation can be referred to as a computer system.

Operations

• Arithmetic

- addition +
- subtraction -
- multiplication *
- division /

• Logic

- Boolean: NOT, AND, OR ! & |
- Equal, not equal == !=
- Less than, less or equal < <=
- More than, more or equal > >=

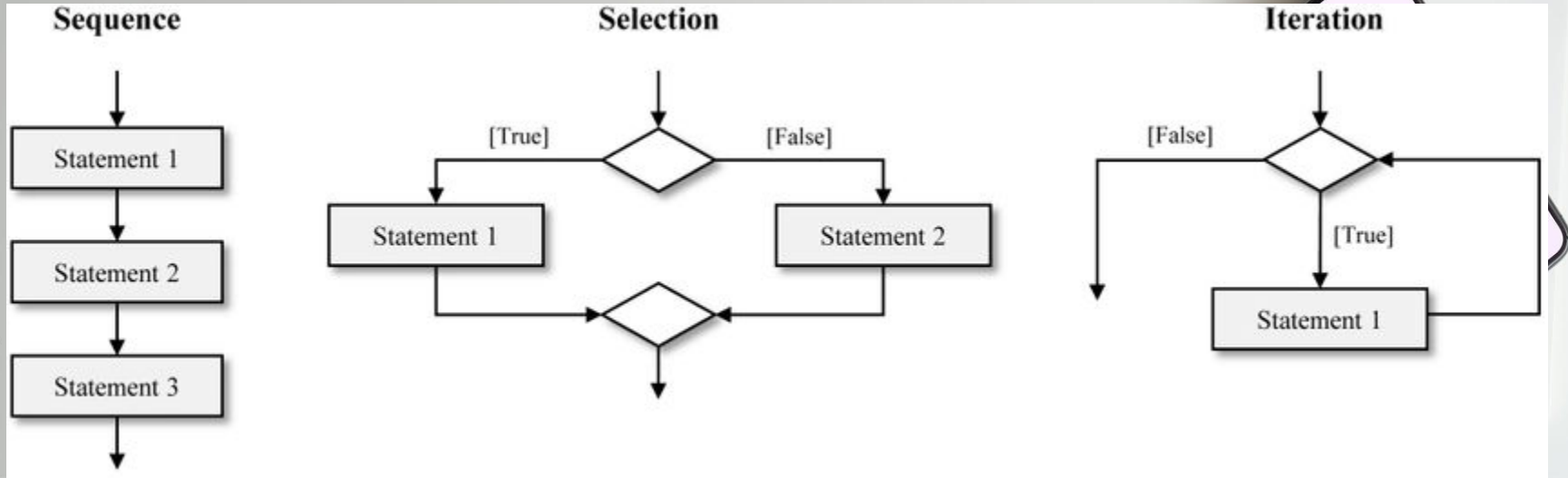
There are more operators, symbols could be different than that



Control structures

- Structures to control the flow of the instructions

Structured program theorem



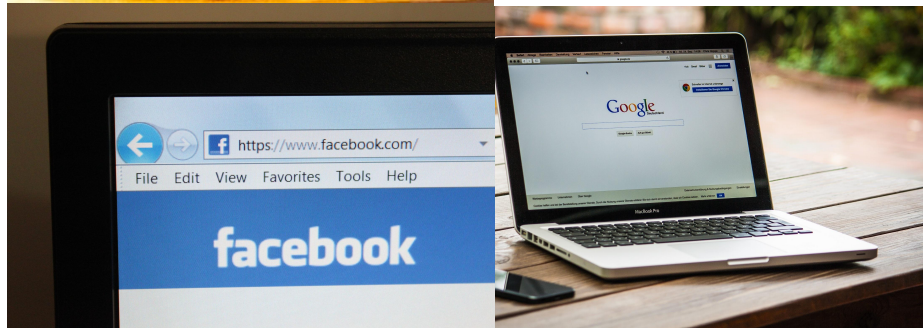
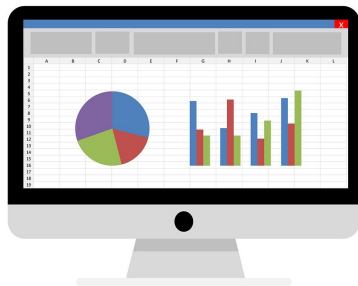
Program example

```
year_born = int(input('Please, insert the year you were born: '))
id = input("What's your ID number: ")
age = 2024 - year_born
if (age >= 18 and id != ""):
    print("Please come in to the VIP discotheque")
else:
    print("Sorry, you can't pass")
```

[See it in action](#)

Computers are for...

— — —



vs

sequences of
arithmetic or
logical
operations
automatically
executed via
programming

Computer yes or no?

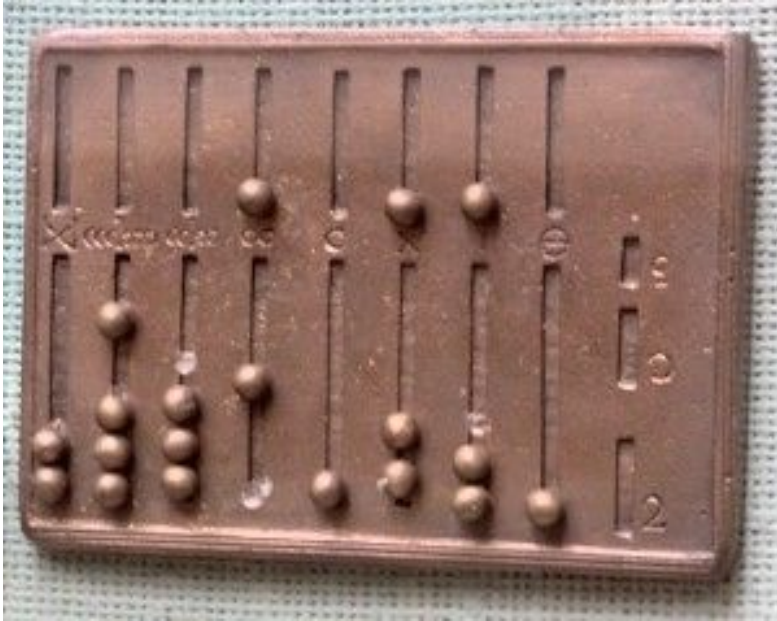


A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' showing. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot is in the top center. Below the laptop is a brown leather folder or notebook. In the bottom right, a white spiral-bound notebook is open, and a smartphone with a pink screen is resting on it.

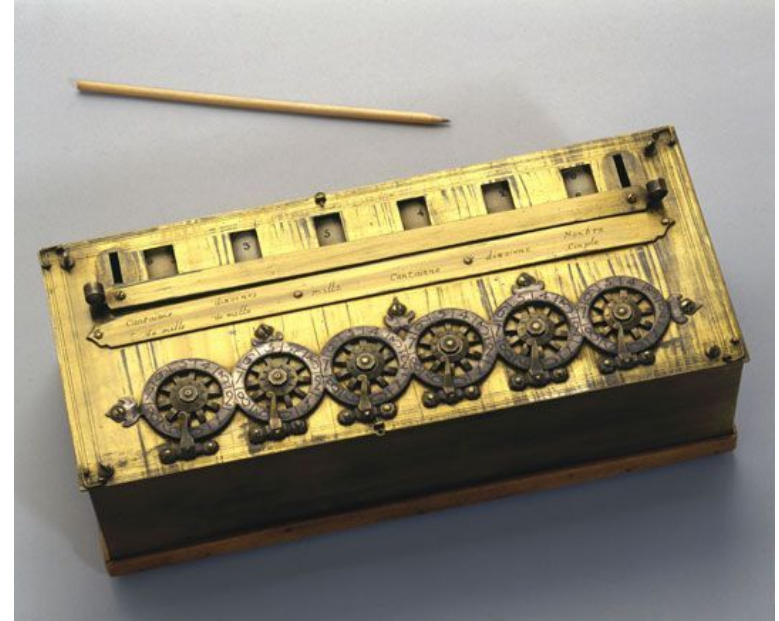
History of computers

Very short version

Before computers



Roman Abacus, 300bC aprox

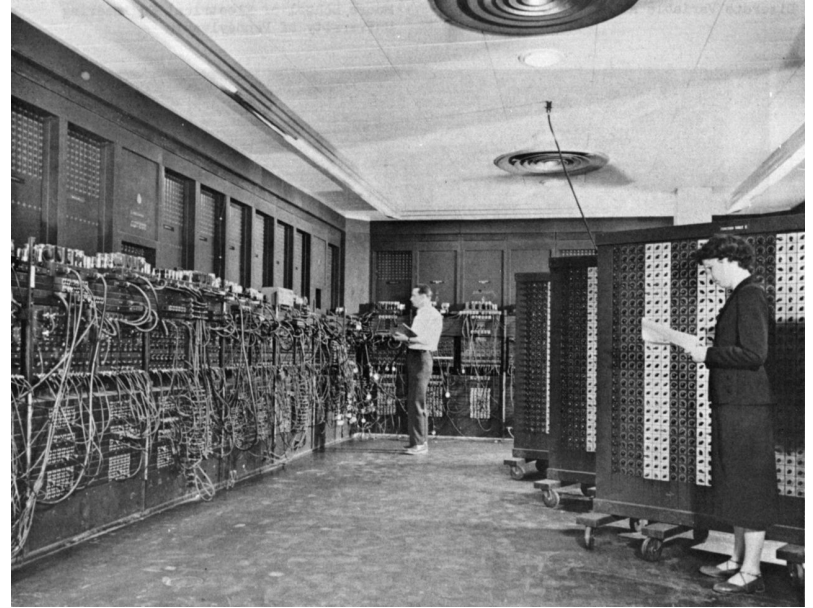


Pascal Calculator, 17th century

1940 – 1956: First Generation – Vacuum Tubes



Harvard Mark I, Howard Aiken,
WWII



ENIAC, WWII

Alan Turing, Enigma and Apple



1976



1977



1998



2001



2007

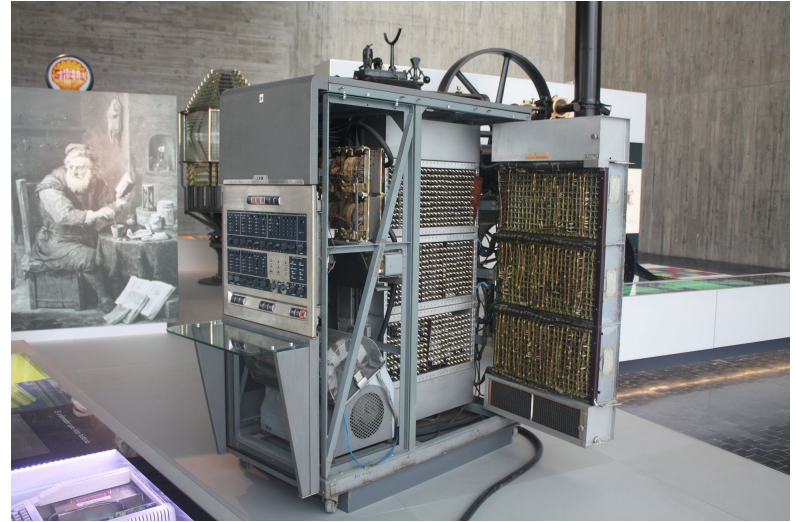


Today

1956 – 1963: Second Generation – Transistors



UNIVAC 1, first transistor
computer

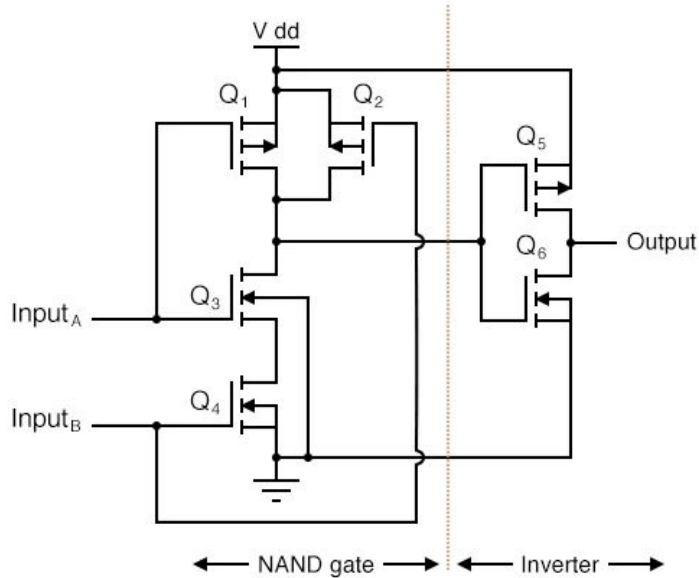


IBM 650, first profitable
computer

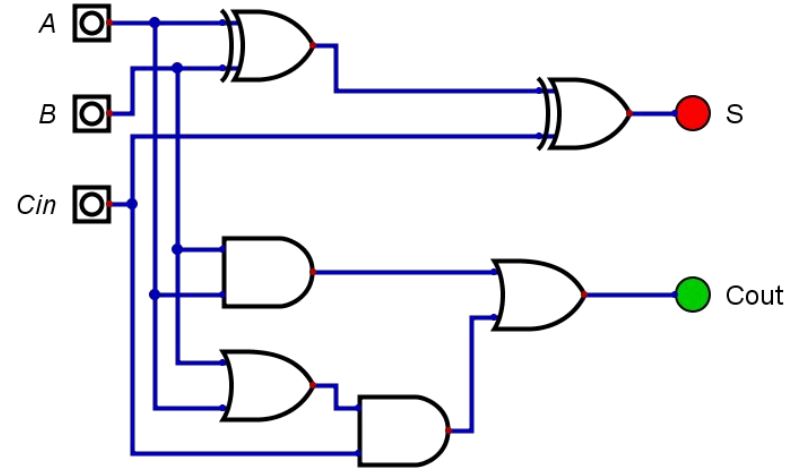
Transistor -> Logic gate -> Circuit -> ALU (I)



CMOS AND Gate

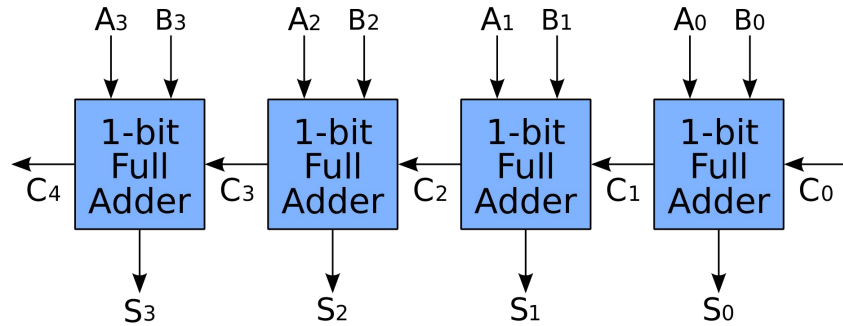


NAND Gate build with 6 transistors (CMOS)

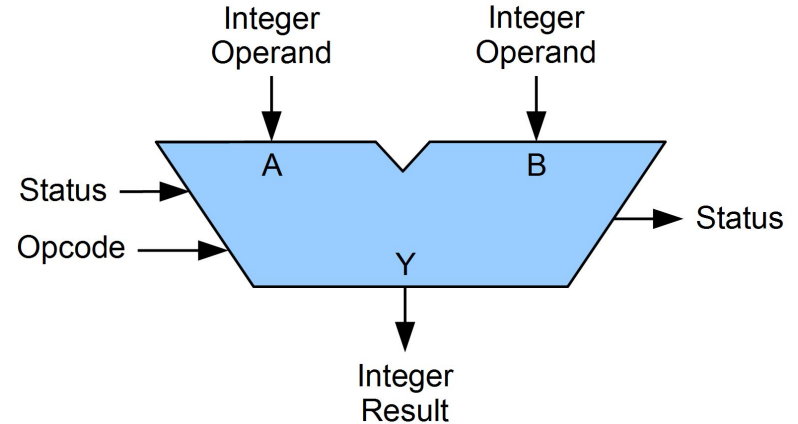


Full adder (3 bits), made with 6 logic gates

Transistor -> Logic gate -> Circuit -> ALU (II)

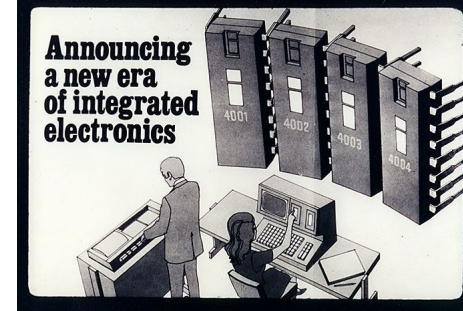


4 bit circuit adder



ALU representation

1964 – 1971: Third Gen – Integrated Circuits

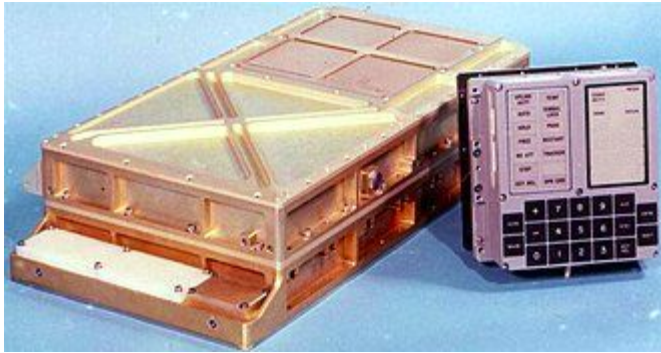


PDP-8, the first computer made by integrated circuit



IBM System/360

Computers to the moon



Apollo Guidance Computer (AGC),
1969



Real-Time Computer Complex
(RTCC) in Houston, Texas, 1969

1972 – today: Fourth Generation – Microprocessors



Apple 2, 1977



IBM Model 5150, 1981

1990-2010: PC era - Apple vs Microsoft



Pirates of Silicon Valley

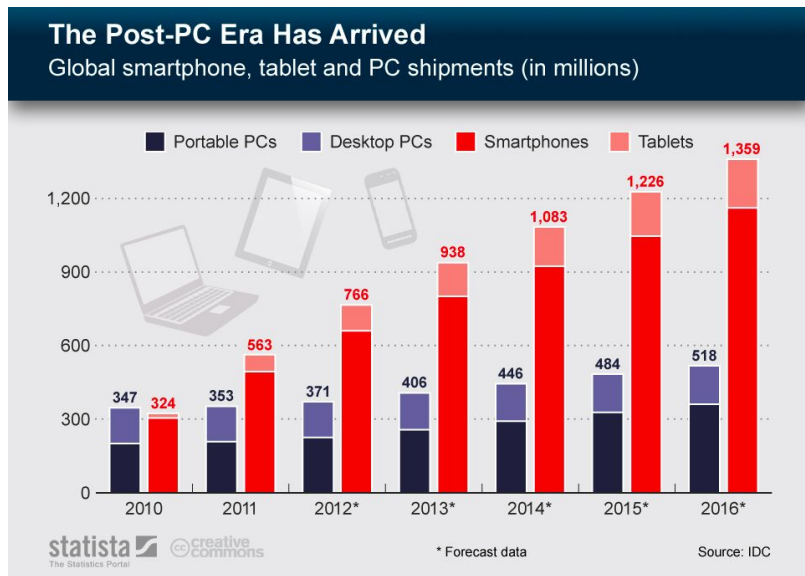


Macintosh Portable,
1989



IBM Thinkpad 300C,
1992

2010 - today: Post-PC era - Smartphones

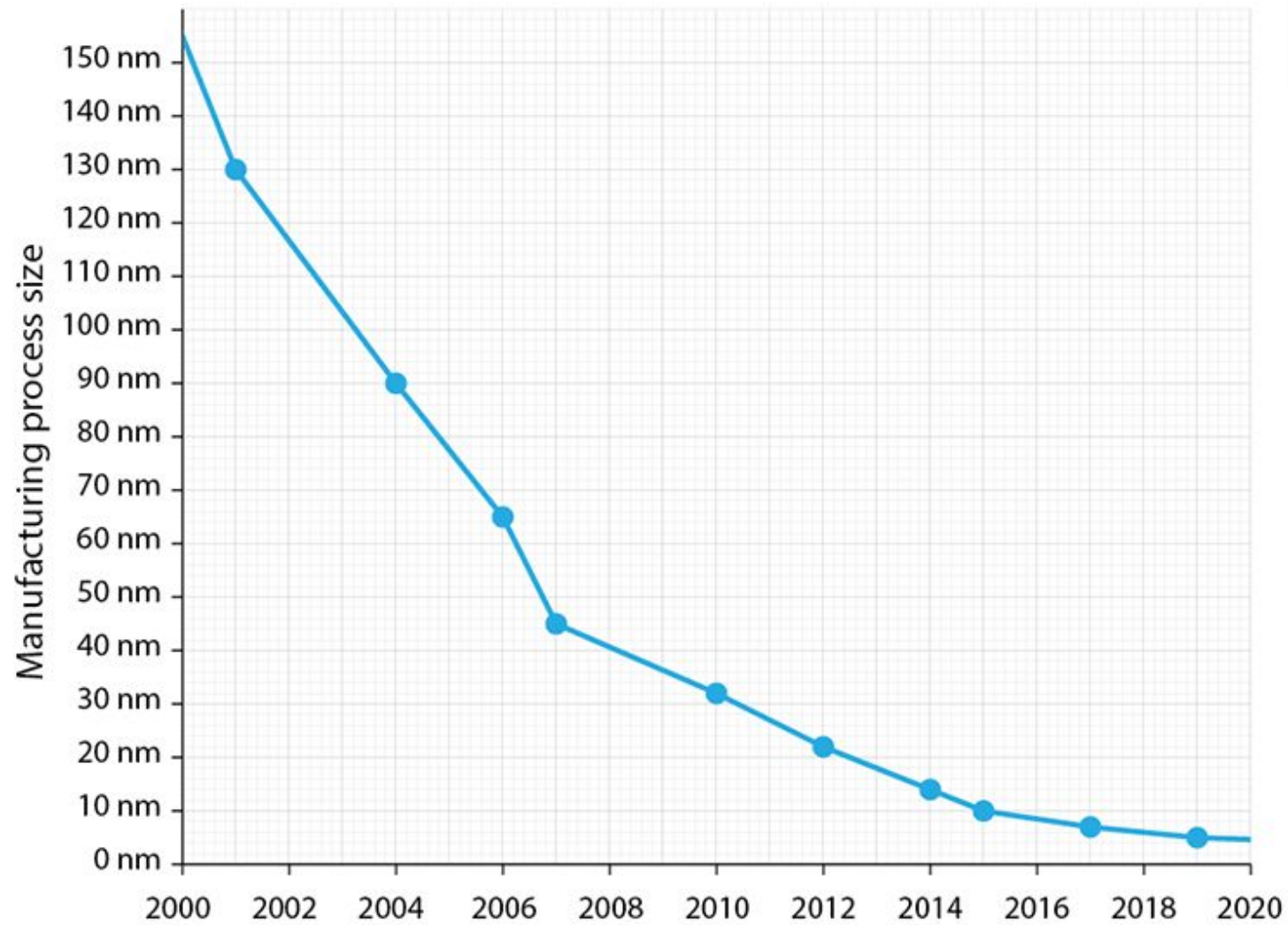


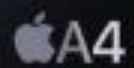
Sales of smartphones and tablets surpassed PCs on 2011



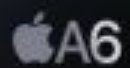
iPhone, the first successful smartphone

Our World
in Data[illegible]

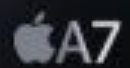


The Apple A4 chip is represented by a white Apple logo followed by the text 'A4'.

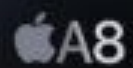
45 nm

The Apple A6 chip is represented by a white Apple logo followed by the text 'A6'.

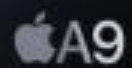
32 nm

The Apple A7 chip is represented by a white Apple logo followed by the text 'A7'.

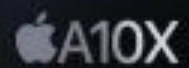
28 nm

The Apple A8 chip is represented by a white Apple logo followed by the text 'A8'.

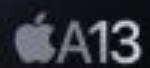
20 nm

The Apple A9 chip is represented by a white Apple logo followed by the text 'A9'.

16 nm

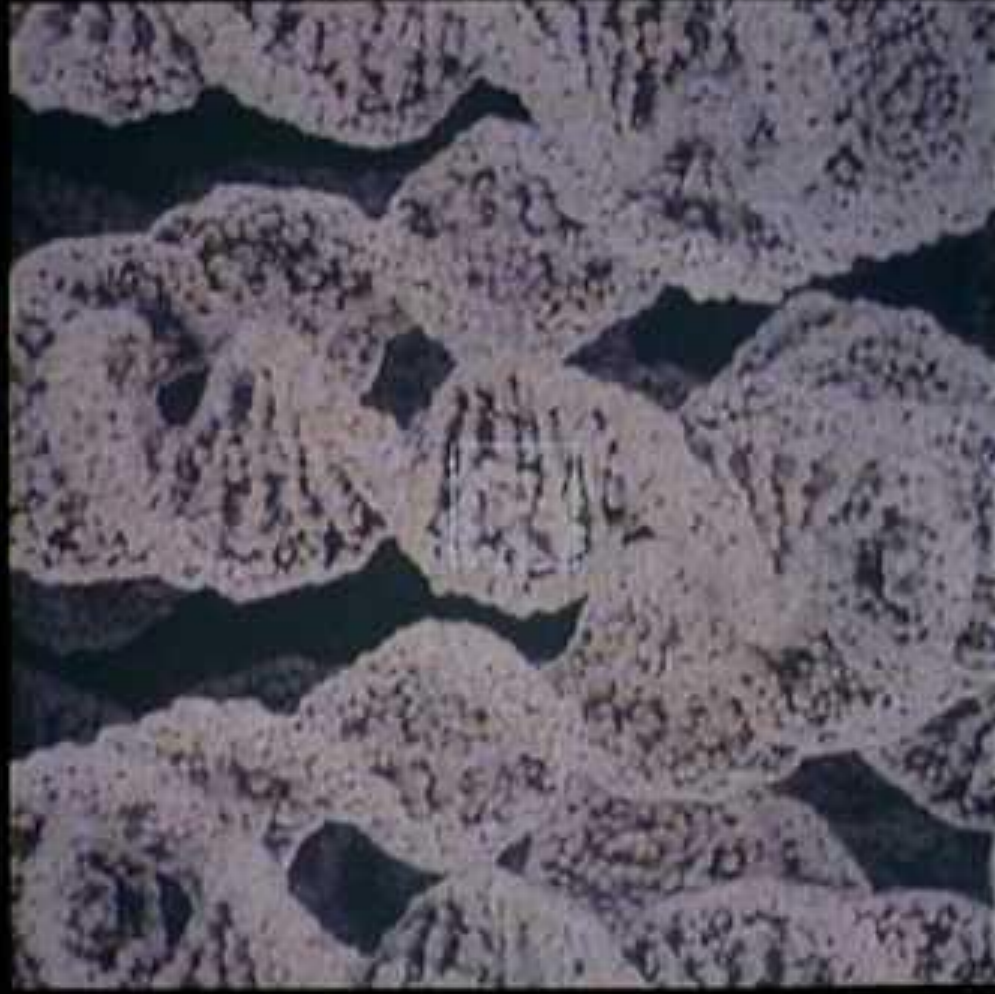
The Apple A10X chip is represented by a white Apple logo followed by the text 'A10X'.

10 nm

The Apple A13 chip is represented by a white Apple logo followed by the text 'A13'.

7 nm

100 ångstroms



-8
10
meters

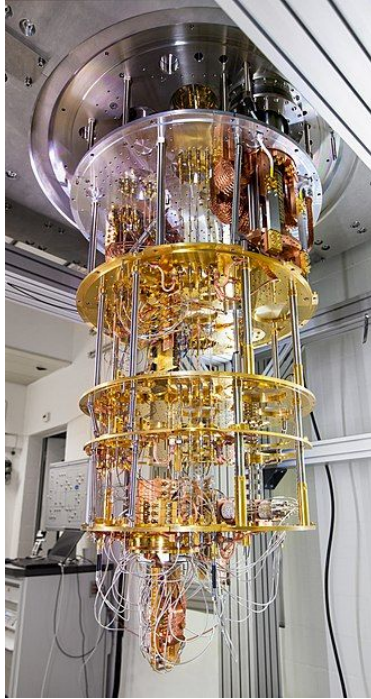
A top-down view of a workspace. In the upper right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V' showing. Next to it is a white cup of dark coffee on a wooden coaster. To the left of the cup is a small wooden tray with green leaves. Below the laptop is a brown leather folder or notebook. In the lower right, a white spiral-bound notebook is open, and a black smartphone with a light pink screen is resting on it.

I think it's fair to say that personal computers have become the most empowering tool we've ever created. They're tools of communication, they're tools of creativity, and they can be shaped by their user.

— BILL GATES, COFOUNDER, MICROSOFT

Will Quantum Computing break Silicon Computers?

— — —

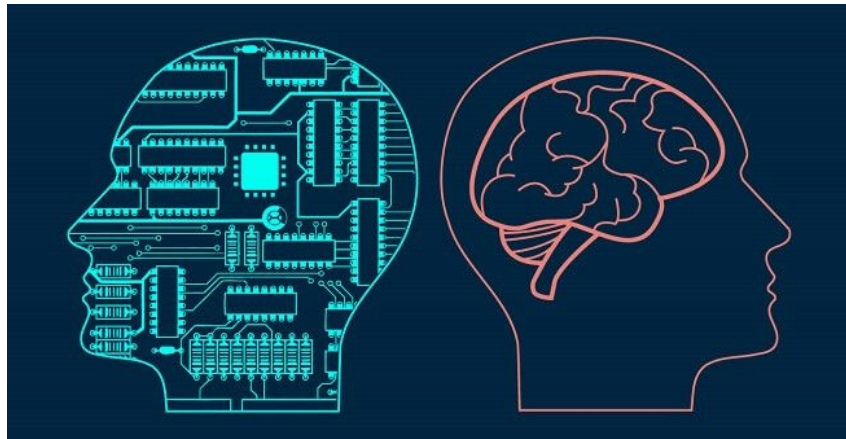


Quantum computing

Present

— — —

- Artificial Intelligence
 - LLM – Large Language Models
- Internet of Things
- Industry 4.0
- Huge connectivity (5G)
- Artificial Vision (Lidar)
- Blockchain
- Brain chips (Neuralink)
- ...



Some authors argue we are in a 5th generation, but there is no consensus

Hype Cycle for Emerging Technologies, 2020



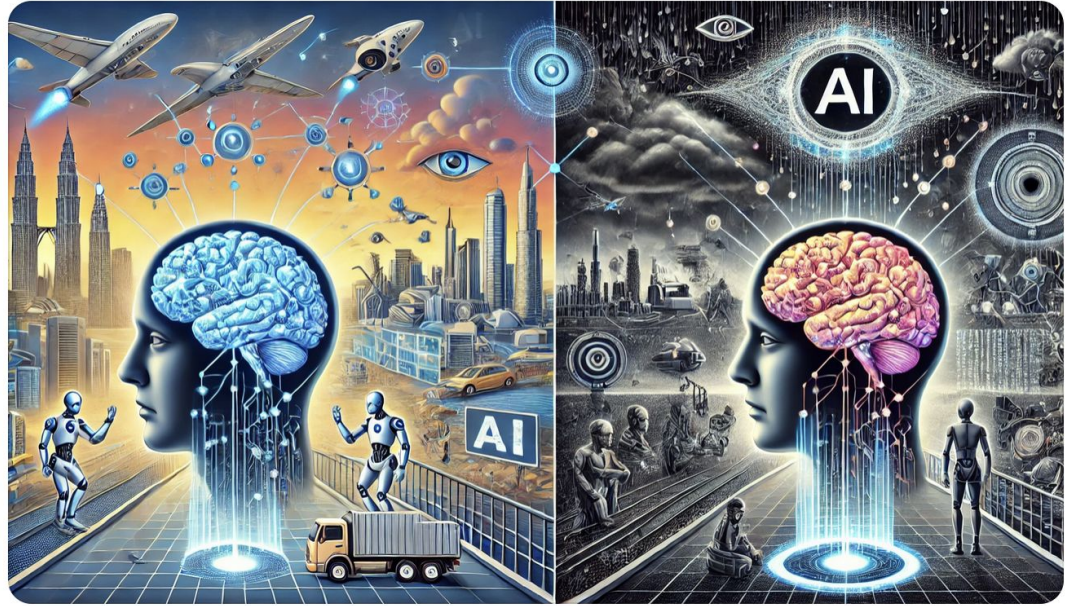
gartner.com/SmarterWithGartner

Source: Gartner
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Gartner

Future

— — —



Here is the regenerated illustration representing the concept of Artificial General Intelligence (AGI) and its potential consequences. On the left, you can see a futuristic cityscape with advanced technologies such as flying cars, robots, and holograms, symbolizing the positive impact of AGI on society. On the right, there is a dystopian scenario with AI controlling humans, surveillance drones, and a dark, oppressive environment, symbolizing the potential negative consequences. In the center, a human brain is connected to an AI brain via a glowing neural network, representing the integration of human intelligence with AGI. The contrasting colors emphasize the duality of outcomes.

Bibliography

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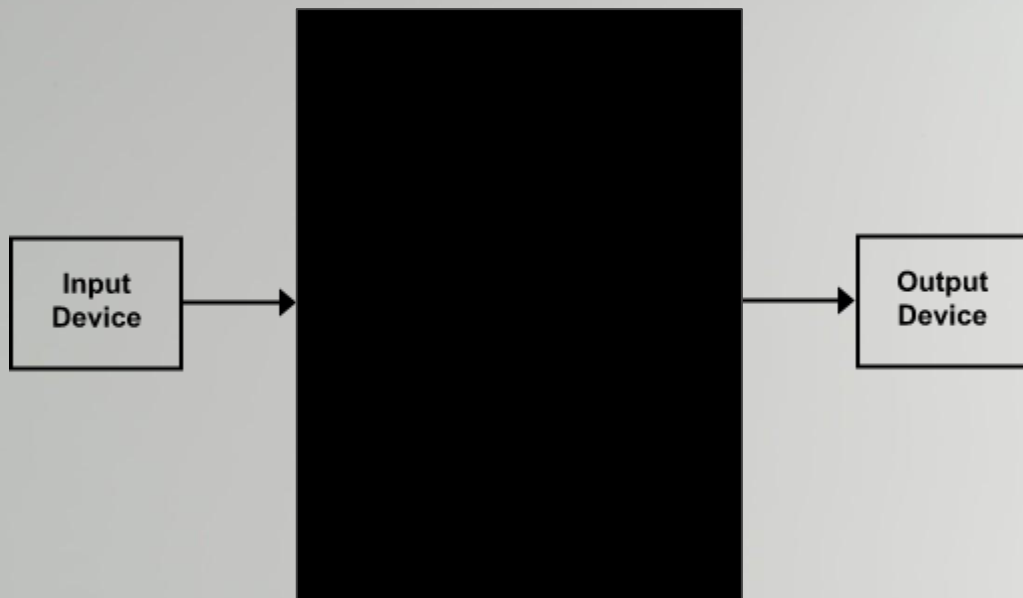
- <https://btob.co.nz/business-news/five-generations-computers/>
- <https://www.britannica.com/technology/computer/The-hobby-market-expands#ref235877>
- <http://people.bu.edu/baws/brief%20computer%20history.html>
- <https://www.computerhistory.org/timeline/computers/>
- <https://www.computerhistory.org/timeline/2015/>
- https://en.wikipedia.org/wiki/Computer#History_of_computing_hardware
- <https://historyofcomputingap.weebly.com/>
- http://news.bbc.co.uk/2/hi/in_depth/sci_tech/2007/the_computing_age/default.stm
- ...

A top-down view of a workspace. In the top right corner, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' showing. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot is also visible. Below the laptop is a brown leather folder or notebook. In the bottom right, a white spiral-bound notebook is open, and a black smartphone with a pink screen is resting on it.

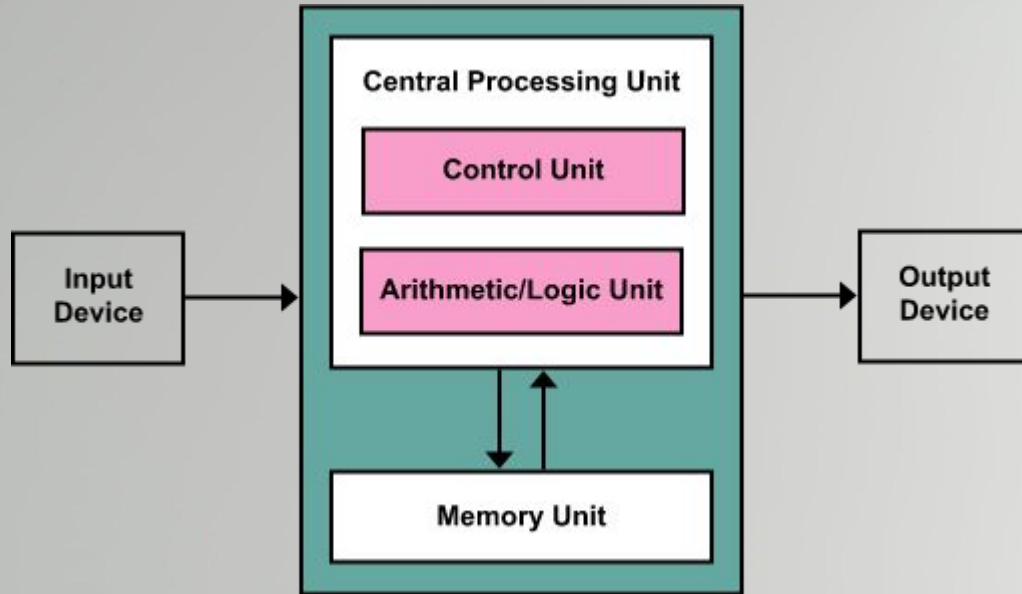
Computers

From a functional perspective

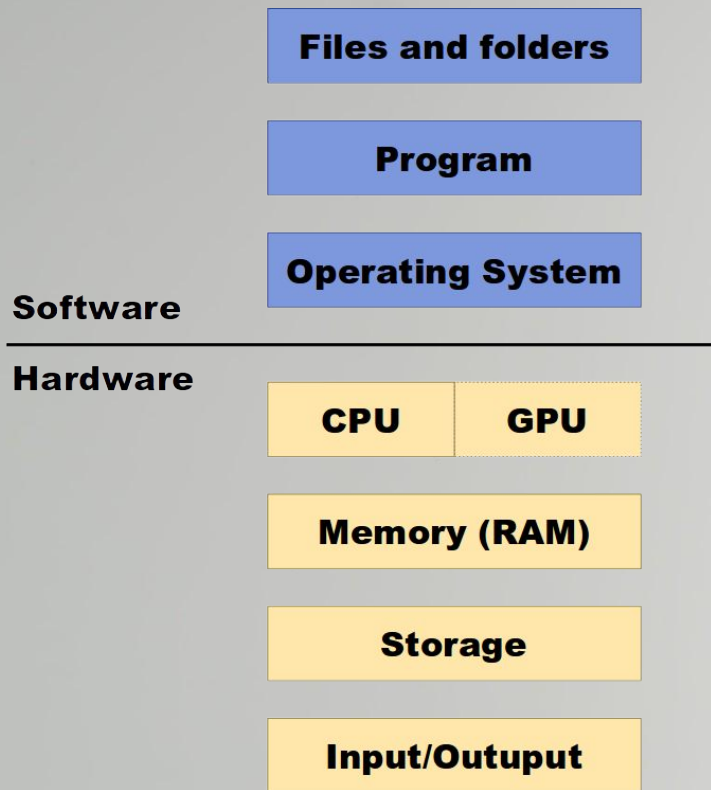
System



Computer system, the Von Neumann architecture

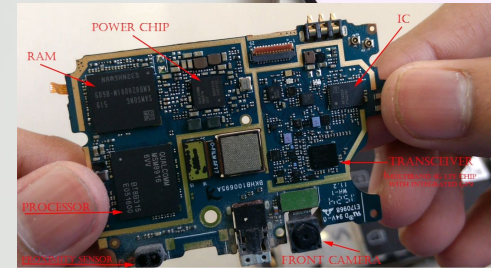
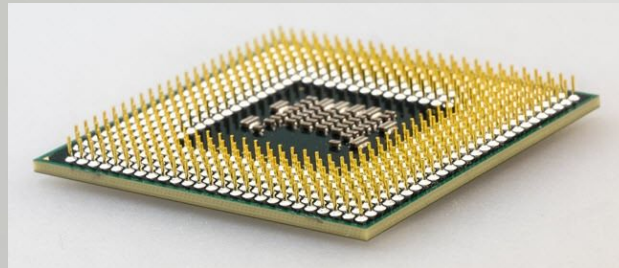


Computer system, Isaac González architecture



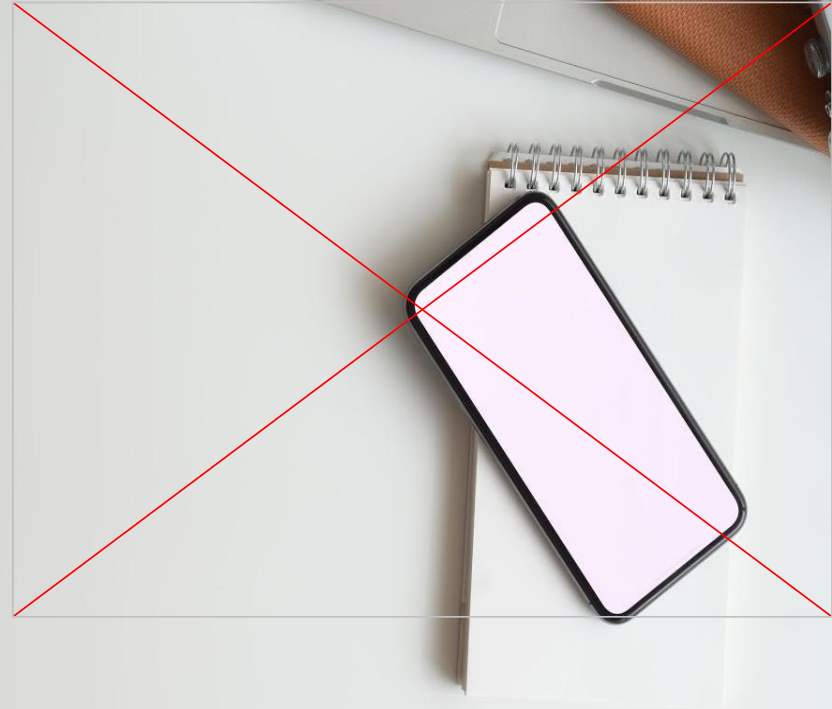
Processor or CPU

- It makes calculations (ALU)
- It controls almost everything on hardware (CU)
- Main features with a [real example](#):
 - **Clock:** 3.5GHz = 3500000000 Hertz ($\approx$ simple operations per second)
 - **Cores:** 16 (ALUs inside)
 - Architecture: 64bits (some years ago was always 32bits)
 - Manufacturer, model, serie

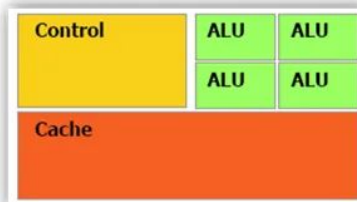


GPU - Graphical Process Unit

- It makes vectorial calculations (CPU are scalar)
- Main features
 - **Clock:** 1785 Mhz
 - **Memory:** 4GB GDDR5 (GHz)
 - **Resolution:** 1080p
 - **Power:** 150W/500W
 - **Computation:** 5.1 Tflops
 - Manufacturer, model, serie

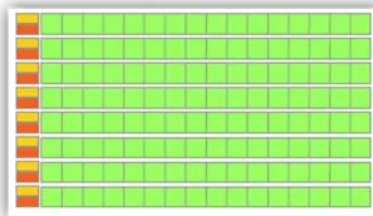


CPU



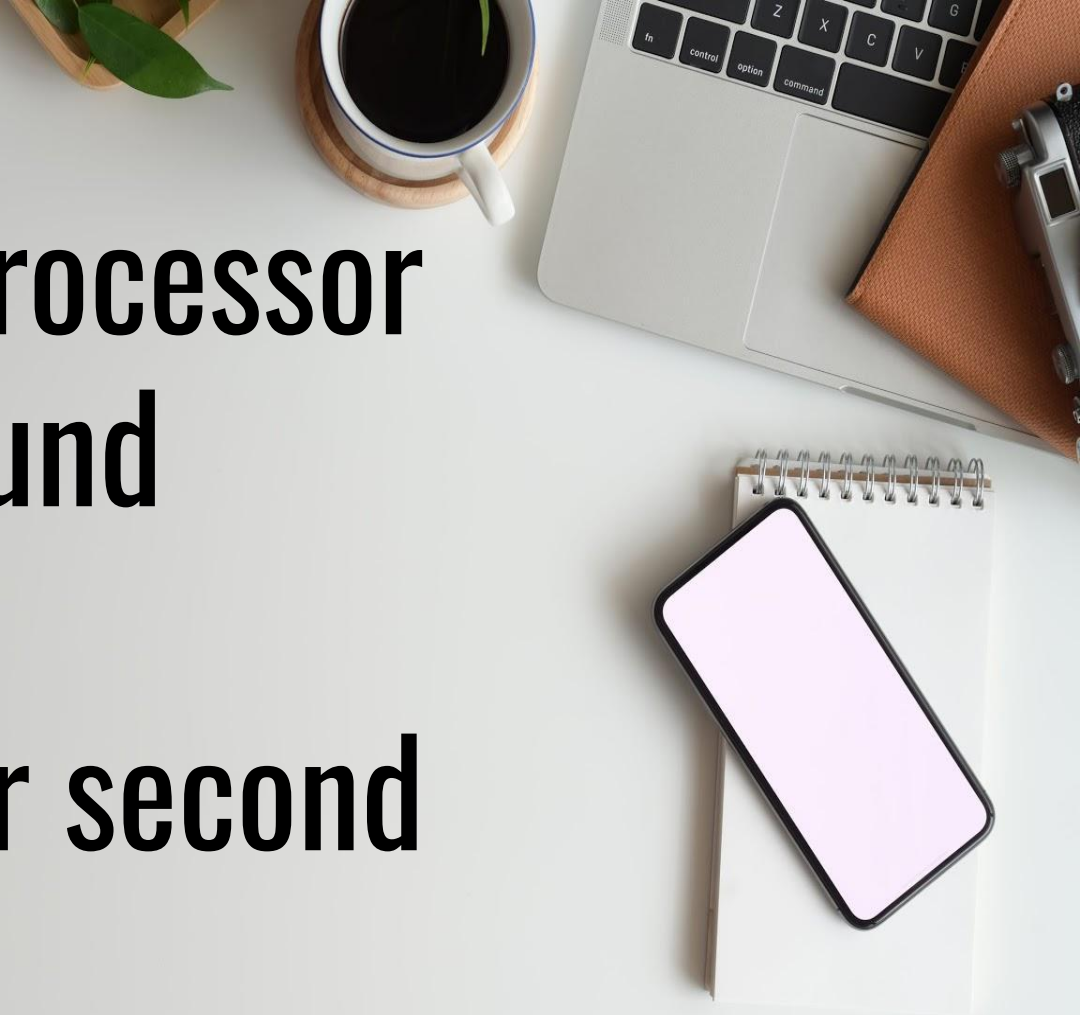
- * Low compute density
- * Complex control logic
- * Large caches (L1\$/L2\$, etc.)
- * Optimized for serial operations
 - Fewer execution units (ALUs)
 - Higher clock speeds
- * Shallow pipelines (<30 stages)
- * Low Latency Tolerance
- * Newer CPUs have more parallelism

GPU



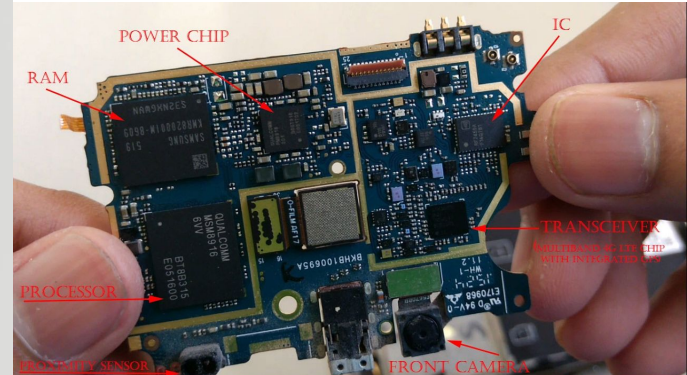
- * High compute density
- * High Computations per Memory Access
- * Built for parallel operations
 - Many parallel execution units (ALUs)
 - Graphics is the best known case of parallelism
- * Deep pipelines (hundreds of stages)
- * High Throughput
- * High Latency Tolerance
- * Newer GPUs:
 - Better flow control logic (becoming more CPU-like)
 - Scatter/Gather Memory Access
 - Don't have one-way pipelines anymore

Nowadays a processor
can make around
60 Billion
operations per second



Memory (RAM)

- It stores the data (binary) while the processor is running
- Other names:
 - RAM: Random Memory Access
 - Volatile memory
 - Memory
- Features and [real example](#):
 - Size (Bytes): 16 GB
 - Technology: DDR4



Input/Output (I)

Some parts of the computer provide data, receive data or both. It's common to call them peripherals.

Please, tell me some of them you know and if they are input, output or input-output devices.



Input/output (II)

Input:

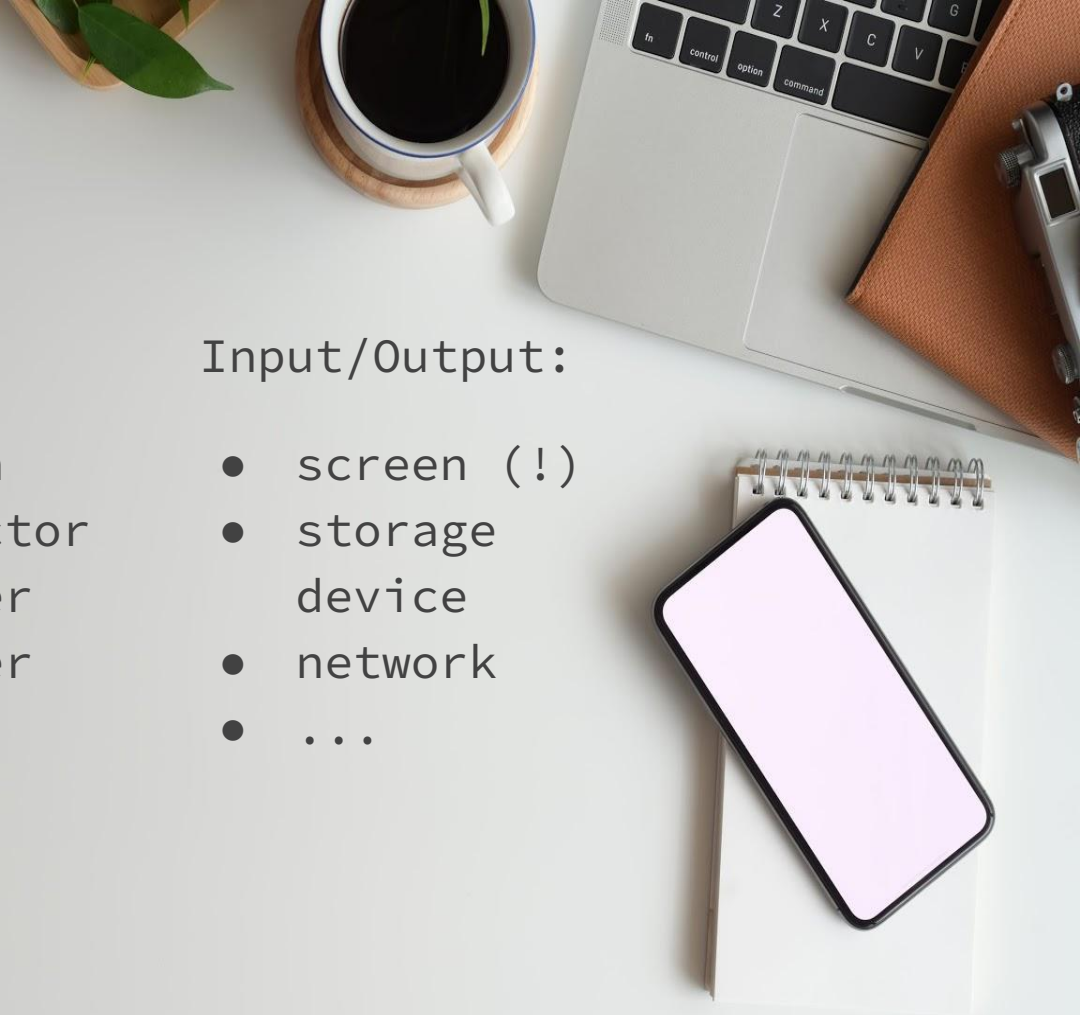
- keyboard
- pointing devices
- camera
- scanner
- microphone
- biometric devices
- ...

Output:

- screen
- projector
- printer
- speaker
- ...

Input/Output:

- screen (!)
- storage device
- network
- ...



Keyboard

— — —



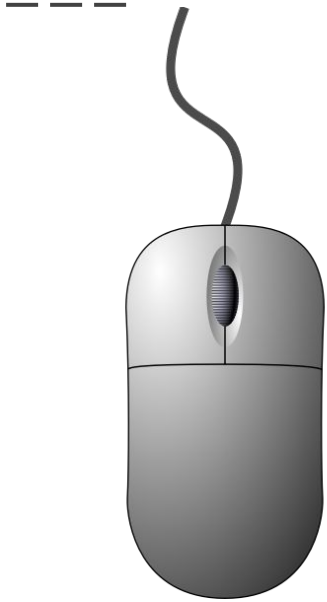
- | | | |
|--|--|---|
|  Typewriter keys |  Function keys |  Enter keys |
|  System keys |  Numeric keypad |  Other |
|  Application key |  Cursor control keys | |

Typing the right way

— — —



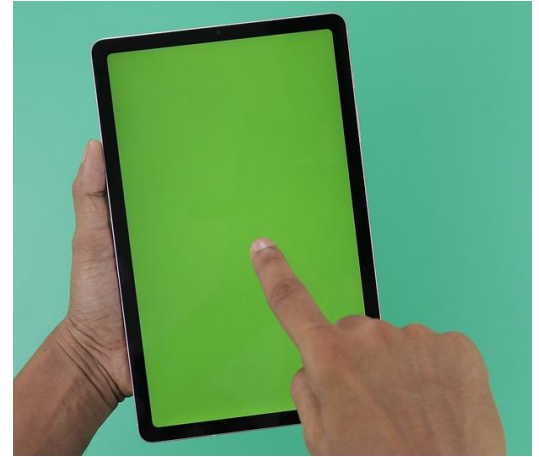
Pointing devices



Mouse



Touchpad



Screen

Other inputs

— — —

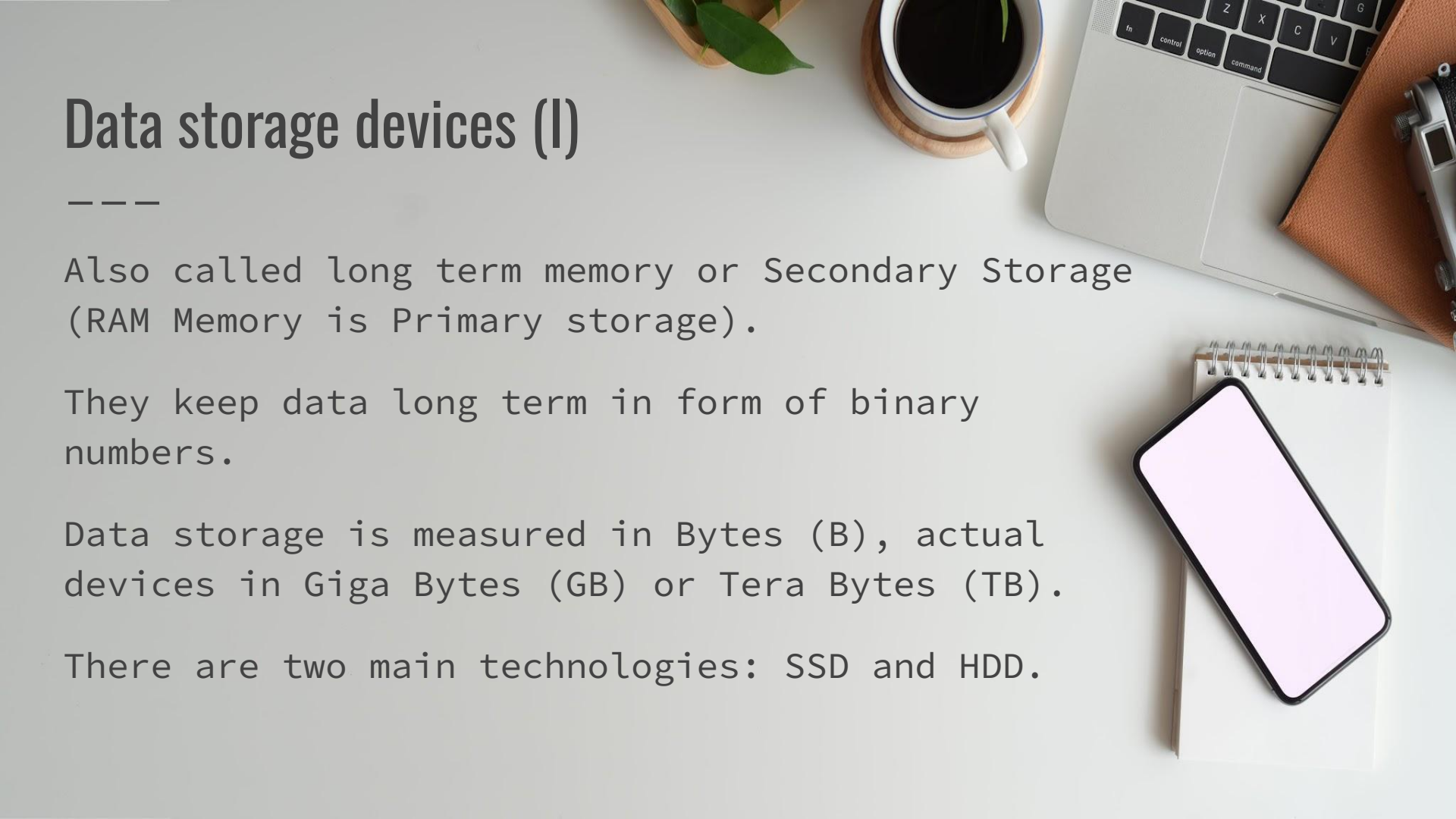


Outputs

— — —



Data storage devices (I)

A top-down view of a desk setup. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V' shown. Next to it is a white cup of black coffee on a wooden coaster. To the left of the cup is a small wooden planter with green leaves. Below the laptop is a brown leather folder or notebook. In the bottom right, a white spiral-bound notebook is open, and a black smartphone with a pink screen is resting on it.

Also called long term memory or Secondary Storage
(RAM Memory is Primary storage).

They keep data long term in form of binary numbers.

Data storage is measured in Bytes (B), actual devices in Giga Bytes (GB) or Tera Bytes (TB).

There are two main technologies: SSD and HDD.

Input/Output - Storage devices (II)

SSD VS HDD

(Solid State Drive)

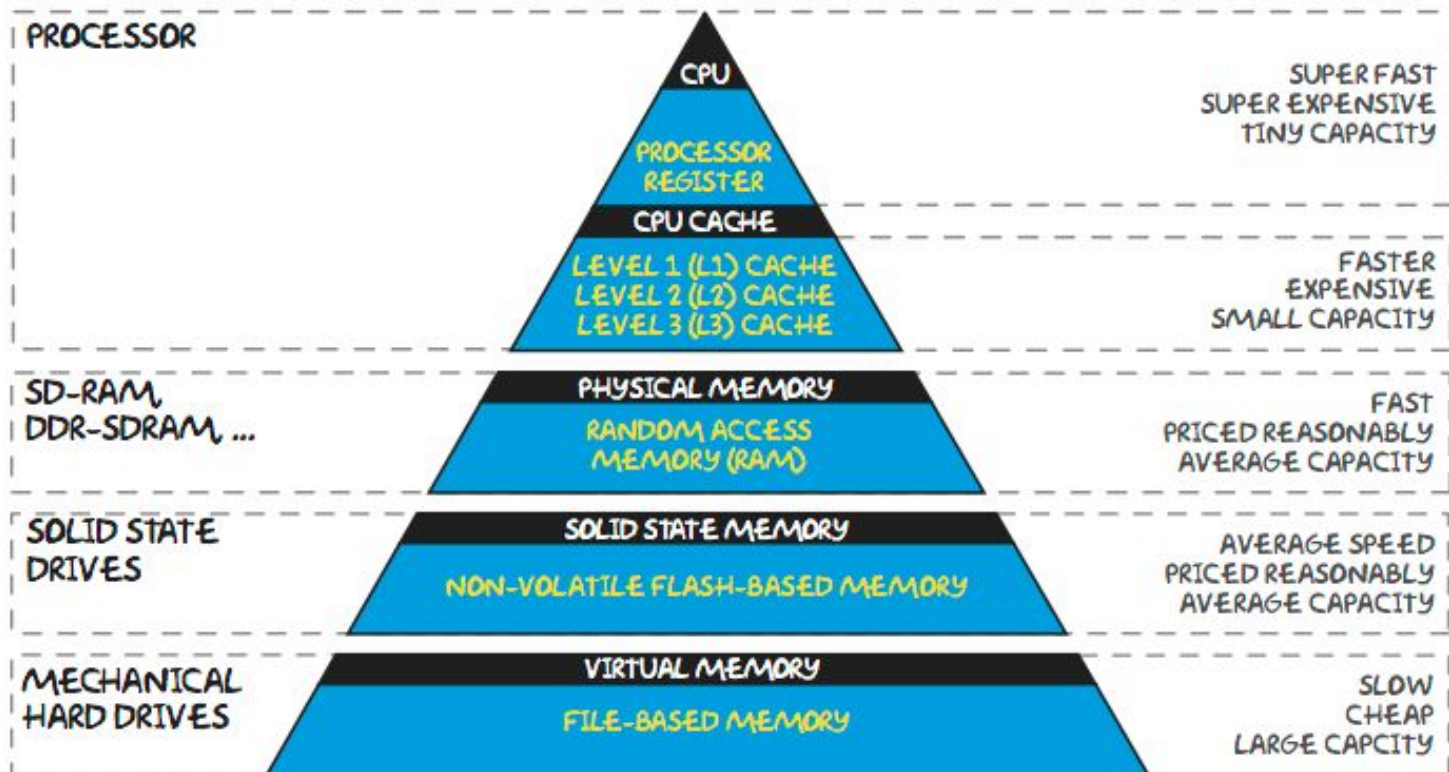
(Hard Disk Drive)



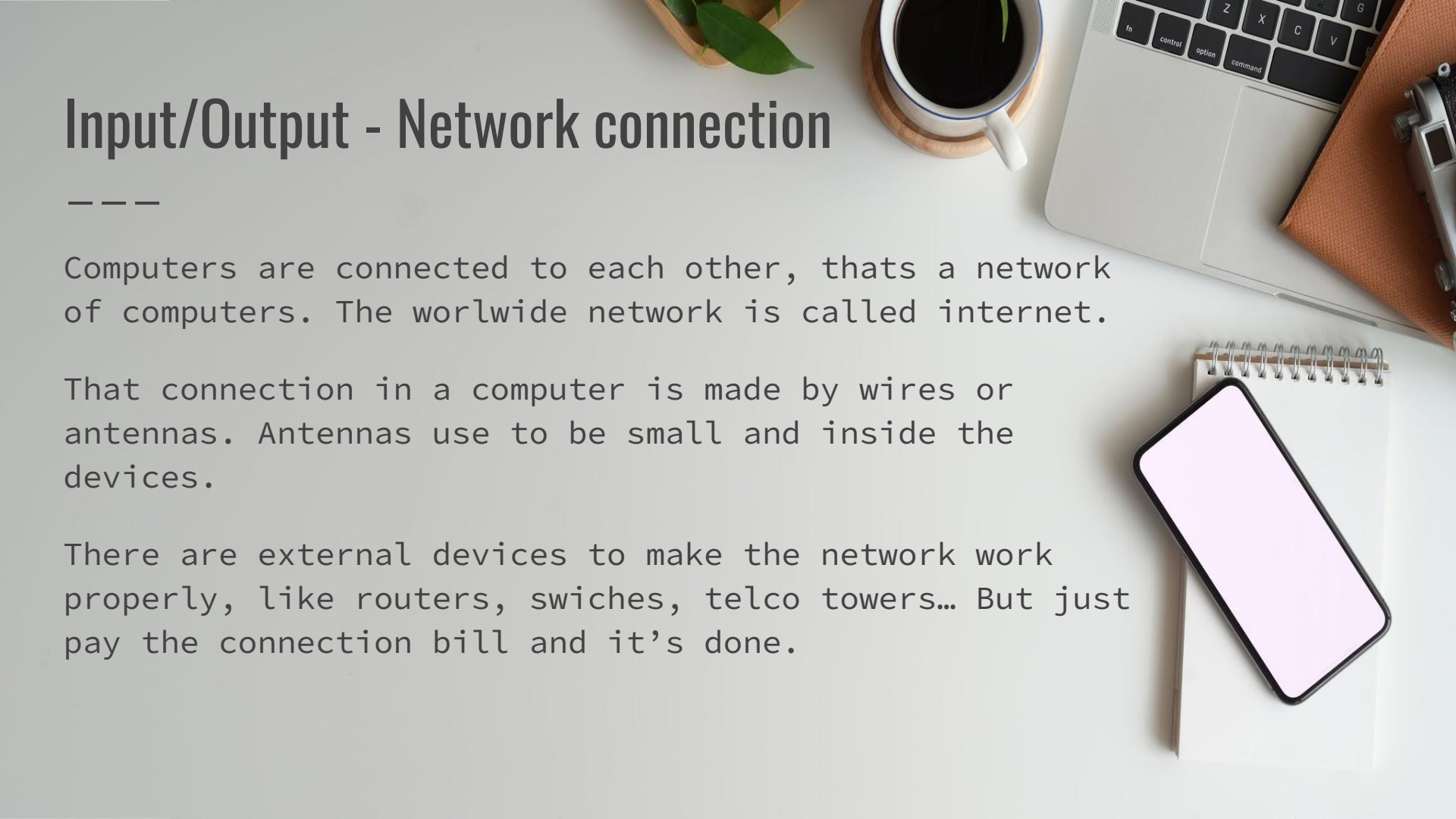
- smaller
240GB - 1TB
- faster
- expensive
- noiseless

- bigger
1TB - 10TB
- slower
- cheaper
- noisy

THE MEMORY HIERARCHY



Input/Output - Network connection

A top-down view of a workspace. In the upper right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot sits above the cup. In the lower right, a smartphone with a pink screen lies on a white spiral-bound notebook. The background is a light gray surface.

Computers are connected to each other, that's a network of computers. The worldwide network is called internet.

That connection in a computer is made by wires or antennas. Antennas used to be small and inside the devices.

There are external devices to make the network work properly, like routers, switches, telco towers... But just pay the connection bill and it's done.

Input/Output - Network devices

— — —



A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot is in the top center. Below the laptop is a brown leather folder. In the bottom right, a white spiral-bound notebook has a black smartphone with a pink screen resting on it.

Computer

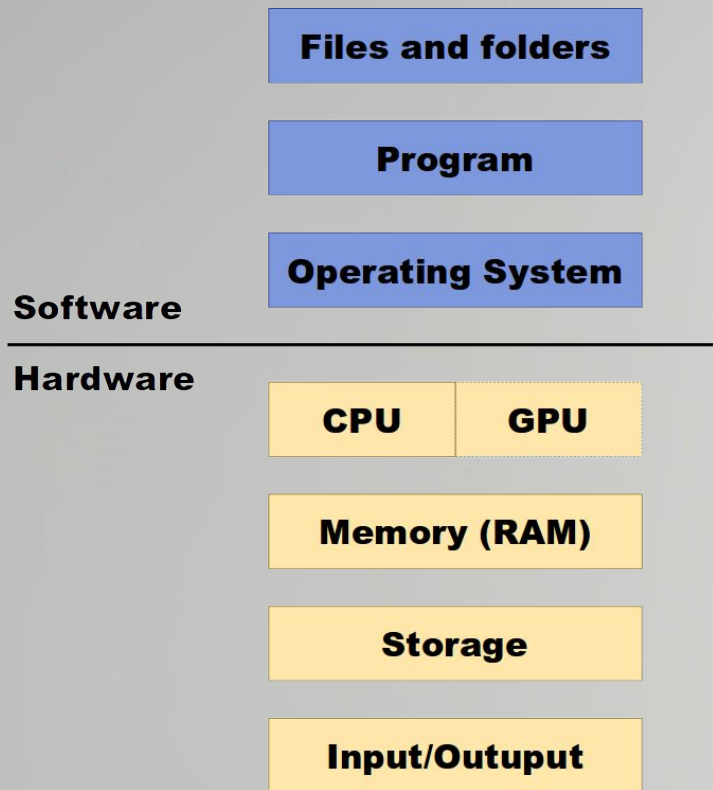
In practice

What happend when

- Type a letter
- Move the mouse
- Open a program
- Print a document
- Open a website in a browser
- ...



Computer system, Isaac González architecture





A top-down view of a workspace. In the upper right, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white coffee cup with a blue rim, filled with dark liquid, sitting on a wooden coaster. To the left of the cup is a small wooden planter with green leaves. In the lower right, a smartphone with a pink screen is placed on a white surface.

Information

A top-down view of a workspace. In the top right corner, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white coffee cup with a blue rim on a wooden coaster. A small potted plant with green leaves is in the top center. A brown leather folder or notebook is on the right. In the bottom right, a spiral-bound notebook is open with a pink smartphone resting on it.

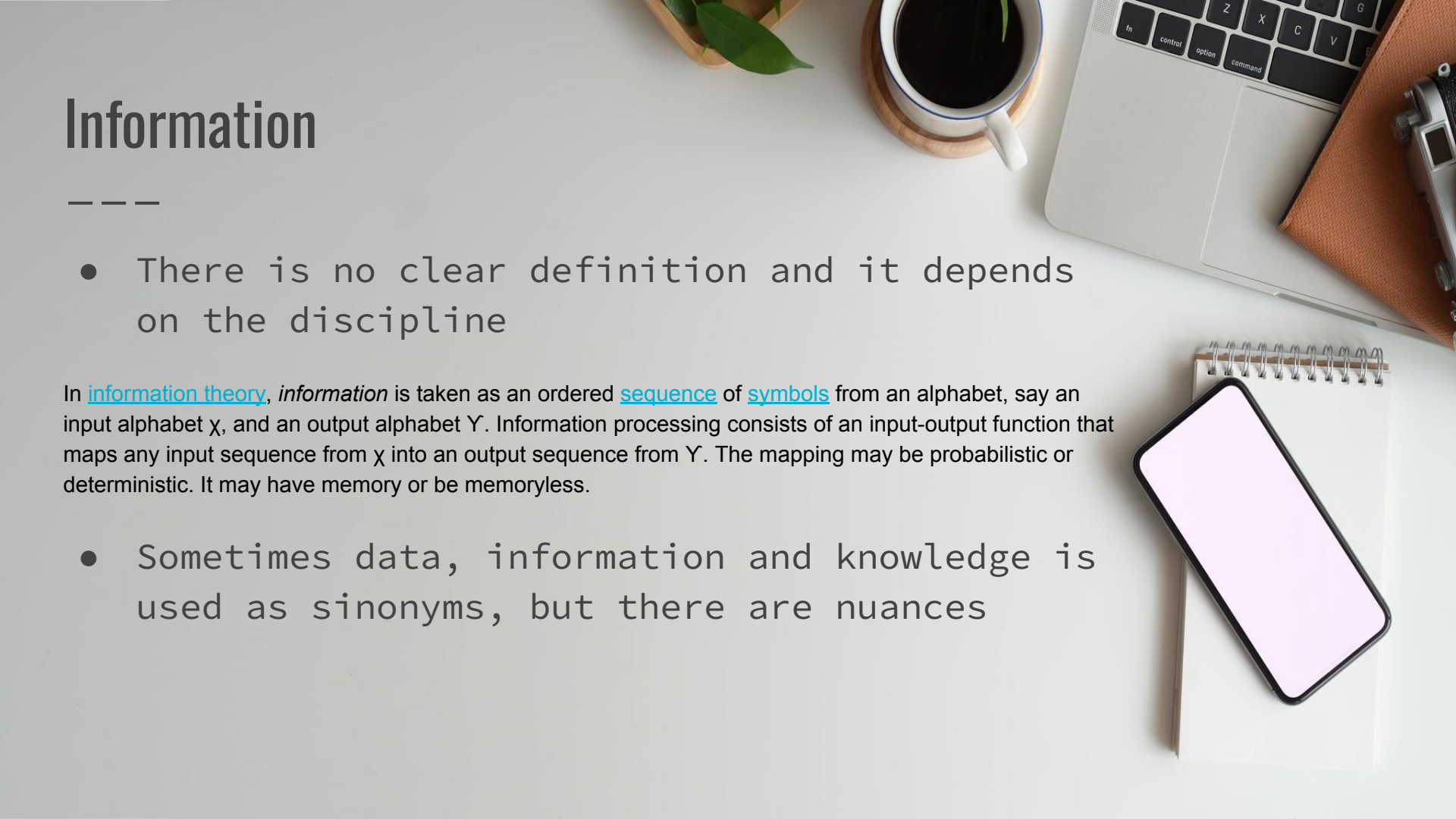
Information

Definition

A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter is in the top center. Below the laptop is a brown leather folder or notebook. In the bottom right, a spiral-bound notebook is open, and a smartphone with a pink screen is resting on it.

What is information?

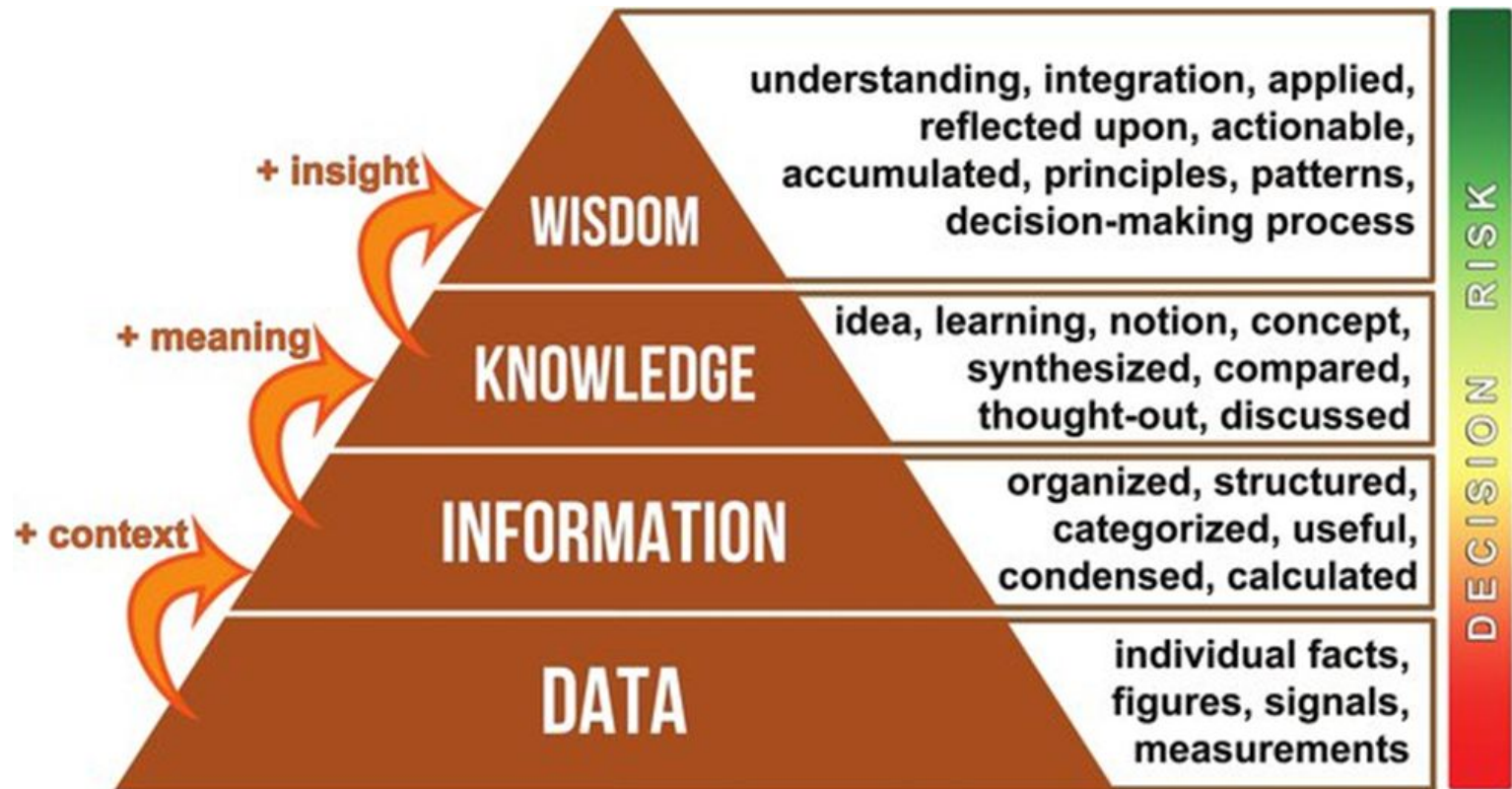
Information

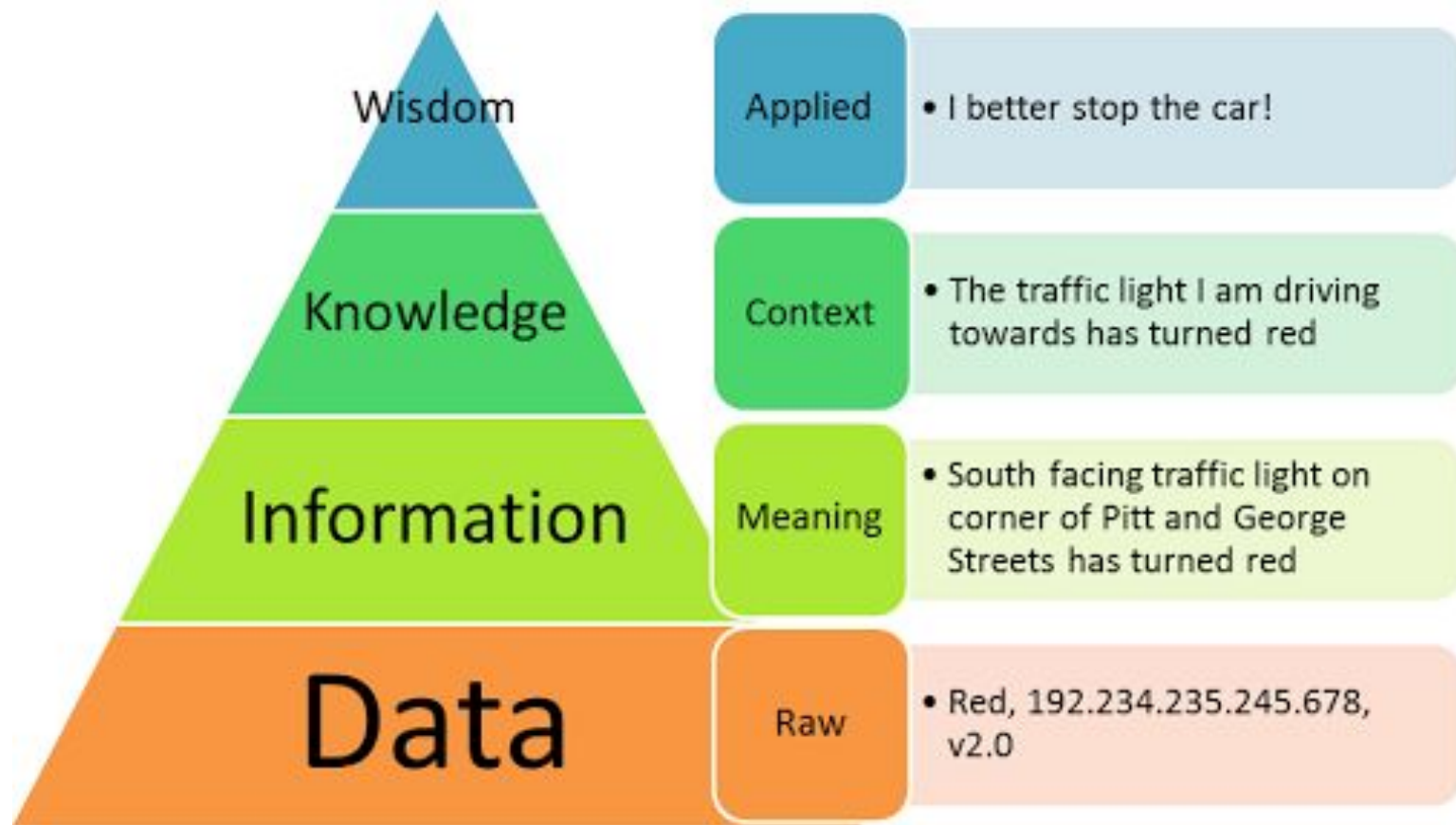
A top-down view of a desk with a laptop, a cup of coffee, a small potted plant, a spiral notebook, and a smartphone. The laptop is silver and partially visible on the right. A white cup with a blue rim sits on a wooden coaster. A small potted plant with green leaves is in the top left. A spiral notebook is open, and a smartphone with a pink screen is resting on it.

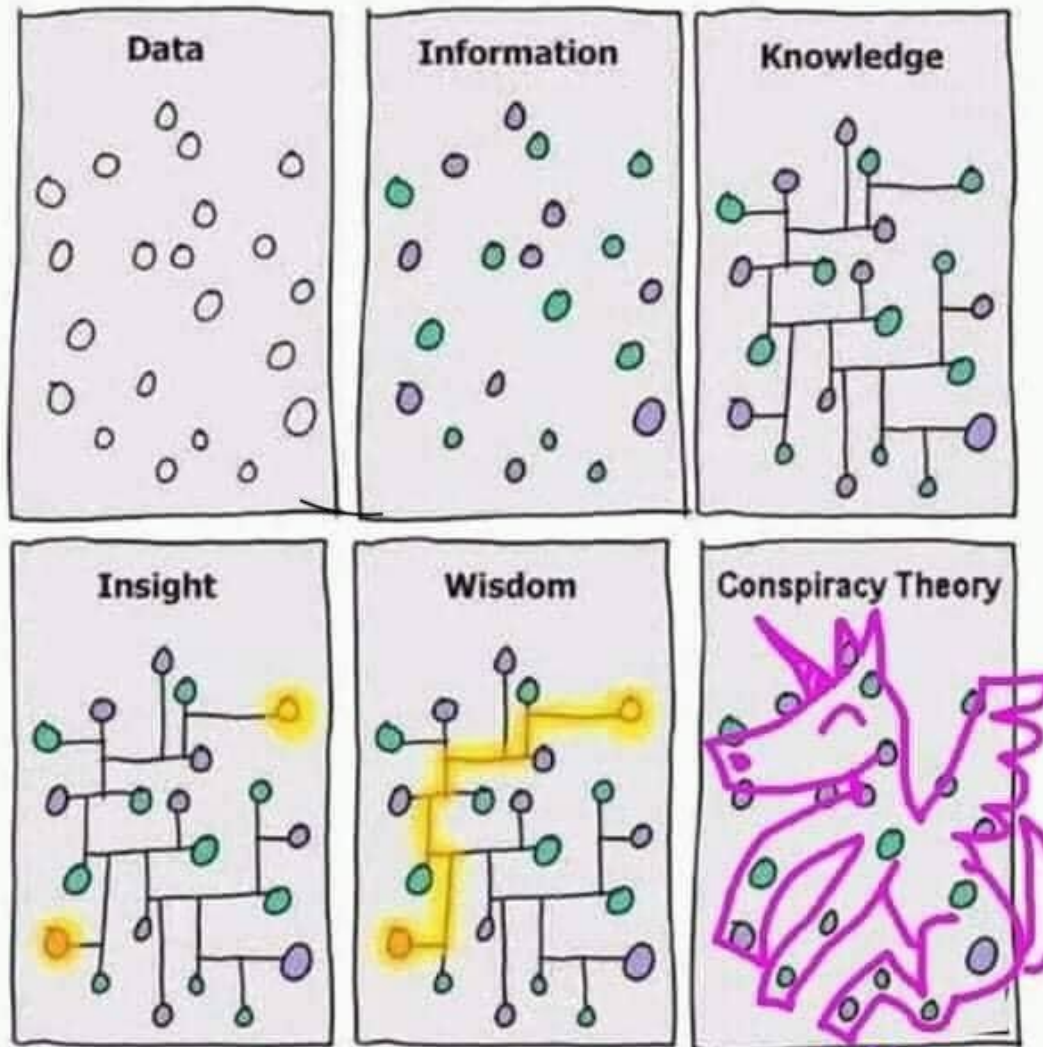
- There is no clear definition and it depends on the discipline

In [information theory](#), *information* is taken as an ordered [sequence](#) of [symbols](#) from an alphabet, say an input alphabet X , and an output alphabet Y . Information processing consists of an input-output function that maps any input sequence from X into an output sequence from Y . The mapping may be probabilistic or deterministic. It may have memory or be memoryless.

- Sometimes data, information and knowledge is used as synonyms, but there are nuances







aka QAnon

What type of
information you use in
a computer?



Information

— — —

Humans:

- Numbers
- Texts
- Images (Colors)
- Audio
- Video
- Time
- Place on earth
- Documents
- Websites
- ...

Computer:

- Binary numbers



Do you know how to
count numbers?





000000





000000



Binary to denary

16 8 4 2 1
1 0 1 1 1

Denary to binary

21

Hexadecimal

Number	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111
Hexadecimal	0	1	2	3	4	5	6	7

Number	8	9	10	11	12	13	14	15
Binary	1000	1001	1010	1011	1100	1101	1110	1111
Hexadecimal	8	9	A	B	C	D	E	F

Binary - Decimal - Hexadecimal

1010110001110010

Binary - Decimal - Hexadecimal

1010110001110010

44146

Binary - Decimal - Hexadecimal

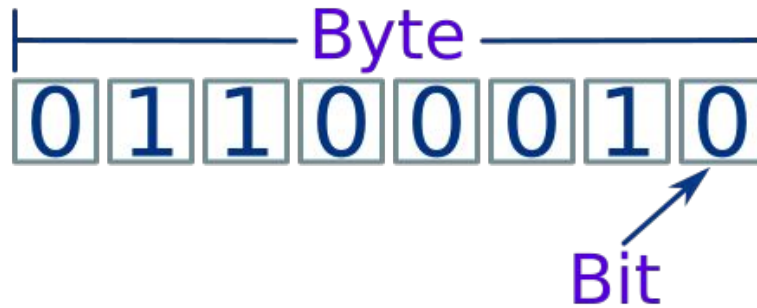
1010110001110010

44146

AC72

Bits and Bytes

— — —



How many bits and how many Bytes you see?

1010110001110010

How many bits and how many Bytes you see?

1010110001110010

16 bits

2 Bytes

International System Prefix

— — —

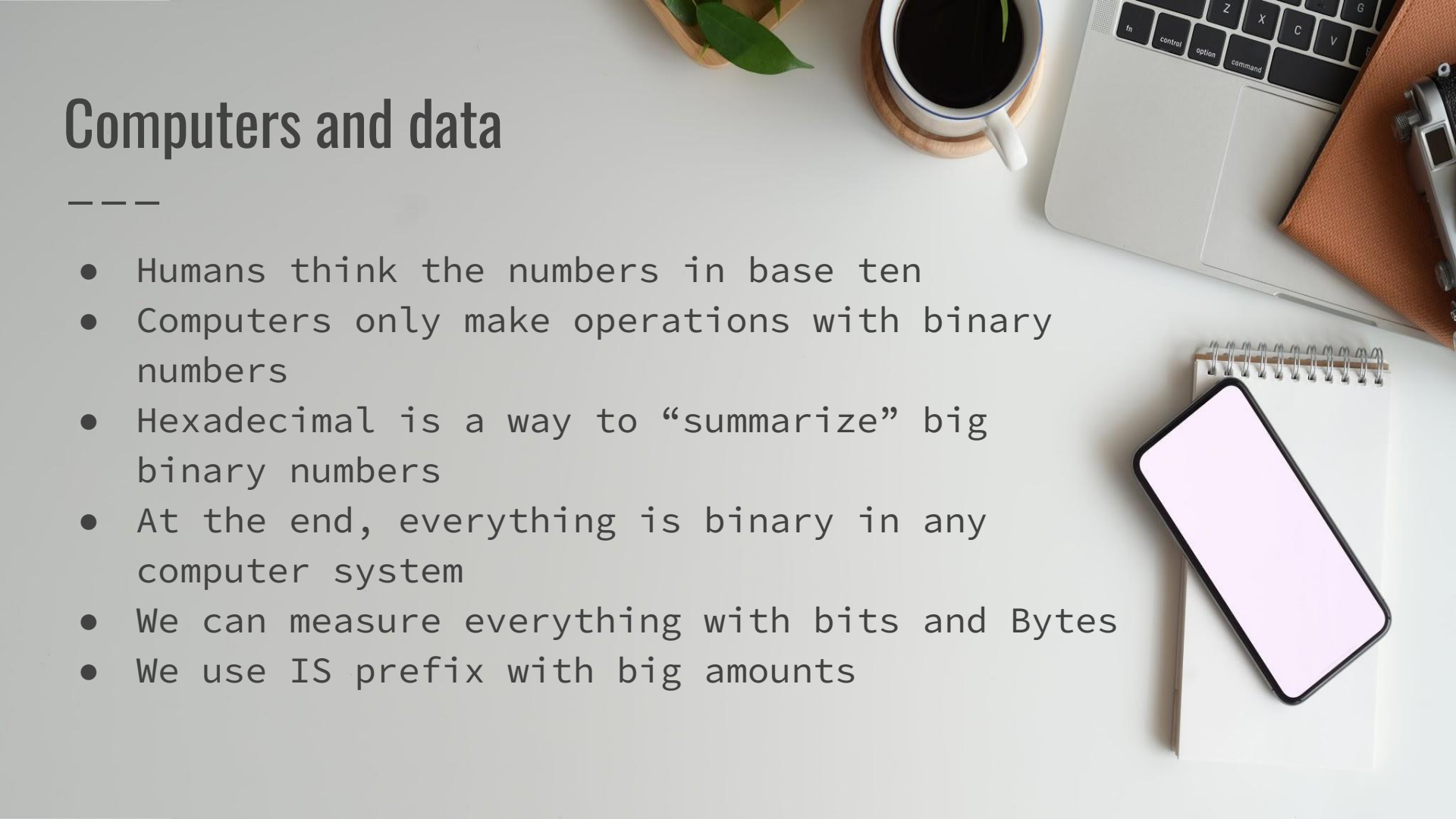
Abbr.	Prefix name	Decimal size	Size in thousands
K	kilo-	10^3	$1,000$
M	mega-	10^6	$1,000^2$
G	giga-	10^9	$1,000^3$
T	tera-	10^{12}	$1,000^4$
P	peta-	10^{15}	$1,000^5$
E	exa-	10^{18}	$1,000^6$

IS Prefix in Computer Science

— — —

Abbr.	Prefix name	Decimal size	Size in thousands	Binary approximation	Address variable size
K	kilo-	10^3	1,000	$1,024 = 2^{10}$	10
M	mega-	10^6	$1,000^2$	$1,024^2 = 2^{20}$	20
G	giga-	10^9	$1,000^3$	$1,024^3 = 2^{30}$	30
T	tera-	10^{12}	$1,000^4$	$1,024^4 = 2^{40}$	40
P	peta-	10^{15}	$1,000^5$	$1,024^5 = 2^{50}$	50
E	exa-	10^{18}	$1,000^6$	$1,024^6 = 2^{60}$	60

Computers and data

A top-down view of a desk with a laptop, a cup of coffee, a small potted plant, a spiral notebook, and a smartphone. The laptop is silver and partially visible on the right. A white cup with a blue rim sits on a wooden coaster. A small potted plant with green leaves is in the top left. A spiral notebook is open, and a smartphone with a pink screen is resting on it.

- Humans think the numbers in base ten
- Computers only make operations with binary numbers
- Hexadecimal is a way to “summarize” big binary numbers
- At the end, everything is binary in any computer system
- We can measure everything with bits and Bytes
- We use IS prefix with big amounts

How to represent all
the human information
only with binary
numbers?



Information

Humans:

- Numbers
- Texts
- Rich texts
- Images (Colors)
- Audio
- Video
- Time
- Place on earth
- Websites
- ...



Computer:

- Binary numbers



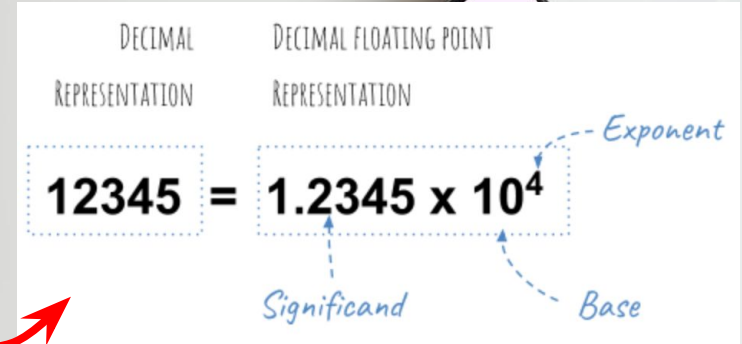
Numbers



Numbers

- We already know to write numbers for computers, right?
- Actual computers use 64bits (8Bytes) for each number in arithmetical operations
- Type of numbers:
 - Integer 15
 - Positive/Negative -15
 - Real 1.5
 - Scientific notation **6.02E23**

How to transform to binary?



Unsigned integer, signed integer

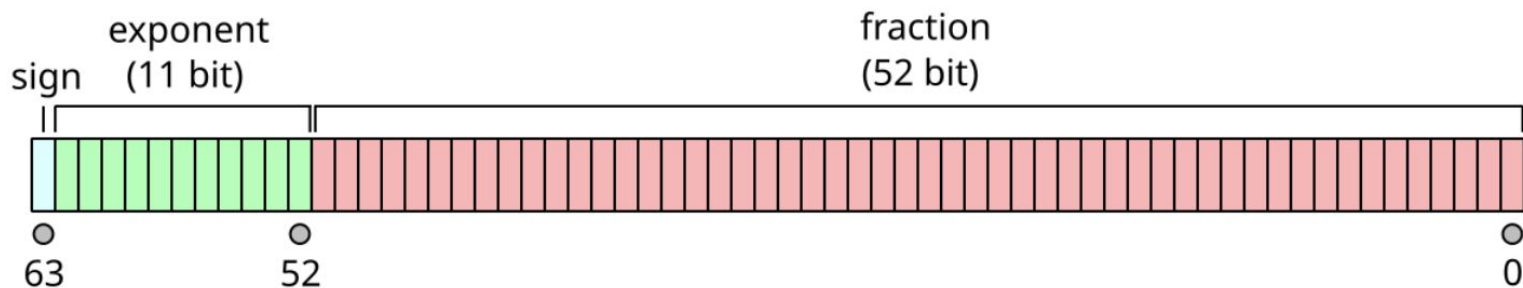
Unsigned Binary

Signed Binary

00100101 ₂ = 37	00100101 ₂ = +37
10100101 ₂ = 165	10100101 ₂ = -37

Floating Point (IEEE-754)

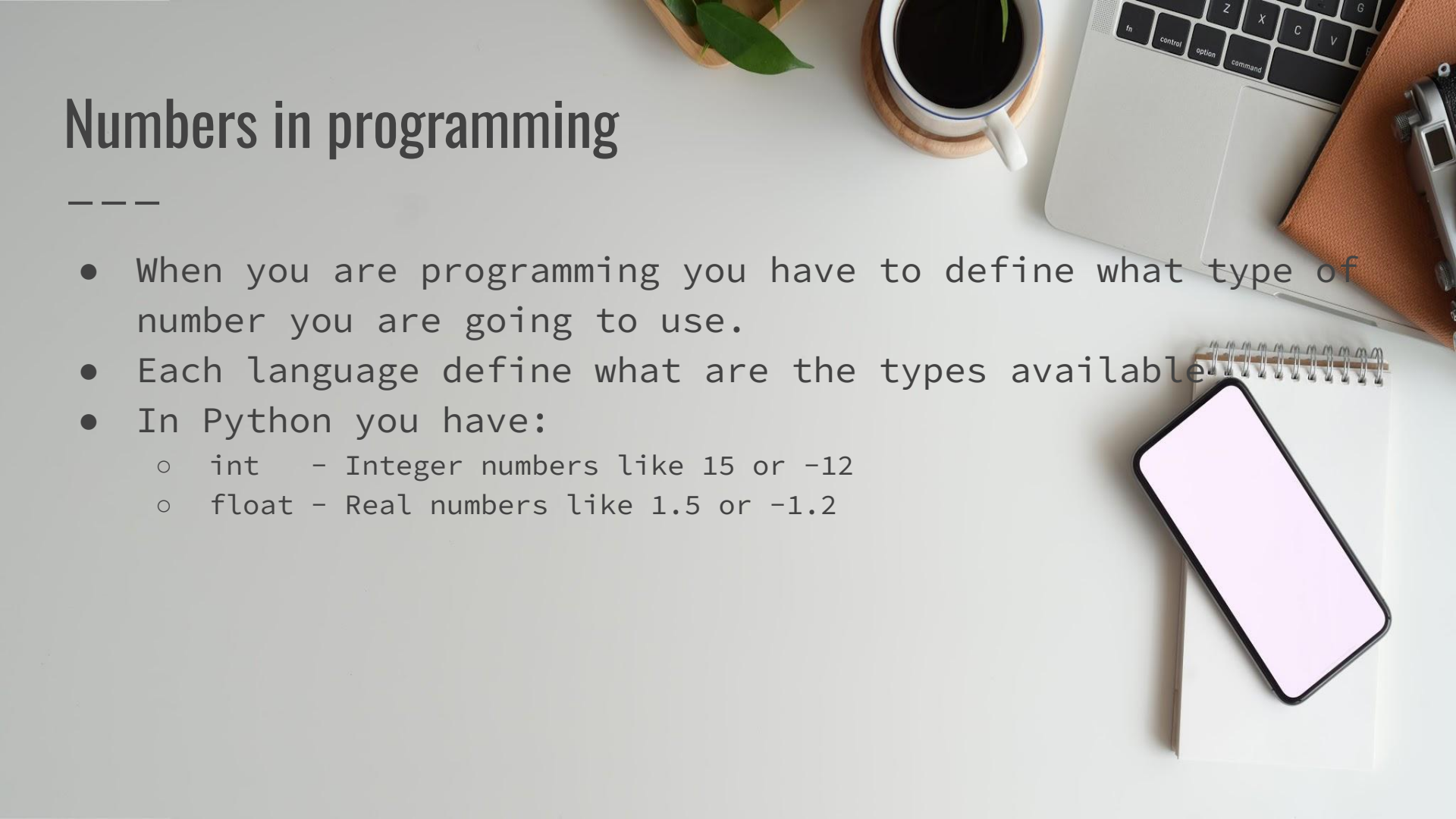
— — —



The real value assumed by a given 64-bit double-precision datum with a given **biased exponent** e and a 52-bit fraction is

$$(-1)^{\text{sign}} (1.b_{51}b_{50}\dots b_0)_2 \times 2^{e-1023}$$

Numbers in programming

A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' showing. Next to it is a white cup of black coffee on a wooden coaster. To the left of the cup is a small wooden tray with green leaves. Below the laptop is a brown leather folder or notebook. In the bottom right, a black smartphone with a pink screen lies on a white spiral-bound notebook.

- When you are programming you have to define what type of number you are going to use.
- Each language define what are the types available
- In Python you have:
 - `int` - Integer numbers like 15 or -12
 - `float` - Real numbers like 1.5 or -1.2

A top-down view of a workspace. In the top right corner, there is a silver laptop with a black keyboard. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot is also visible. Below the laptop is a brown leather folder or notebook. In the bottom right corner, there is a white spiral-bound notebook with a black smartphone resting on it. The phone's screen is a solid light pink color.

Texts

Plain text

Texts in binary numbers?



Hello
my name is

01010100 01111001 01110010
01100001 01111001



Texts in binary numbers?

- How to represent a letter in binary?
- How to represent a text in binary?
- What about special characters?
- What about accent marks?
- What about different alphabets?
- What about emojis?
- ...



Decimal - Binary - Octal - Hex – ASCII Conversion Chart

Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII
0	00000000	000	00	NUL	32	00100000	040	20	SP	64	01000000	100	40	@	96	01100000	140	60	`
1	00000001	001	01	SOH	33	00100001	041	21	!	65	01000001	101	41	A	97	01100001	141	61	a
2	00000010	002	02	STX	34	00100010	042	22	"	66	01000010	102	42	B	98	01100010	142	62	b
3	00000011	003	03	ETX	35	00100011	043	23	#	67	01000011	103	43	C	99	01100011	143	63	c
4	00000100	004	04	EOT	36	00100100	044	24	\$	68	01000100	104	44	D	100	01100100	144	64	d
5	00000101	005	05	ENQ	37	00100101	045	25	%	69	01000101	105	45	E	101	01100101	145	65	e
6	00000110	006	06	ACK	38	00100110	046	26	&	70	01000110	106	46	F	102	01100110	146	66	f
7	00000111	007	07	BEL	39	00100111	047	27	'	71	01000111	107	47	G	103	01100111	147	67	g
8	00001000	010	08	BS	40	00101000	050	28	(	72	01001000	110	48	H	104	01101000	150	68	h
9	00001001	011	09	HT	41	00101001	051	29	)	73	01001001	111	49	I	105	01101001	151	69	i
10	00001010	012	0A	LF	42	00101010	052	2A	*	74	01001010	112	4A	J	106	01101010	152	6A	j
11	00001011	013	0B	VT	43	00101011	053	2B	+	75	01001011	113	4B	K	107	01101011	153	6B	k
12	00001100	014	0C	FF	44	00101100	054	2C	,	76	01001100	114	4C	L	108	01101100	154	6C	l
13	00001101	015	0D	CR	45	00101101	055	2D	-	77	01001101	115	4D	M	109	01101101	155	6D	m
14	00001110	016	0E	SO	46	00101110	056	2E	.	78	01001110	116	4E	N	110	01101110	156	6E	n
15	00001111	017	0F	SI	47	00101111	057	2F	/	79	01001111	117	4F	O	111	01101111	157	6F	o
16	00010000	020	10	DLE	48	00110000	060	30	0	80	01010000	120	50	P	112	01110000	160	70	p
17	00010001	021	11	DC1	49	00110001	061	31	1	81	01010001	121	51	Q	113	01110001	161	71	q
18	00010010	022	12	DC2	50	00110010	062	32	2	82	01010010	122	52	R	114	01110010	162	72	r
19	00010011	023	13	DC3	51	00110011	063	33	3	83	01010011	123	53	S	115	01110011	163	73	s
20	00010100	024	14	DC4	52	00110100	064	34	4	84	01010100	124	54	T	116	01110100	164	74	t
21	00010101	025	15	NAK	53	00110101	065	35	5	85	01010101	125	55	U	117	01110101	165	75	u
22	00010110	026	16	SYN	54	00110110	066	36	6	86	01010110	126	56	V	118	01110110	166	76	v
23	00010111	027	17	ETB	55	00110111	067	37	7	87	01010111	127	57	W	119	01110111	167	77	w
24	00011000	030	18	CAN	56	00111000	070	38	8	88	01011000	130	58	X	120	01111000	170	78	x
25	00011001	031	19	EM	57	00111001	071	39	9	89	01011001	131	59	Y	121	01111001	171	79	y
26	00011010	032	1A	SUB	58	00111010	072	3A	:	90	01011010	132	5A	Z	122	01111010	172	7A	z
27	00011011	033	1B	ESC	59	00111011	073	3B	;	91	01011011	133	5B	[	123	01111011	173	7B	{
28	00011100	034	1C	FS	60	00111100	074	3C	<	92	01011100	134	5C	\	124	01111100	174	7C	
29	00011101	035	1D	GS	61	00111101	075	3D	=	93	01011101	135	5D	]	125	01111101	175	7D	}
30	00011110	036	1E	RS	62	00111110	076	3E	>	94	01011110	136	5E	^	126	01111110	176	7E	~
31	00011111	037	1F	US	63	00111111	077	3F	?	95	01011111	137	5F	_	127	01111111	177	7F	DEL

6.2. ISO 8859 family

Year	Norm	Description	Variant
1987	ISO 8859-1	Western European: German, French, Italian, ...	cp1252
1987	ISO 8859-2	Central European: Croatian, Polish, Czech, ...	cp1250
1988	ISO 8859-3	South European: Turkish and Esperanto	•
1988	ISO 8859-4	North European -	
1988	ISO 8859-5	Latin/Cyrillic: Macedonian, Russian, ...	KOI family
1987	ISO 8859-6	Latin/Arabic: Arabic language characters	cp1256
1987	ISO 8859-7	Latin/Greek: modern Greek language	cp1253
1988	ISO 8859-8	Latin/Hebrew: modern Hebrew alphabet	cp1255
1989	ISO 8859-9	Turkish: Largely the same as ISO 8859-1	cp1254
1992	ISO 8859-10	Nordic: a rearrangement of Latin-4	•
2001	ISO 8859-11	Latin/Thai: Thai language	TIS 620, cp874
1998	ISO 8859-13	Baltic Rim: Baltic languages	cp1257
1998	ISO 8859-14	Celtic: Gaelic, Breton	•
1999	ISO 8859-15	Revision of 8859-1: euro sign	cp1252
2001	ISO 8859-16	South-Eastern European	•

ISO/IEC 8859-1																
	_0	_1	_2	_3	_4	_5	_6	_7	_8	_9	_A	_B	_C	_D	_E	_F
0_0																
1_16																
2_32	SP 0020	! 0021	" 0022	# 0023	\$ 0024	% 0025	& 0026	' 0027	(0028	) 0029	* 002A	+ 002B	, 002C	- 002D	. 002E	/ 002F
3_48	0 0030	1 0031	2 0032	3 0033	4 0034	5 0035	6 0036	7 0037	8 0038	9 0039	: 003A	; 003B	< 003C	= 003D	> 003E	? 003F
4_64	@ 0040	A 0041	B 0042	C 0043	D 0044	E 0045	F 0046	G 0047	H 0048	I 0049	J 004A	K 004B	L 004C	M 004D	N 004E	O 004F
5_80	P 0050	Q 0051	R 0052	S 0053	T 0054	U 0055	V 0056	W 0057	X 0058	Y 0059	Z 005A	[005B	\ 005C	] 005D	^ 005E	_ 005F
6_96	` 0060	a 0061	b 0062	c 0063	d 0064	e 0065	f 0066	g 0067	h 0068	i 0069	j 006A	k 006B	l 006C	m 006D	n 006E	o 006F
7_112	p 0070	q 0071	r 0072	s 0073	t 0074	u 0075	v 0076	w 0077	x 0078	y 0079	z 007A	{ 007B	 007C	} 007D	~ 007E	
8_128																
9_144																
A_160	NBSP 00A0	ì 00A1	¢ 00A2	£ 00A3	¤ 00A4	¥ 00A5	¦ 00A6	§ 00A7	¨ 00A8	© 00A9	ª 00AA	« 00AB	¬ 00AC	SHY 00AD	® 00AE	¯ 00AF
B_176	° 00B0	± 00B1	² 00B2	³ 00B3	´ 00B4	µ 00B5	¶ 00B6	· 00B7	¸ 00B8	¹ 00B9	º 00BA	» 00BB	¼ 00BC	½ 00BD	¾ 00BE	¿ 00BF
C_192	À 00C0	Á 00C1	Â 00C2	Ã 00C3	Ä 00C4	Å 00C5	Æ 00C6	Ç 00C7	È 00C8	É 00C9	Ê 00CA	Ë 00CB	Ì 00CC	Í 00CD	Î 00CE	Ï 00CF
D_208	Ð 00D0	Ñ 00D1	Ò 00D2	Ó 00D3	Ô 00D4	Õ 00D5	Ö 00D6	× 00D7	Ø 00D8	Ù 00D9	Ú 00DA	Û 00DB	Ü 00DC	Ý 00DD	Þ 00DE	ß 00DF
E_224	à 00E0	á 00E1	â 00E2	ã 00E3	ä 00E4	å 00E5	æ 00E6	ç 00E7	è 00E8	é 00E9	ê 00EA	ë 00EB	ì 00EC	í 00ED	î 00EE	ï 00EF
F_240	ð 00F0	ñ 00F1	ò 00F2	ó 00F3	ô 00F4	õ 00F5	ö 00F6	÷ 00F7	ø 00F8	ù 00F9	ú 00FA	û 00FB	ü 00FC	ý 00FD	þ 00FE	ÿ 00FF

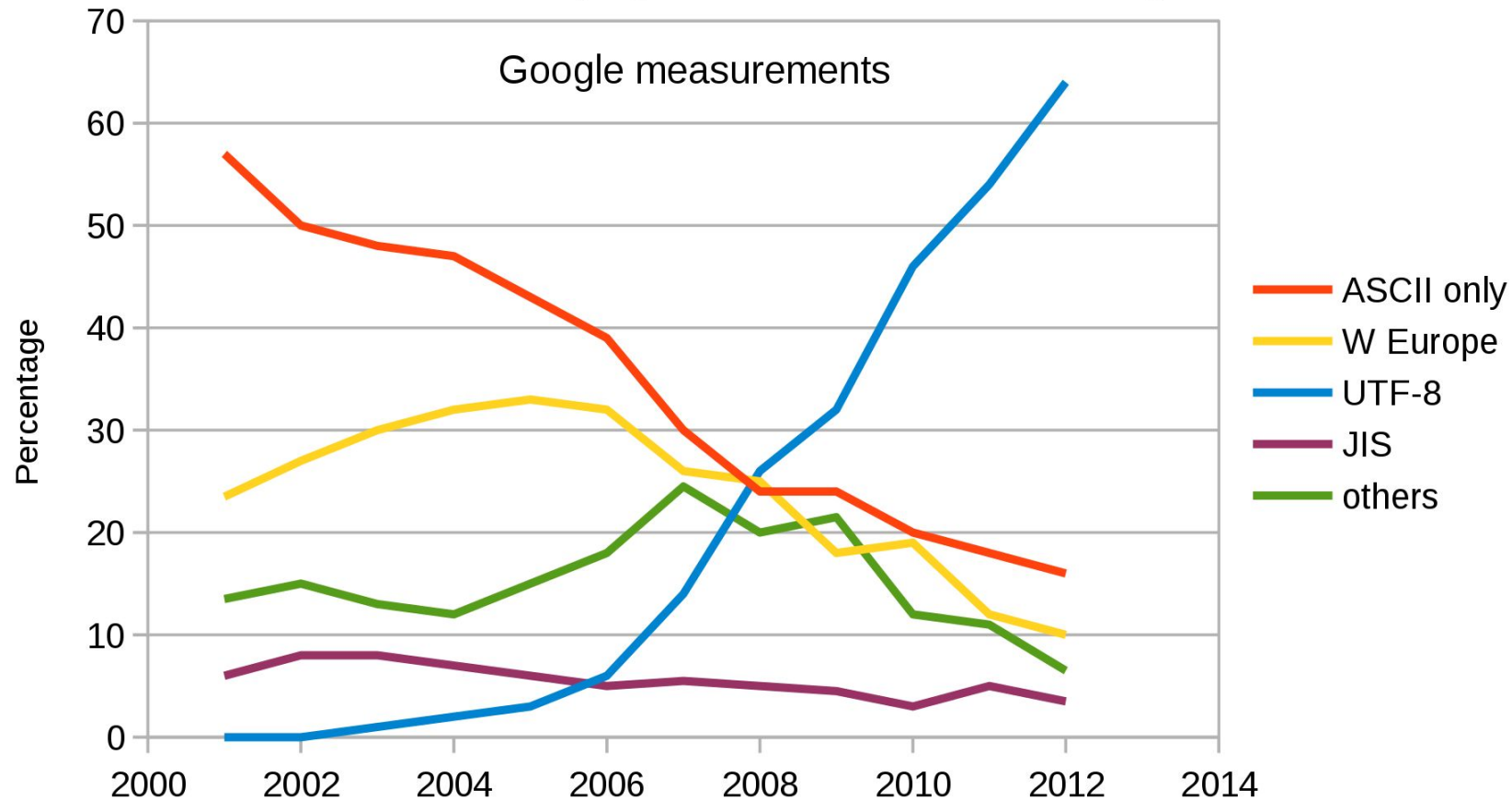


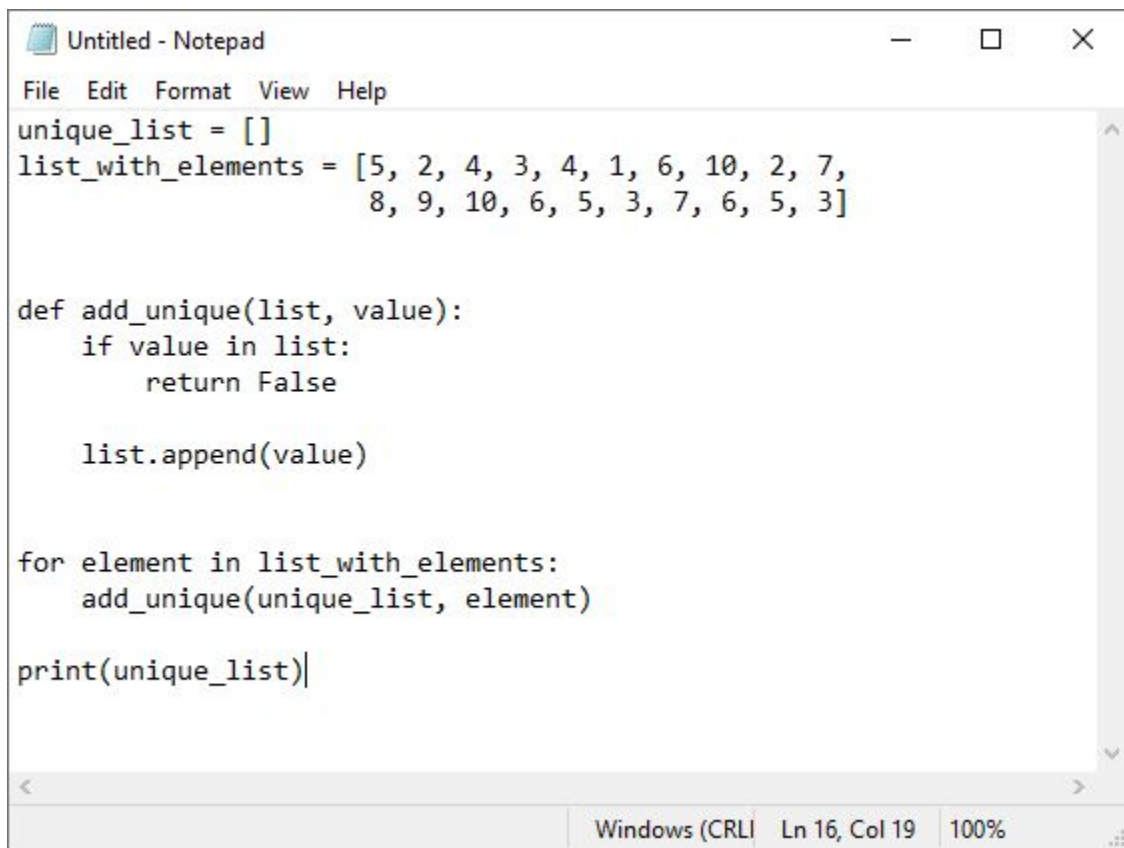
UTF-8																
	_0	_1	_2	_3	_4	_5	_6	_7	_8	_9	_A	_B	_C	_D	_E	_F
0_	NUL 0000	SOH 0001	STX 0002	ETX 0003	EOT 0004	ENQ 0005	ACK 0006	BEL 0007	BS 0008	HT 0009	LF 000A	VT 000B	FF 000C	CR 000D	SO 000E	SI 000F
1_	DLE 0010	DC1 0011	DC2 0012	DC3 0013	DC4 0014	NAK 0015	SYN 0016	ETB 0017	CAN 0018	EM 0019	SUB 001A	ESC 001B	FS 001C	GS 001D	RS 001E	US 001F
2_	SP 0020	! 0021	" 0022	# 0023	\$ 0024	% 0025	& 0026	' 0027	(0028	) 0029	* 002A	+ 002B	, 002C	- 002D	. 002E	/ 002F
3_	0 0030	1 0031	2 0032	3 0033	4 0034	5 0035	6 0036	7 0037	8 0038	9 0039	: 003A	; 003B	< 003C	= 003D	> 003E	? 003F
4_	@ 0040	A 0041	B 0042	C 0043	D 0044	E 0045	F 0046	G 0047	H 0048	I 0049	J 004A	K 004B	L 004C	M 004D	N 004E	O 004F
5_	P 0050	Q 0051	R 0052	S 0053	T 0054	U 0055	V 0056	W 0057	X 0058	Y 0059	Z 005A	[005B	\ 005C	] 005D	^ 005E	_ 005F
6_	` 0060	a 0061	b 0062	c 0063	d 0064	e 0065	f 0066	g 0067	h 0068	i 0069	j 006A	k 006B	l 006C	m 006D	n 006E	o 006F
7_	p 0070	q 0071	r 0072	s 0073	t 0074	u 0075	v 0076	w 0077	x 0078	y 0079	z 007A	{ 007B	 007C	} 007D	~ 007E	DEL 007F
8_	• +00	• +01	• +02	• +03	• +04	• +05	• +06	• +07	• +08	• +09	• +0A	• +0B	• +0C	• +0D	• +0E	• +0F
9_	• +10	• +11	• +12	• +13	• +14	• +15	• +16	• +17	• +18	• +19	• +1A	• +1B	• +1C	• +1D	• +1E	• +1F
A_	• +20	• +21	• +22	• +23	• +24	• +25	• +26	• +27	• +28	• +29	• +2A	• +2B	• +2C	• +2D	• +2E	• +2F
B_	• +30	• +31	• +32	• +33	• +34	• +35	• +36	• +37	• +38	• +39	• +3A	• +3B	• +3C	• +3D	• +3E	• +3F
2 C_	¿ 0000	¿ 0040	LATIN 0080	LATIN 00C0	LATIN 0100	LATIN 0140	LATIN 0180	LATIN 01C0	LATIN 0200	IPA 0240	IPA 0280	IPA 02C0	ACCENTS 0300	ACCENTS 0340	GREEK 0380	GREEK 03C0
2 D_	CYRIL 0400	CYRIL 0440	CYRIL 0480	CYRIL 04C0	CYRIL 0500	ARMENI 0540	HEBREW 0580	HEBREW 05C0	ARABIC 0600	ARABIC 0640	ARABIC 0680	ARABIC 06C0	SYRIAC 0700	ARABIC 0740	THAANA 0780	N'Ko 07C0
3 E_	INDIC 0800	MISC. 1000	SYMBOL 2000	KANA.. 3000	CJK 4000	CJK 5000	CJK 6000	CJK 7000	CJK 8000	CJK 9000	ASIAN A000	HANGUL B000	HANGUL C000	HANGUL D000	PUA E000	FORMS F000
4 F_	SMP.. 10000	□ 40000	□ 80000	SSP.. C0000	SPU.. 100000	4 140000	4 180000	4 1C0000	5 200000	5 280000	5 300000	5 340000	6 4000000	6 4800000		

1. Emoticons (1F601 - 1F64F)

Native ^[1]	Apple ^[2]	Android ^[3]	Android ^[3]	Symbola ^[4]	Twitter ^[5]	Unicode	Bytes (UTF-8)	Description
						U+1F601	\xF0\x9F\x98\x81	grinning face with smiling eyes
						U+1F602	\xF0\x9F\x98\x82	face with tears of joy
						U+1F603	\xF0\x9F\x98\x83	smiling face with open mouth
						U+1F604	\xF0\x9F\x98\x84	smiling face with open mouth and smiling eyes
						U+1F605	\xF0\x9F\x98\x85	smiling face with open mouth and cold sweat
						U+1F606	\xF0\x9F\x98\x86	smiling face with open mouth and tightly-closed eyes
						U+1F609	\xF0\x9F\x98\x89	winking face
						U+1F60A	\xF0\x9F\x98\x8A	smiling face with smiling eyes
						U+1F60B	\xF0\x9F\x98\x8B	face savouring delicious food
						U+1F60C	\xF0\x9F\x98\x8C	relieved face

Share of web pages with different encodings





```

def add_unique(list, value):
    if value in list:
        return False

    list.append(value)

unique_list = []
list_with_elements = [5, 2, 4, 3, 4, 1, 6, 10, 2, 7,
                      8, 9, 10, 6, 5, 3, 7, 6, 5, 3]

for element in list_with_elements:
    add_unique(unique_list, element)

print(unique_list)

```

Windows (CRLI) Ln 16, Col 19 100%

Texts

Rich text



Document1 - Word

File Home Insert Design Layout References Mailings Review View Tell me... Olenna M... Share

Clipboard Paste Font Paragraph Styles Editing

Calibri (Body) 11

B I U abc x₂ x² A

A aly A Aa A A

AaBbCcDc AaBbCcDc AaBbCc

Normal No Spacing Heading 1

1 2 3 4 5 6 7

- **.doc** and **.docx** - [Microsoft Word](#) file
- **.odt** - [OpenOffice](#) Writer document file
- **.pdf** - [PDF](#) file
- **.rtf** - [Rich Text Format](#)
- **.tex** - A LaTeX document file
- **.txt** - Plain text file

Page 1 of 1 0 words 90%

Serif

Sans Serif

Hand Writing

Monospace

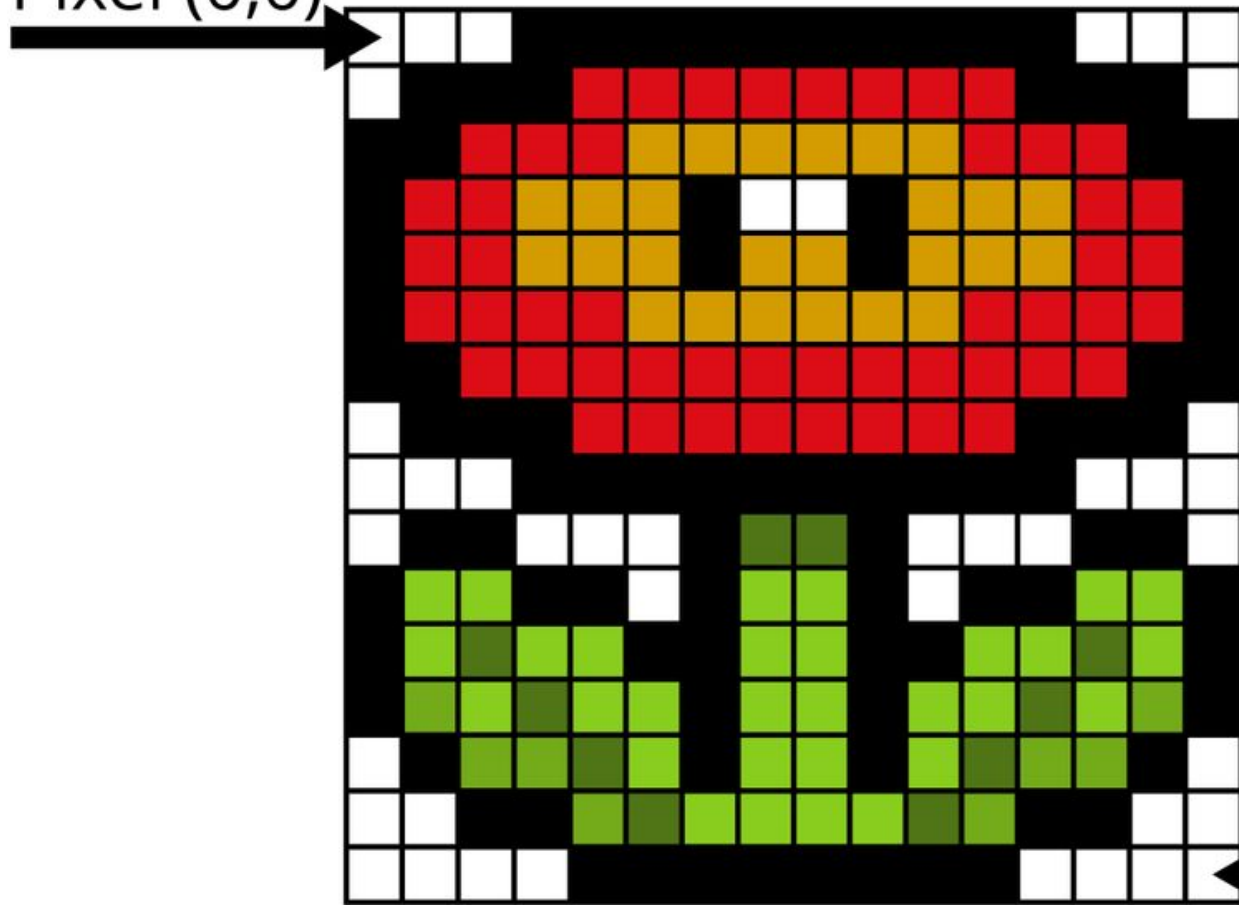
DISPLAY

A top-down view of a workspace. In the top right corner, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' visible. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter is also in the top right. A brown leather folder or notebook is next to the laptop. In the bottom right, a spiral-bound notebook is open, and a smartphone with a black frame and a light pink screen is resting on it.

Images

Bitmap and vector

Pixel (0,0)



Pixel (15,15)

RGB



Digital, Web

CMYK



Print

Lime
#A4C400
RGB(164,
196, 0)

Green
#60A917
RGB(96,
169, 23)

Emerald
#008A00
RGB(0,
138, 0)

Teal
#00ABA9
RGB(0,
171, 169)

Cyan
#1BA1E2
RGB(27,
161, 226)

Cobalt
#0050EF
RGB(0, 80,
239)

Indigo
#6A00FF
RGB(106,
0, 255)

Violet
#AA00FF
RGB(170,
0, 255)

Pink
#F472D0
RGB(244,
114, 208)

Magenta
#D80073
RGB(216,
0, 115)

Crimson
#A20025
RGB(162,
0, 37)

Red
#E51400
RGB(229,
20, 0)

Orange
#FA6800
RGB(250,
104, 0)

Amber
#F0A30A
RGB(240,
163, 10)

Yellow
#E3C800
RGB(227,
200, 0)

Brown
#825A2C
RGB(130,
90, 44)

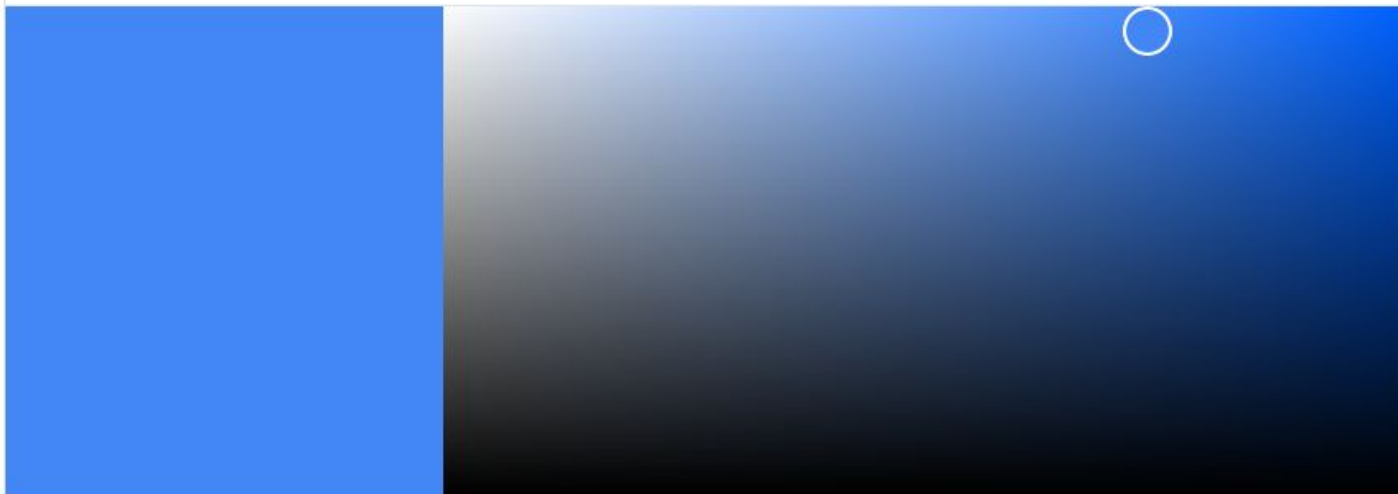
Olive
#6D8764
RGB(109,
135, 100)

Steel
#647687
RGB(100,
118, 135)

Mauve
#76608A
RGB(118,
96, 138)

Taupe
#87794E
RGB(135,
121, 78)

Colour picker



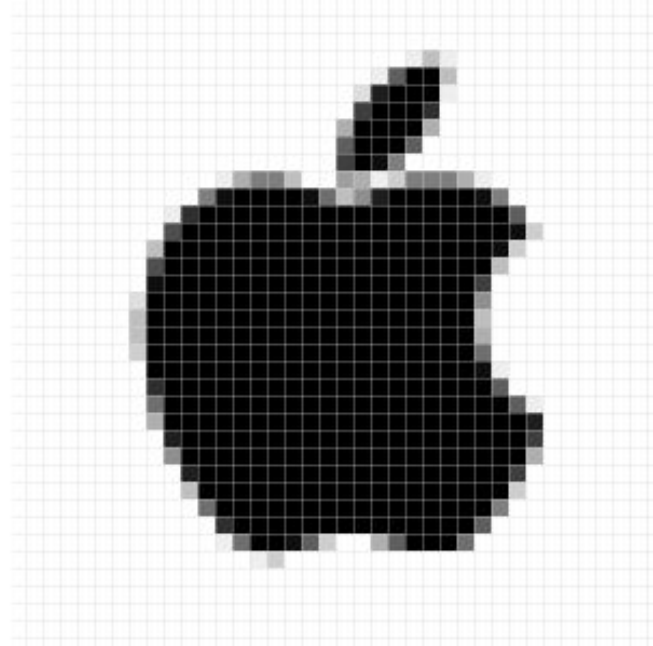
HEX
#4287f5

RGB
66, 135, 245

CMYK
73%, 45%, 0%, 4%

HSV
217°, 73%, 96%

HSL
217°, 90%, 61%



Vector @100%



Bitmap @100%



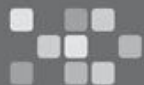
Vector @300%



Bitmap @300%



Image file formats



Raster images

Pixel-based graphics
Resolution dependent
Photos & web graphics

JPG

Web & print
photos and
quick previews

GIF

Animation &
transparency in
limited colors

PNG

Transparency
with millions
of colors

TIFF

High quality
print graphics
and scans

RAW

Unprocessed
data from
digital cameras

PSD

Layered Adobe
Photoshop
design files



Vector images

Curve-based graphics
Resolution independent
Logos, icons, & type

PDF

Print files and
web-based
documents

EPS

Individual
vector design
elements

AI

Original Adobe
Illustrator
design files

SVG

Vector files
for web
publishing

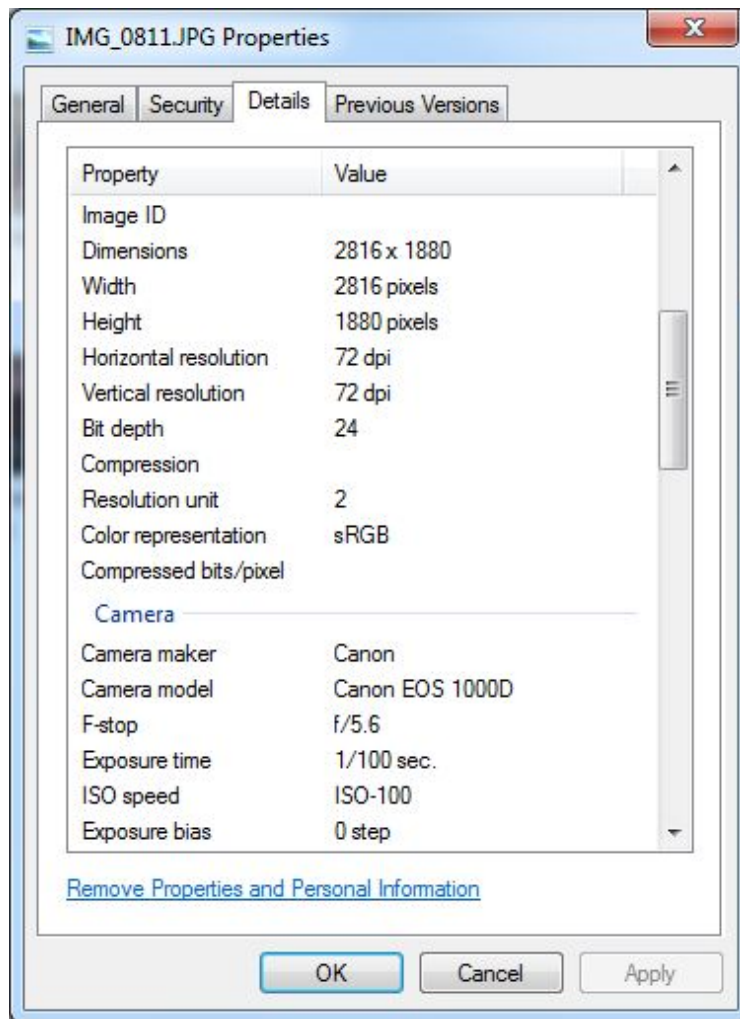
Image color formats

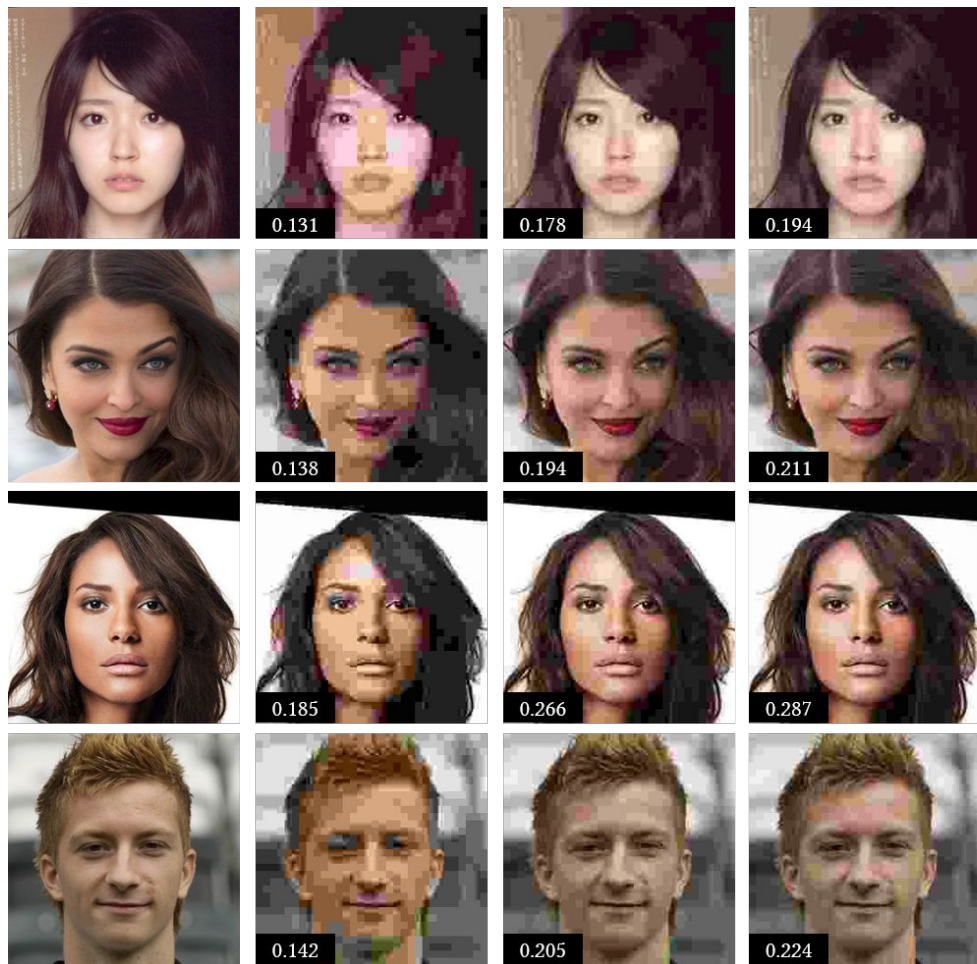
C M Y K

CMYK is a 4-color printing process that stands for cyan, magenta, yellow and key (black).

R G B

RGB is a light-based color model that stands for red, green and blue.





(a) input

(b) JPEG (qp=4)

(c) JPEG (qp=7)

(d) JPEG (qp=8)

Audio

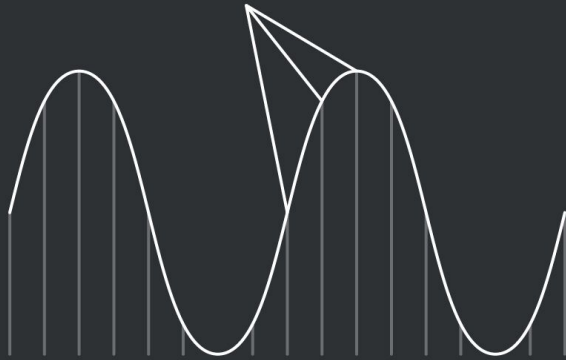
Waves to bits



Analog Wave

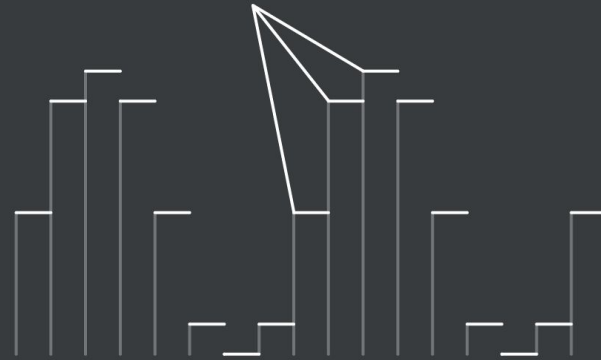
Digital Result

samples taken at
these points

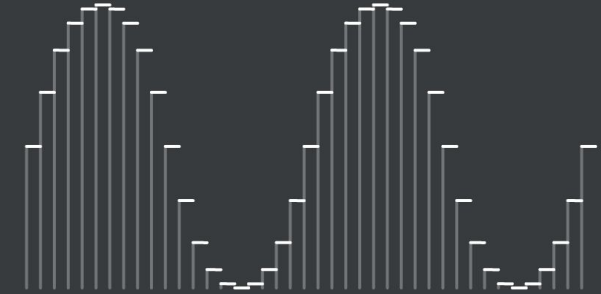
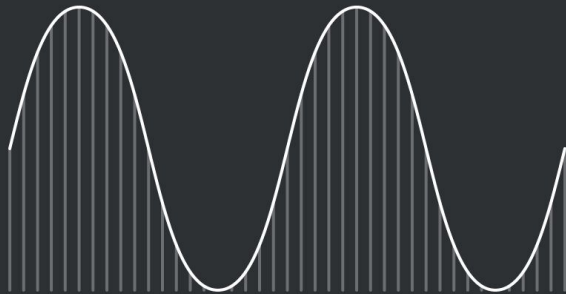


a
→

digital samples



b
→



12 AUDIO FILE FORMATS

.WAV

.FLAC

.MP3

.OGG

.M3U

.ACC

.WMA

.MIDI

.AIF

.M4A

.MPA

.PLS

Bit Rate	Bit Size(bit)	Channel	Sample Rate	File Size(Uncompressed)	Bandwidth
24kbps	16	Stereo	44.1KHz	630MB	10.8MB/hr
56kbps	16	Stereo	44.1KHz	630MB	25.2MB/hr
56kbps	24	Stereo	44.1KHz	948MB	25.2MB/hr
96kbps	16	Stereo	44.1KHz	630MB	43.2MB/hr
128kbps	24	Stereo	44.1KHz	948MB	57.6MB/hr
128kbps	16	Stereo	44.1KHz	691MB	57.6MB/hr
128kbps	16	Mono	48KHz	342MB	57.6MB/hr
128kbps	16	Stereo	22KHz	316MB	57.6MB/hr
196kbps	16	Stereo	44.1KHz	630MB	88.2MB/hr
196kbps	16	Mono	44.1KHz	342MB	88.2MB/hr

A top-down view of a minimalist workspace. In the top right corner, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white ceramic cup of black coffee on a light wooden coaster. A small potted plant with green leaves sits above the cup. To the right of the laptop is a brown leather-bound notebook with a silver metal clasp. In the bottom right, a white spiral-bound notebook lies flat, with a black smartphone resting on it. The phone's screen is a solid, bright pink color. The entire scene is set against a plain, light gray background.

Video

Repeat and mix

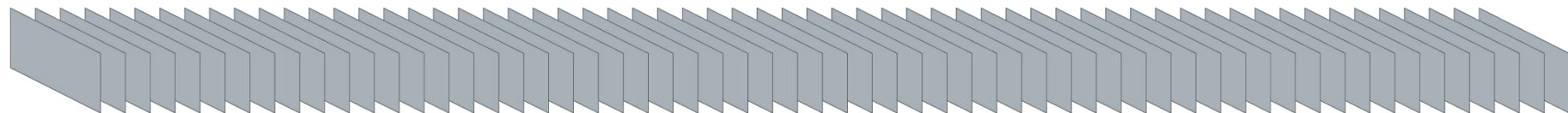
4K 3840x2160px (16:9)

2K 2560x1440px (16:9)

FHD 1920x1080px (16:9)

HD 1280x720px (16:9)

Aspect Ratio	Uses	TVs
4:3	Standard Channels	Old TV's
16:9	HD Channels	HDTV'S, Laptops
21:9	Most Movies	Most Theatres
14:10	IMAX films	IMAX Cinemas
19:10	IMAX Digital	IMAX Theatres

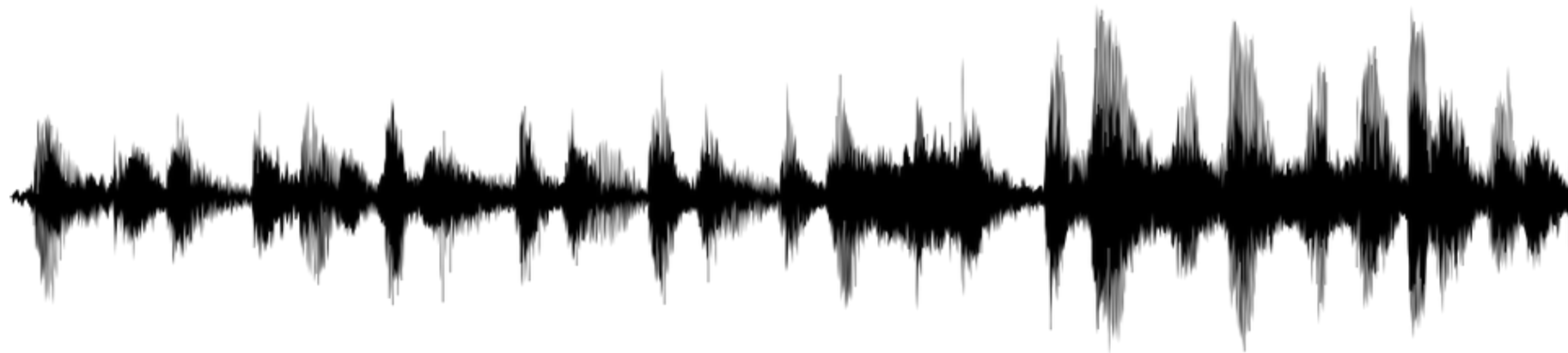


60fps



One second

24fps



MKV



Free and open source. It also offers a plethora of extra features.

MP4



Popularized by Apple in the iTunes Store, widely supported by a variety of the most popular digital devices on the market.

AVCHD



It leverages the power of H.264/MPEG-4 video compression technology, and supports 3D video.



AVI

AUDIO VIDEO INTERLEAVE

Used to absolutely dominate the video container market

WMV

WINDOWS MEDIA VIDEO



Exclusively designed for streaming applications

Produce less compression than competing formats

Other benefits of AVI and WMV include their wide compatibility with both PC and Mac systems.

Other drawbacks include an inability to select aspect ratios manually, which can be quite exasperating for some users.



MOV

Designed explicitly to save feature-length movies

QT



N/A

These two file extensions are commonly saved together.

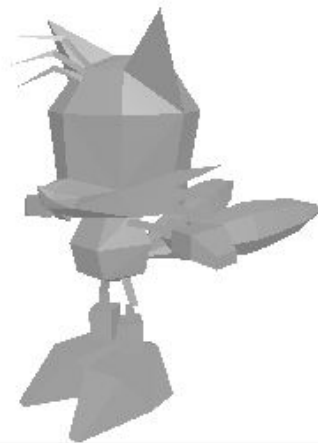
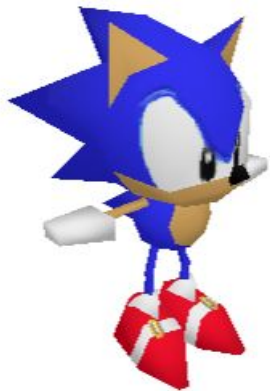
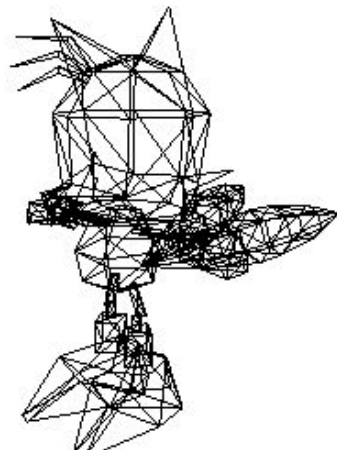
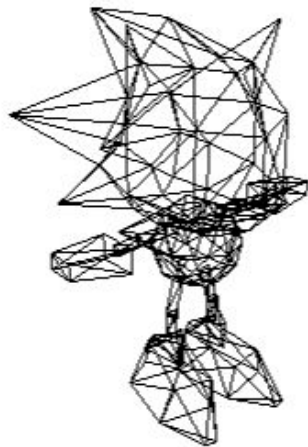
Both formats employ a multimedia container file that can be separated into a variety of tracks.

Users can devote each track to the storage of particular type of data (video, audio, effects, and/or text)

3D



3D Models



A top-down view of a minimalist desk. In the top right corner, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white coffee cup with a blue rim, filled with dark coffee, sitting on a wooden coaster. A small green plant in a wooden planter is in the top left. An orange textured notebook with a silver clip is on the right. In the bottom right, a spiral-bound notebook with a white cover has a black smartphone with a pink screen resting on it.

Time

Define a moment

Dates, the correct way

PUBLIC SERVICE ANNOUNCEMENT:

OUR DIFFERENT WAYS OF WRITING DATES AS NUMBERS CAN LEAD TO ONLINE CONFUSION. THAT'S WHY IN 1988 ISO SET A GLOBAL STANDARD NUMERIC DATE FORMAT.

THIS IS ***THE*** CORRECT WAY TO WRITE NUMERIC DATES:

2013-02-27

THE FOLLOWING FORMATS ARE THEREFORE DISCOURAGED:

02/27/2013 02/27/13 27/02/2013 27/02/13

20130227 2013.02.27 27.02.13 27-02-13

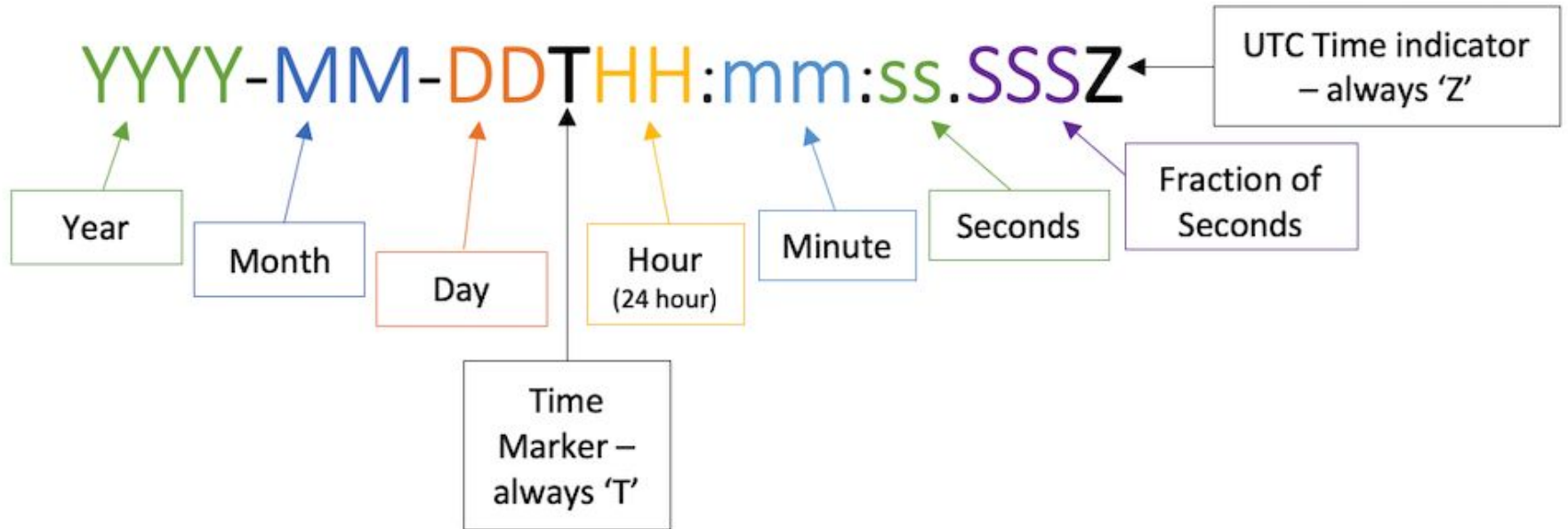
27.2.13 2013.II.27. $27\frac{1}{2}$ -13 2013.158904109

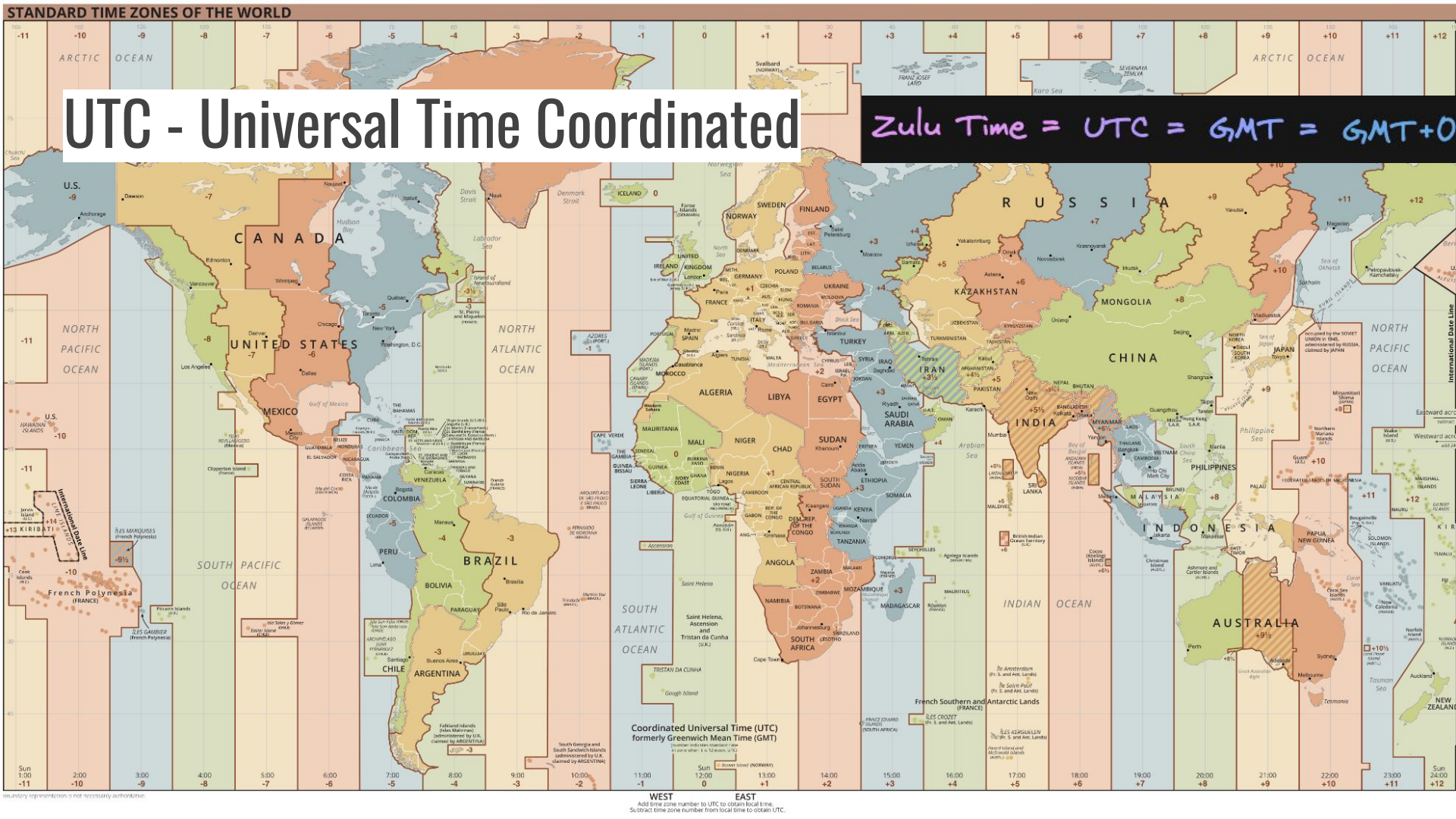
MMXIII-II-XXVII MMXIII ^{LVII}/_{CCCLXV} 1330300800

$((3+3) \times (111+1) - 1) \times 3 / 3 - 1 / 3^3$ ~~2013~~  hiss

10/11011/1101 02/27/20/13 $\begin{matrix} 2 & 3 & 1 & 4 \\ 0 & 1 & 2 & 3 & 7 \\ 5 & 6 & 7 & 8 \end{matrix}$

Time format - ISO 8601





Timestamp - Unix time

The Current Epoch Unix Timestamp

Enter a Timestamp

Supports Unix timestamps in seconds, milliseconds, microseconds and nanoseconds.

Convert →

1695551353

SECONDS SINCE JAN 01 1970. (UTC)

12:29:15 PM

Copy

Enter a Date & Time

Year

Month

Day

Hour (24 hour)

Minutes

Seconds

Convert →

The current epoch translates to

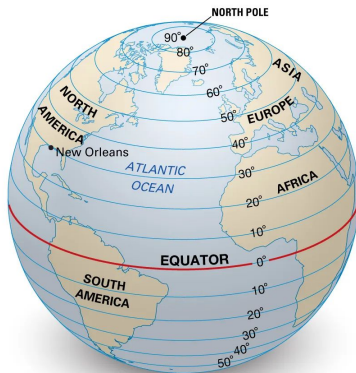
Date	
09/24/2023 @ 10:28am	UTC
2023-09-24T10:28:28+00:00	ISO 8601
Sun, 24 Sep 2023 10:28:28 +0000	RFC 822 , 1036 , 1123 , 2822
Sunday, 24-Sep-23 10:28:28 UTC	RFC 2822
2023-09-24T10:28:28+00:00	RFC 3339

A top-down view of a minimalist desk. In the top right corner, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter sits above the cup. To the right of the laptop is a brown leather folder or notebook. In the bottom right, a spiral-bound notebook with a white cover has a black smartphone with a pink screen resting on it. The entire scene is set against a plain, light gray background.

Place on earth

Define a space

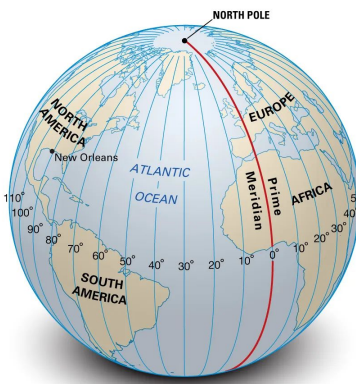
Location - Latitude & Longitude



FACTS ABOUT LINES OF LATITUDE

- Are known as parallels.
- Run in an east-west direction.
- Measure distance north or south from the Equator.
- Are parallel to one another and never meet.
- Cross the prime meridian at right angles.
- Lie in planes that cross the Earth's axis at right angles.
- Get shorter toward the poles, with only the Equator, the longest, a great circle.

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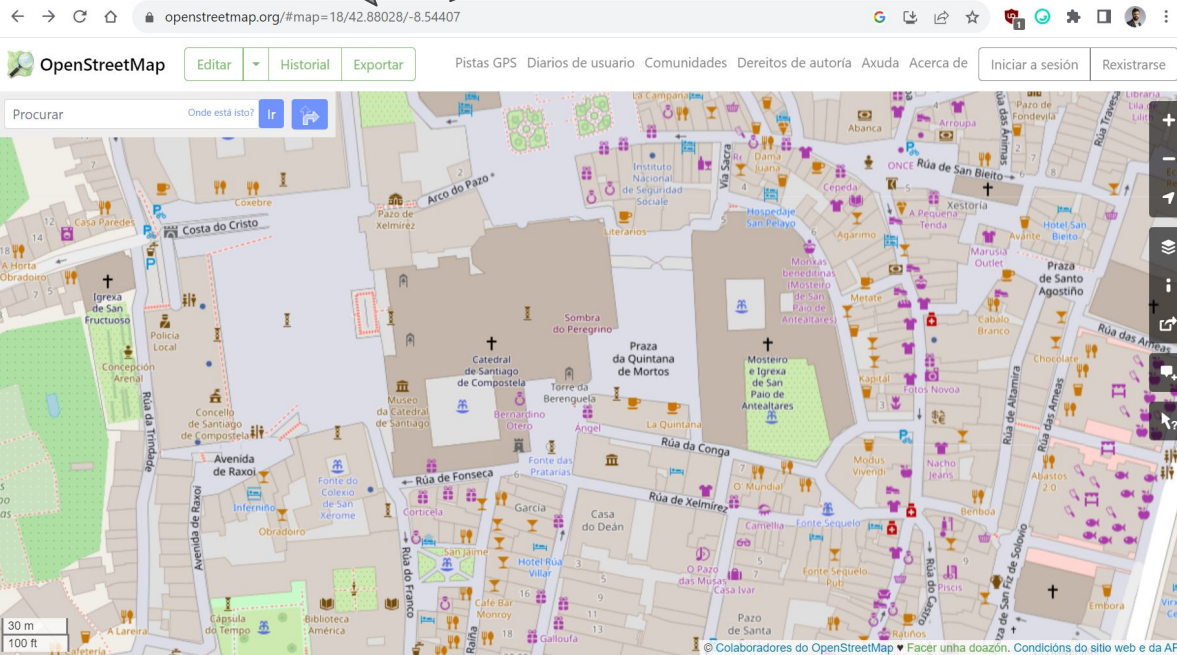
FACTS ABOUT LINES OF LONGITUDE

- Are known as meridians.
- Run in a north-south direction.
- Measure distance east or west of the prime meridian.
- Are farthest apart at the Equator and meet at the poles.
- Cross the Equator at right angles.
- Lie in planes that pass through the Earth's axis.
- Are equal in length.
- Are halves of great circles.

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Latitude

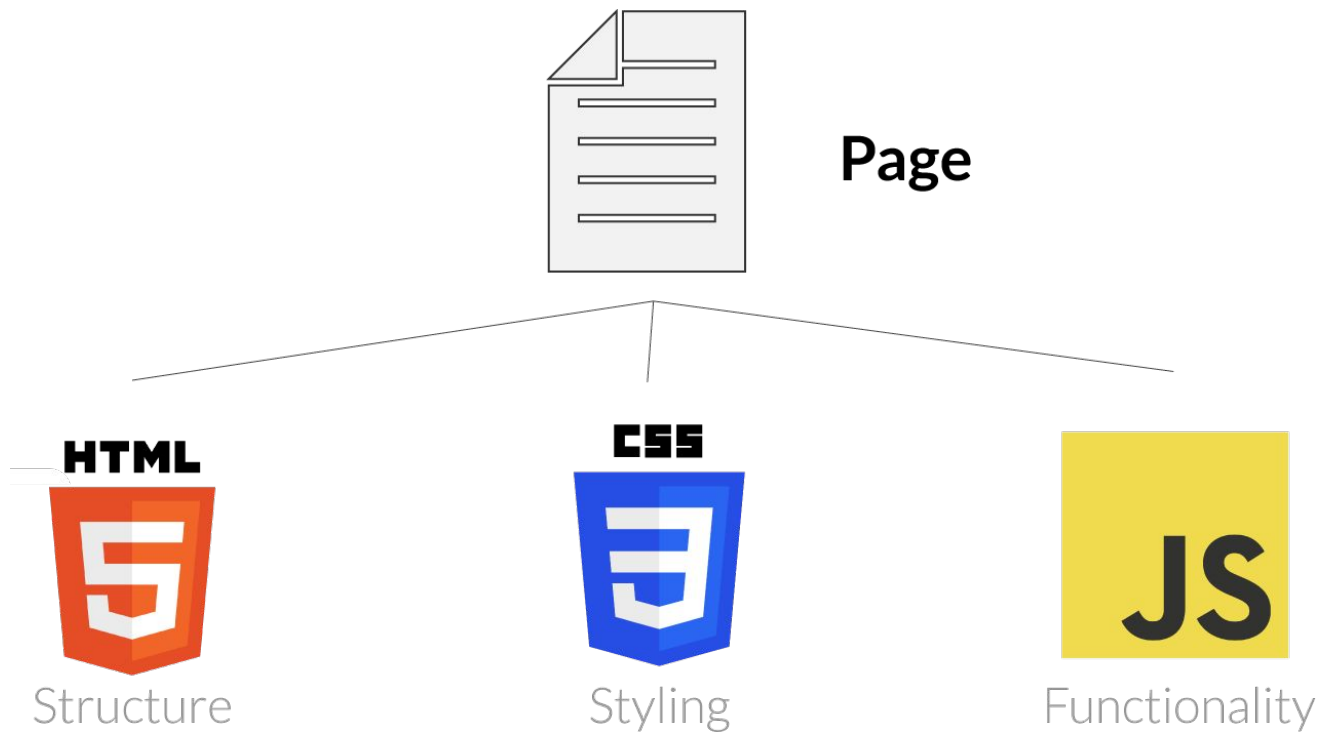
Longitude



A top-down view of a minimalist workspace. In the top right corner, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white ceramic cup of black coffee on a light wooden coaster. A small green plant in a wooden planter sits above the cup. To the right of the laptop is a brown leather folder or notebook. In the bottom right, a spiral-bound notebook with a white cover has a black smartphone resting on it. The smartphone screen is a solid light pink color. The word 'Website' is centered in a large, dark grey, sans-serif font.

Website

Website



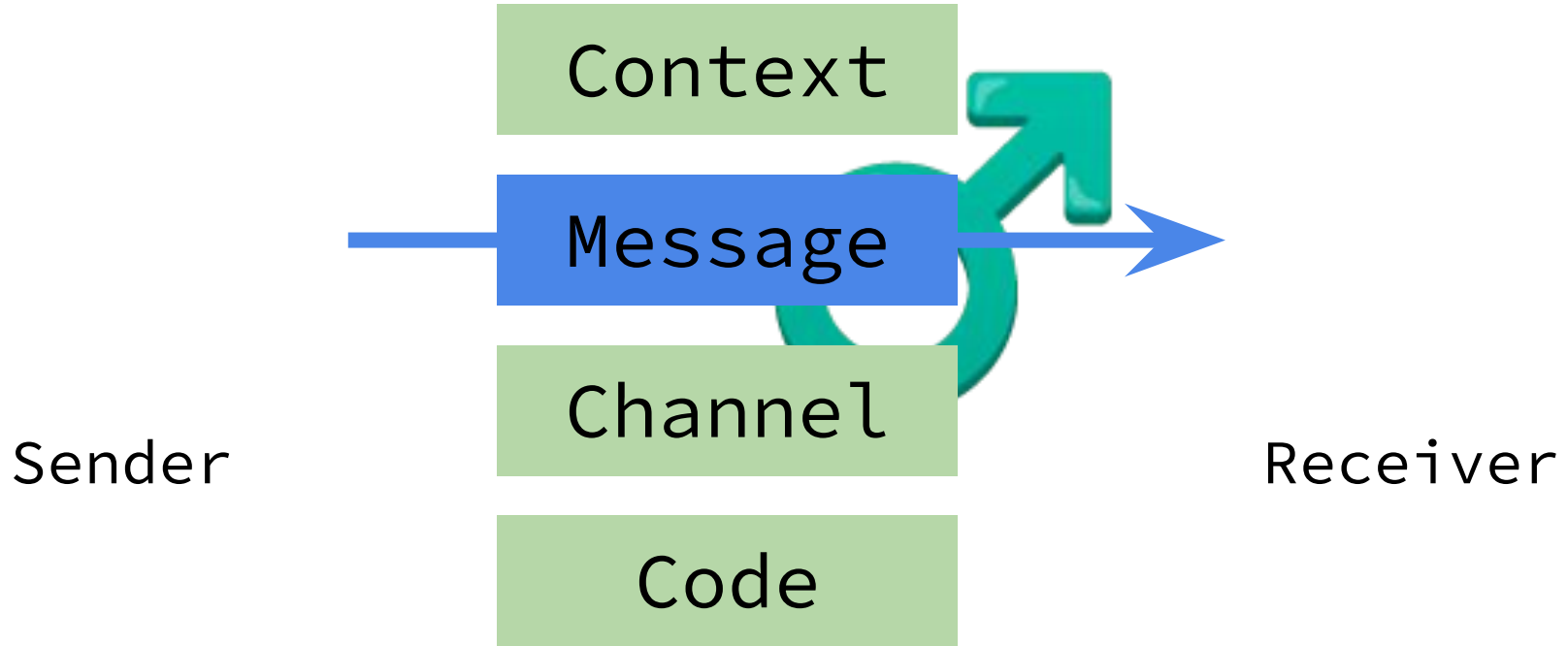
A top-down view of a workspace. In the upper right, a silver laptop is partially visible, showing keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white mug with a blue rim, filled with dark coffee, sitting on a light-colored wooden coaster. To the left of the mug is a small wooden planter with a green leafy plant. In the lower right, a smartphone with a pink screen is placed on a white rectangular object, possibly a notepad or a small stand.

Networks

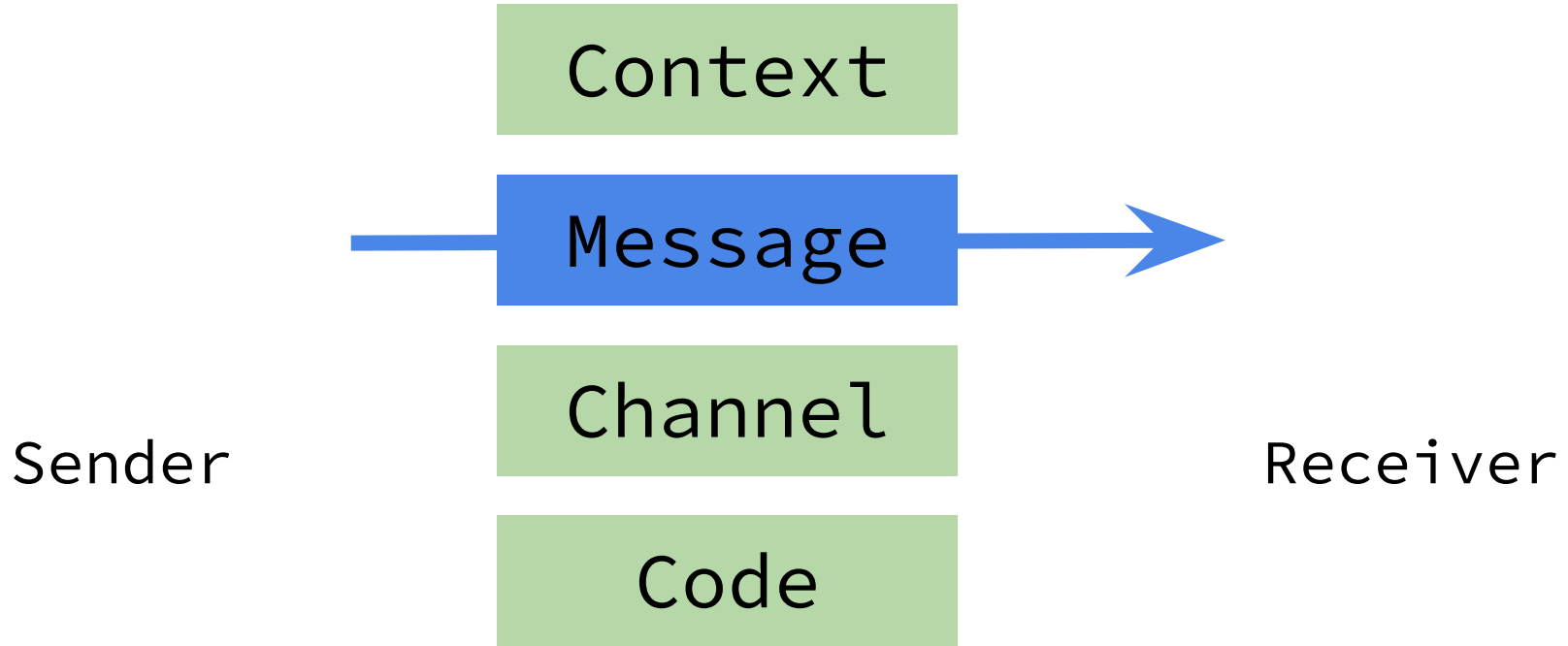
Communication



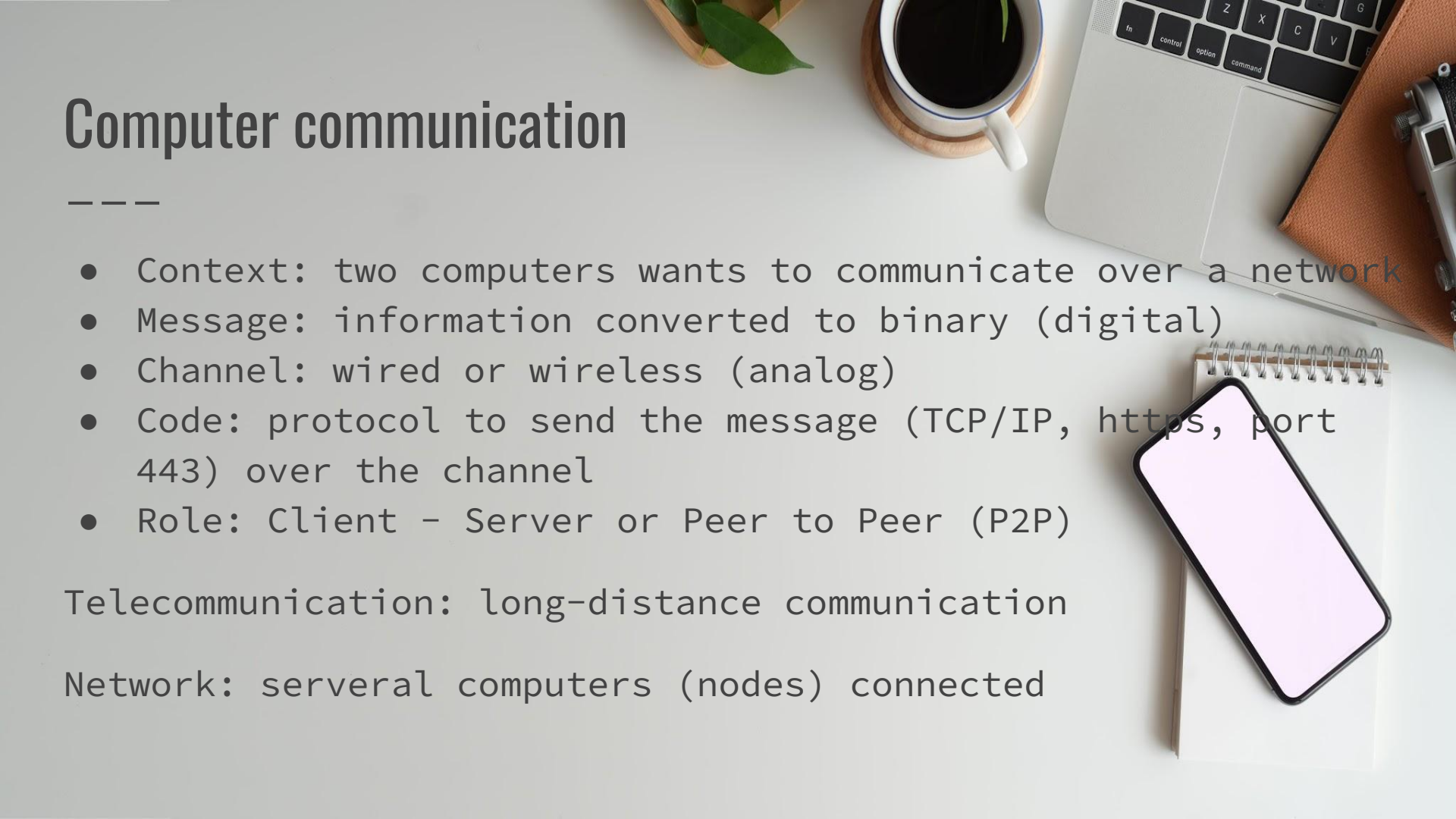
Jakobson's functions of language



Jakobson's functions of language in computers



Computer communication

A top-down view of a desk with a laptop, a cup of coffee, a small potted plant, and a smartphone on a notebook.

- Context: two computers wants to communicate over a network
- Message: information converted to binary (digital)
- Channel: wired or wireless (analog)
- Code: protocol to send the message (TCP/IP, https, port 443) over the channel
- Role: Client - Server or Peer to Peer (P2P)

Telecommunication: long-distance communication

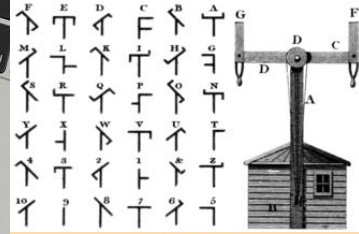
Network: serveral computers (nodes) connected

A top-down view of a workspace. In the top right, a silver laptop is partially visible with its keyboard and trackpad. Next to it is a white cup of black coffee on a wooden coaster. A small potted plant with green leaves is in the top left. A brown leather folder or notebook is on the right. In the bottom right, a smartphone with a pink screen lies on a white spiral-bound notebook. The background is a plain, light-colored surface.

History of Telecommunication

History of telecommunication

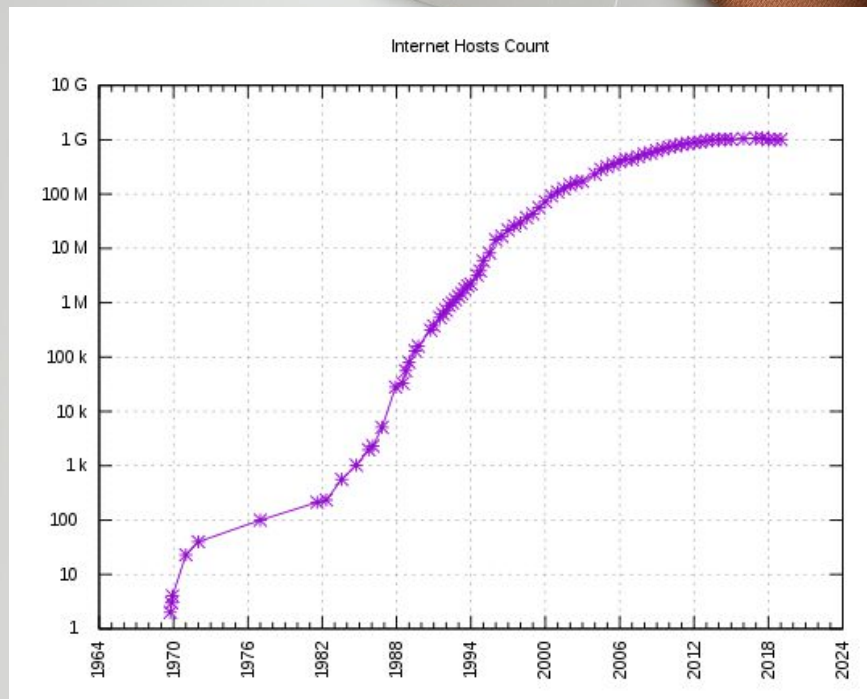
- Ancient systems: fires, smoke signals, drums, horns
- 6th century BCE: Mail, Pidgeon Post, hydraulic semaphore
- 1838: Telegraph & Morse
- 1860: Pony express (10 days US cost to cost)
- 1876: Telephones
- 1896: Radio
- 1927: Television
- 1962: Commercial telecommunications satellite
- 1969: Computer networking (Arpanet)
- 1981: First mobile phone network
- 1983: Internet

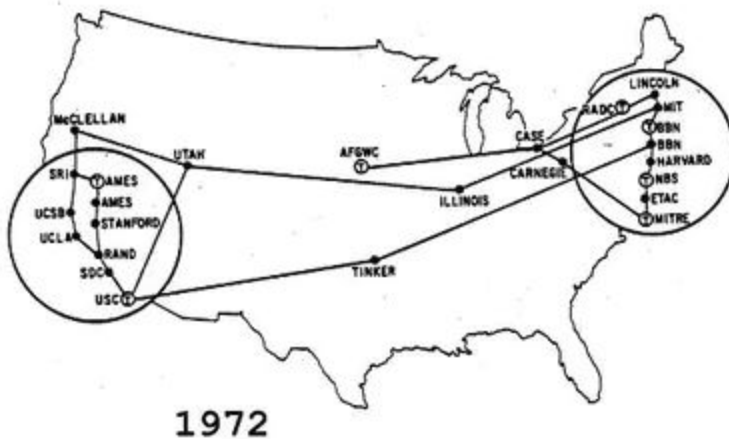
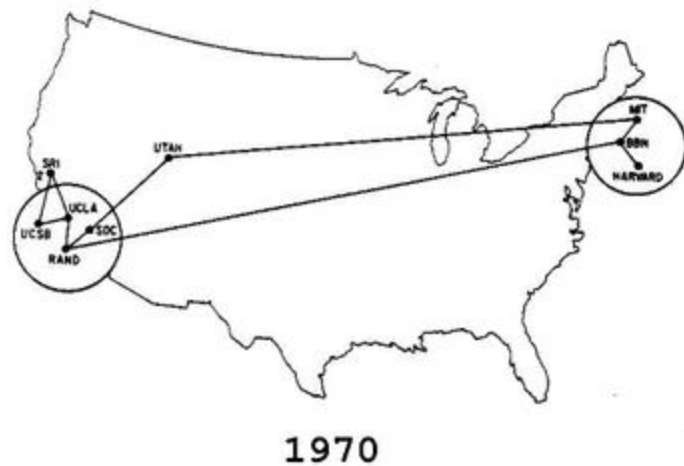
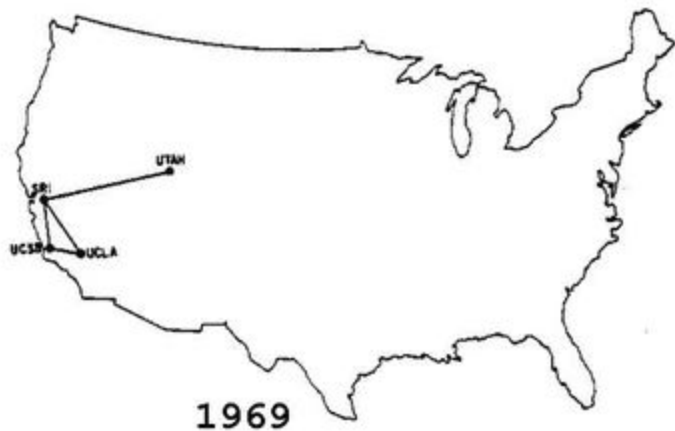


History of Internet

— — —

- 1957: ARPA
- 1961: First email at MIT
- 1963: Packet switching
- 1969: ARPANET starts
- 1974: TCP/IP protocol created
- 1974: ARPANET splits Milnet
- 1987: DNS protocol created
- 1990: ARPANET renamed Internet
- 1991: World Wide Web
- 1993: First Browser
- 1998: Google
- 2001: Wikipedia
- 2004: Facebook



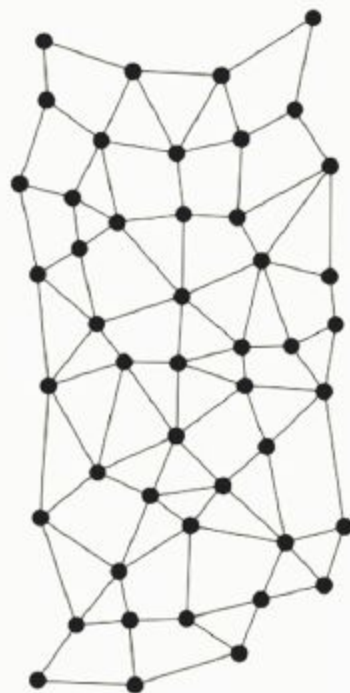




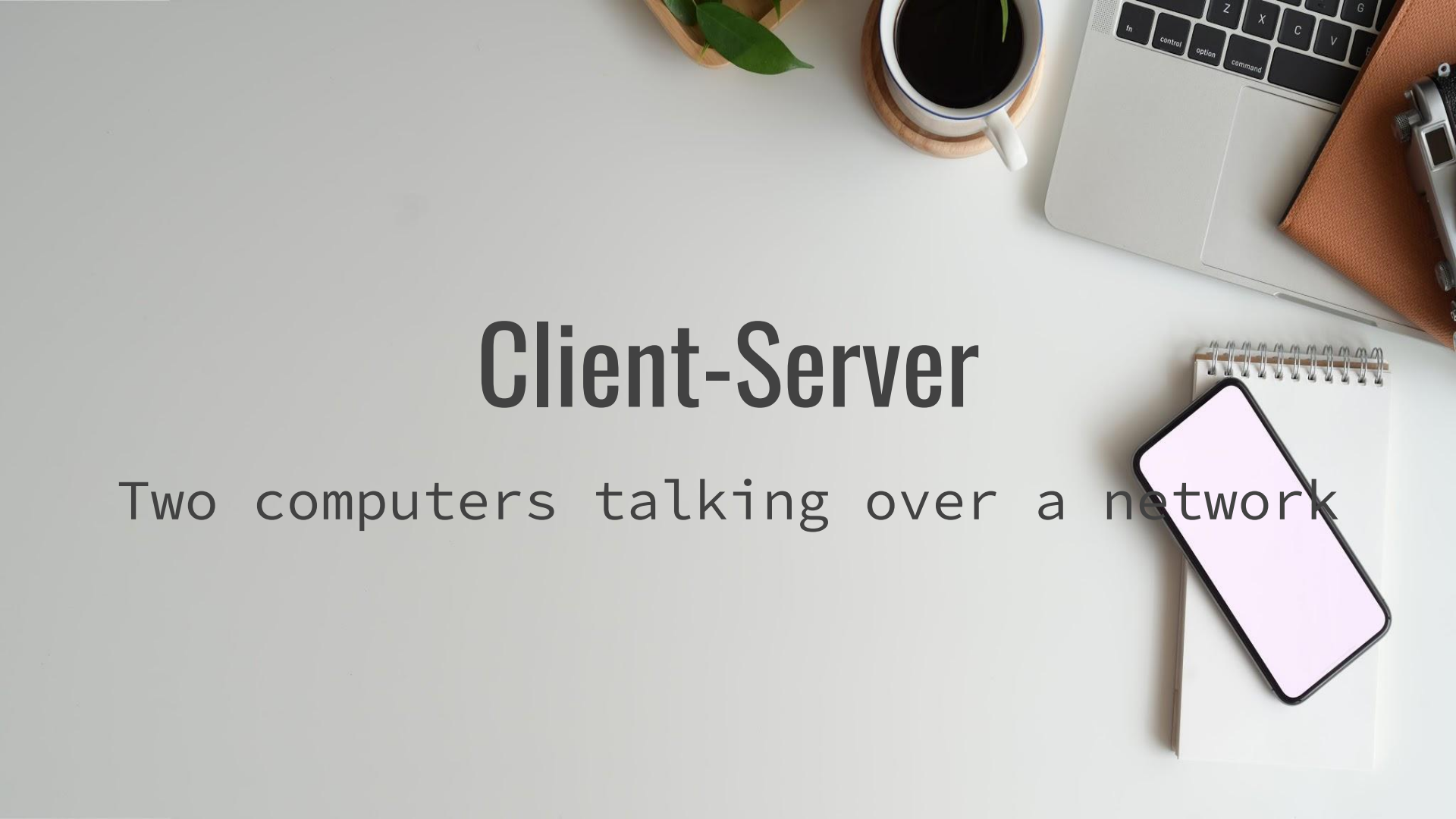
Centralised (A)



Decentralised (B)



Distributed (C)

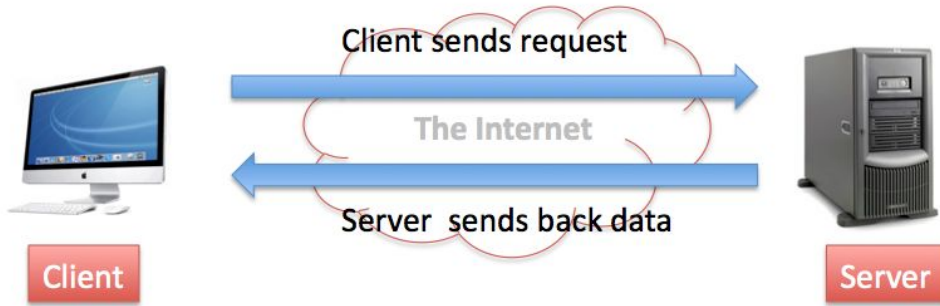
A top-down view of a desk with a laptop, coffee cup, plant, and notebook. The laptop is silver and partially open, showing the keyboard. A white coffee cup with a blue rim sits on a wooden coaster. A small potted plant with green leaves is in the top left. A spiral-bound notebook with a white cover is in the bottom right, with a smartphone resting on it. The phone's screen is a solid light pink color.

Client-Server

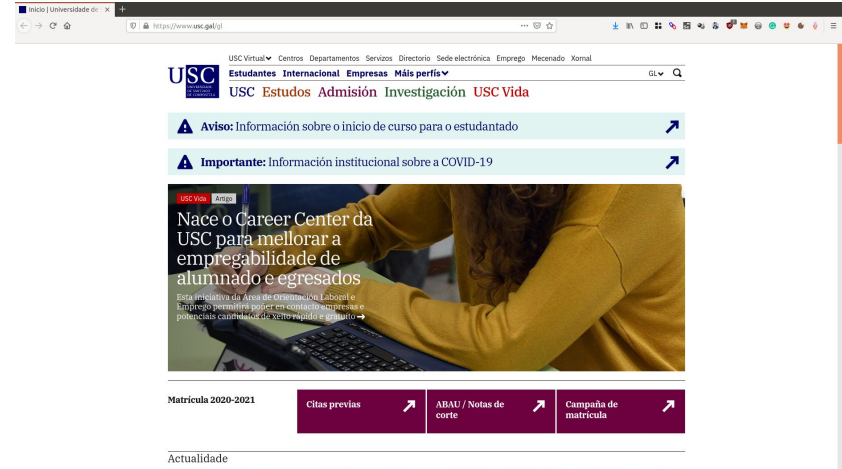
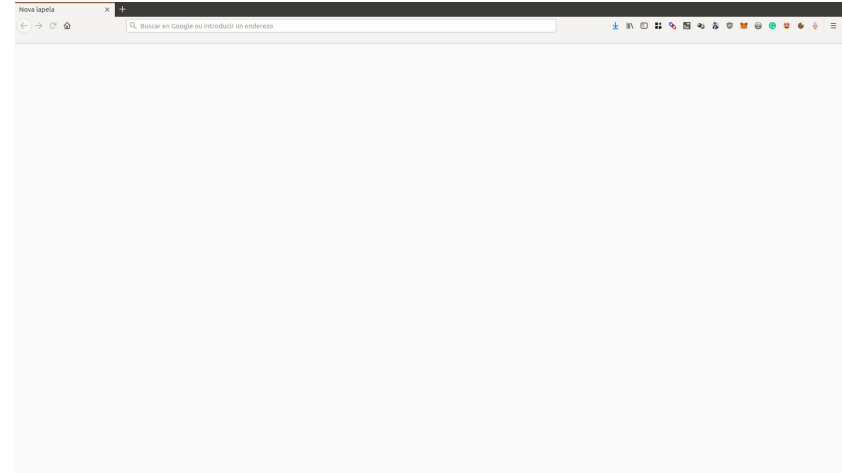
Two computers talking over a network

Client-Server in practice

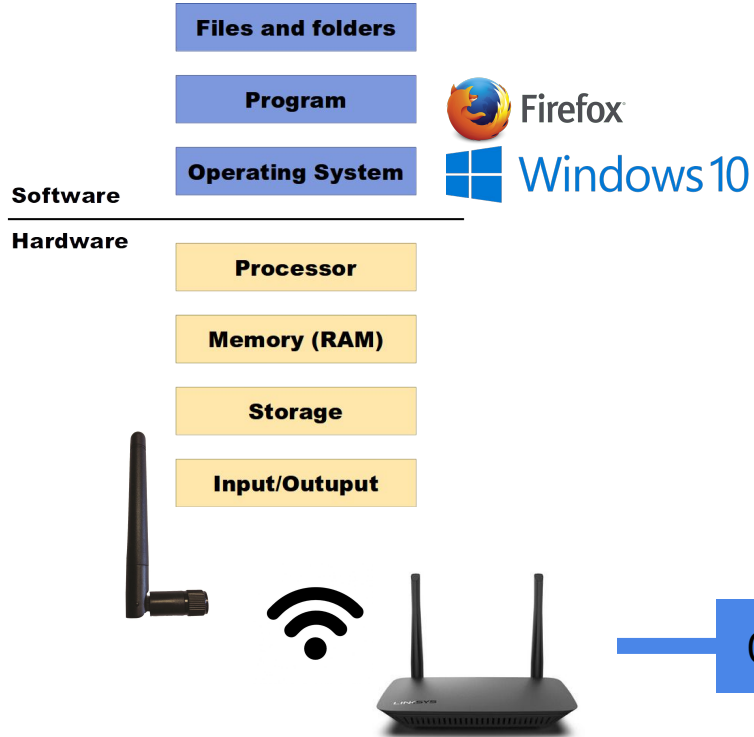
- Type the address `usc.gal` + ENTER



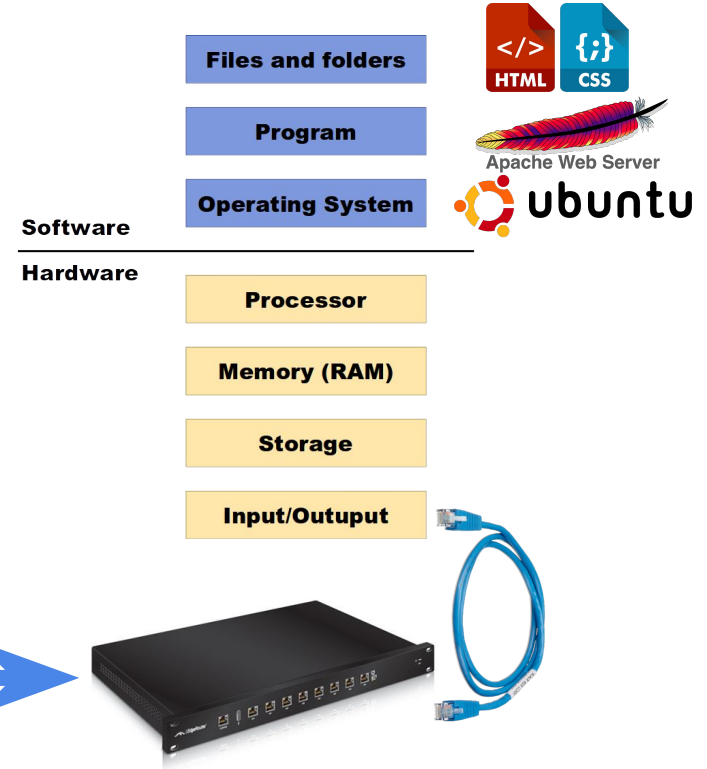
- Browser show the server data



Client

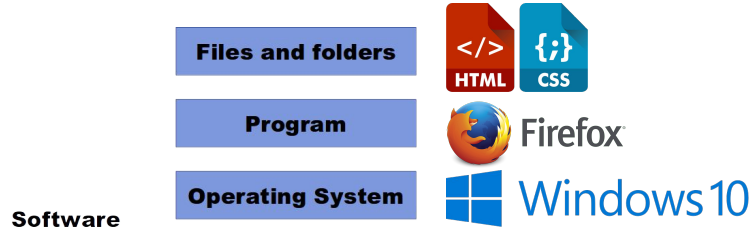


Server

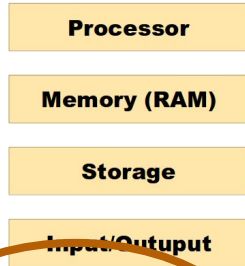


01101101

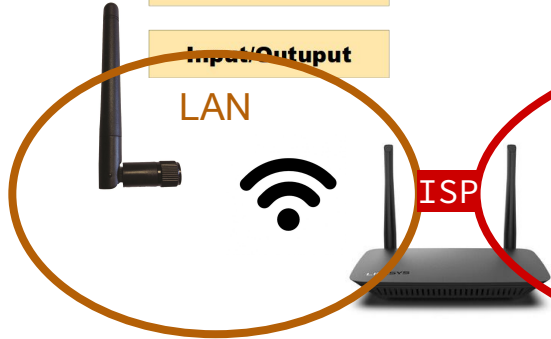
Client



Hardware



LAN



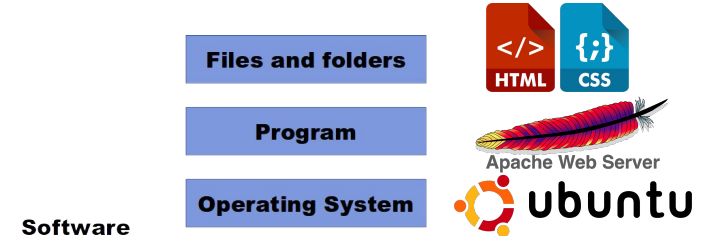
ISP

WAN: Internet

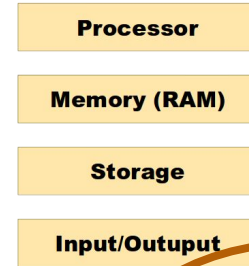
01101101



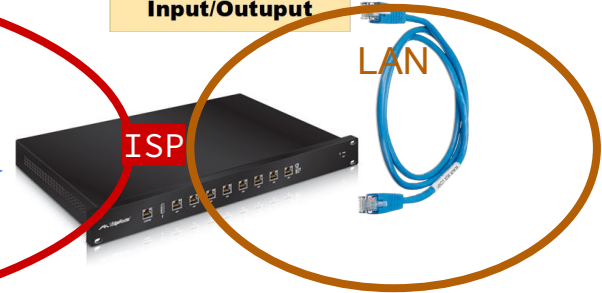
Server



Hardware



LAN



ISP

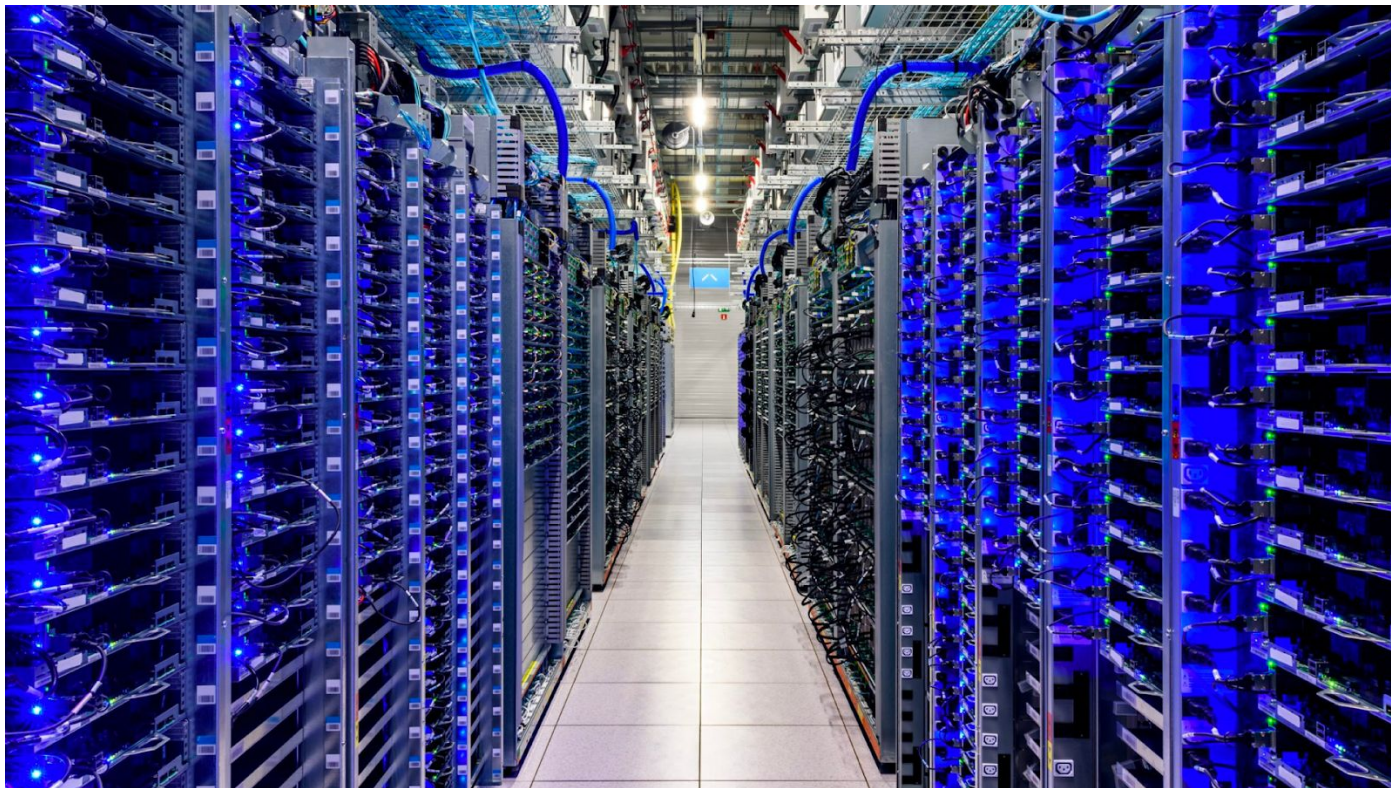
Client



Server



Datacenter



Hosting

Server Virtual Private Server Shared Hosting

BRAND	MODEL	CPU	CORES	THREADS	RAM	DISC	RAID	DELIVERY	RATES FROM
	Dell R240 New model	1x Xeon E-2236 3.4 GHz	6	12	16 GB	From 2 x 1 TB (NVMe or SSD)	1	< 24h	127,67 € / month 159,58 € / month
	Dell R440	2x Xeon Silver 4114	20	40	32 GB	2 x 1 TB SSD	1	< 24h	254,69 € / month 339,58 € / month
	Intel NVM	2x Xeon Silver 4216 2.10 / 3.20 GHz	32	64	64 GB	2 x 1 TB NVMe	1	< 24h	439,50 € / month
	Dell R640	2x Xeon Silver 4116 2.1GHz	24	48	64 GB	2 x 1 TB SSD	1	< 24h	439,58 € / month

DOMAINS HOSTING **SERVERS** RESELLERS MUCH MORE 24/7 SUPPORT

WE ARE DH BLOG

VPS I

The autonomy that your website needs

from **34,20** /month

- 1 vCPU Processor
- 4 GB Guaranteed RAM
- 60 GB Total Size
- ∞ Unlimited transfer
- Discos **SSD NVMe**
- Money-back guarantee: 30 days

VPS II

For those who want more resources

from **64,35** /month

- 2 vCPU Processor
- 8 GB Guaranteed RAM
- 90 GB Total Size
- ∞ Unlimited transfer
- Discos **SSD NVMe**
- Money-back guarantee: 30 days

VPS III

High-performance VPS

from **124,20** /month

- 6 vCPU Processor
- 12 GB Guaranteed RAM
- 200 GB Total Size
- ∞ Unlimited transfer
- Discos **SSD NVMe**
- Money-back guarantee: 30 days

VPS IV

For the most exigent projects

from **135,07** /month

- 8 vCPU Processor
- 16 GB Guaranteed RAM
- 500 GB Total Size
- ∞ Unlimited transfer
- Discos **SSD NVMe**
- Money-back guarantee: 30 days

DOMAINS HOSTING **SERVERS** RESELLERS MUCH MORE 24/7 SUPPORT

WE ARE DH BLOG

EMAIL

Professional mail with your own domain

from **1 €** /month

with annual contract (after 2 €/mo.)

- 50 GB Space
- Unlimited transfer
- 10 E-mail accounts and chat
- NVMe disks, maximum speed!
- Antispam and Antivirus in real time

BASIC

Affordable and simple hosting

from **2,25 €** /month

with annual contract (after 4,50 €/mo.)

- 50 GB Space
- Unlimited transfer
- 20 E-mail accounts and chat
- NVMe disks, maximum speed!
- Antispam and Antivirus in real time
- WordPress, Prestashop, etc. to choose

ADVANCED

The hosting for professional developments

from **4,95 €** /month

with annual contract (after 9,90 €/mo.)

- 100 GB Space
- Unlimited transfer
- Unlimited e-mail accounts and chat
- NVMe disks, maximum speed!
- Antispam and Antivirus in real time
- WordPress, Prestashop, etc. unlimited

MULTIHOSTING

5 hostings with independent panels

from **11,25 €** /month

with annual contract (after 22,50 €/mo.)

- 100 GB Space
- Unlimited transfer
- Unlimited e-mail accounts and chat
- NVMe disks, maximum speed!
- Antispam and Antivirus in real time
- WordPress, Prestashop, etc. unlimited

Cloud Computing

PRODUCT	aws	Microsoft Azure	Google Cloud Platform
Virtual Servers	Instances	VMs	VM Instances
Platform-as-a-Service	Elastic Beanstalk	Cloud Services	App Engine
Serverless Computing	Lambda	Azure Functions	Cloud Functions
Docker Management	ECS	Container Service	Container Engine
Kubernetes Management	EKS	Kubernetes Service	Kubernetes Engine
Object Storage	S3	Block Blob	Cloud Storage
Archive Storage	Glacier	Archive Storage	Coldline
File Storage	EFS	Azure Files	ZFS / Avere
Global Content Delivery	CloudFront	Delivery Network	Cloud CDN
Managed Data Warehouse	Redshift	SQL Warehouse	Big Query

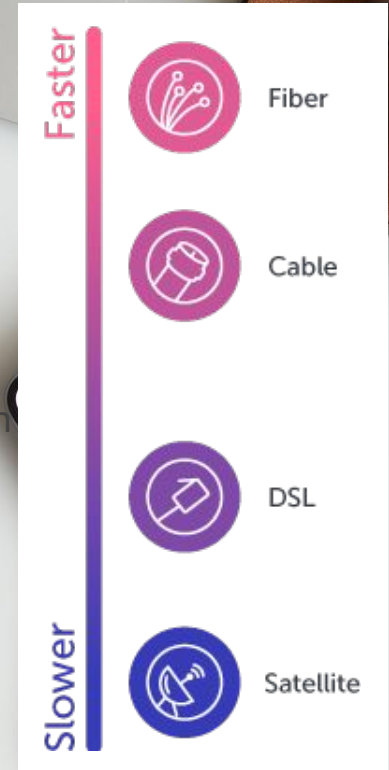
A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', 'V', and 'G' showing. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden planter is in the top center. A brown leather folder or notebook is on the right. In the bottom right, a smartphone with a pink screen lies on a spiral-bound notebook.

Internet Service Provider

The phisycal network

ISP - Internet Service Provider

- TELCO companies offer internet connection
- Actual technologies:
 - Fiber (FTTH) - Best wired connection by far
 - Cable (FTTB/ethernet) - Good wired connection
 - DSL - Old and slow wired connection
 - 3G, 4G, 5G - Good and expensive wireless connection for mobile or when there isn't wired
 - Satellite, Radiolink - Old and slow wireless connection
- ISPs are interconnected by the backbone connection of internet



ISP & city networks

1. World Wide Web

Data from the internet is waiting for you to access it!



2. Hub

Data enters our hub.



3. Utility Poles

Data is pushed from our HUB through fiber optic cables residing on your local utility poles or underground



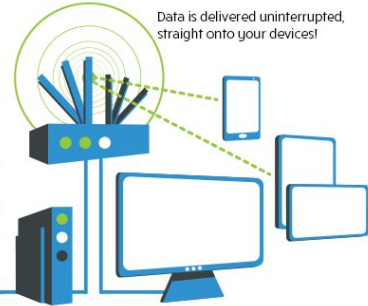
4. Fiber Optic cable is dropped to your home (Fiber-to-the-home)

MAW Communications will give you a fiber drop to your home upon signing the property agreement



5. Fiber Optic cable Installed into your home

Data is delivered uninterrupted, straight onto your devices!

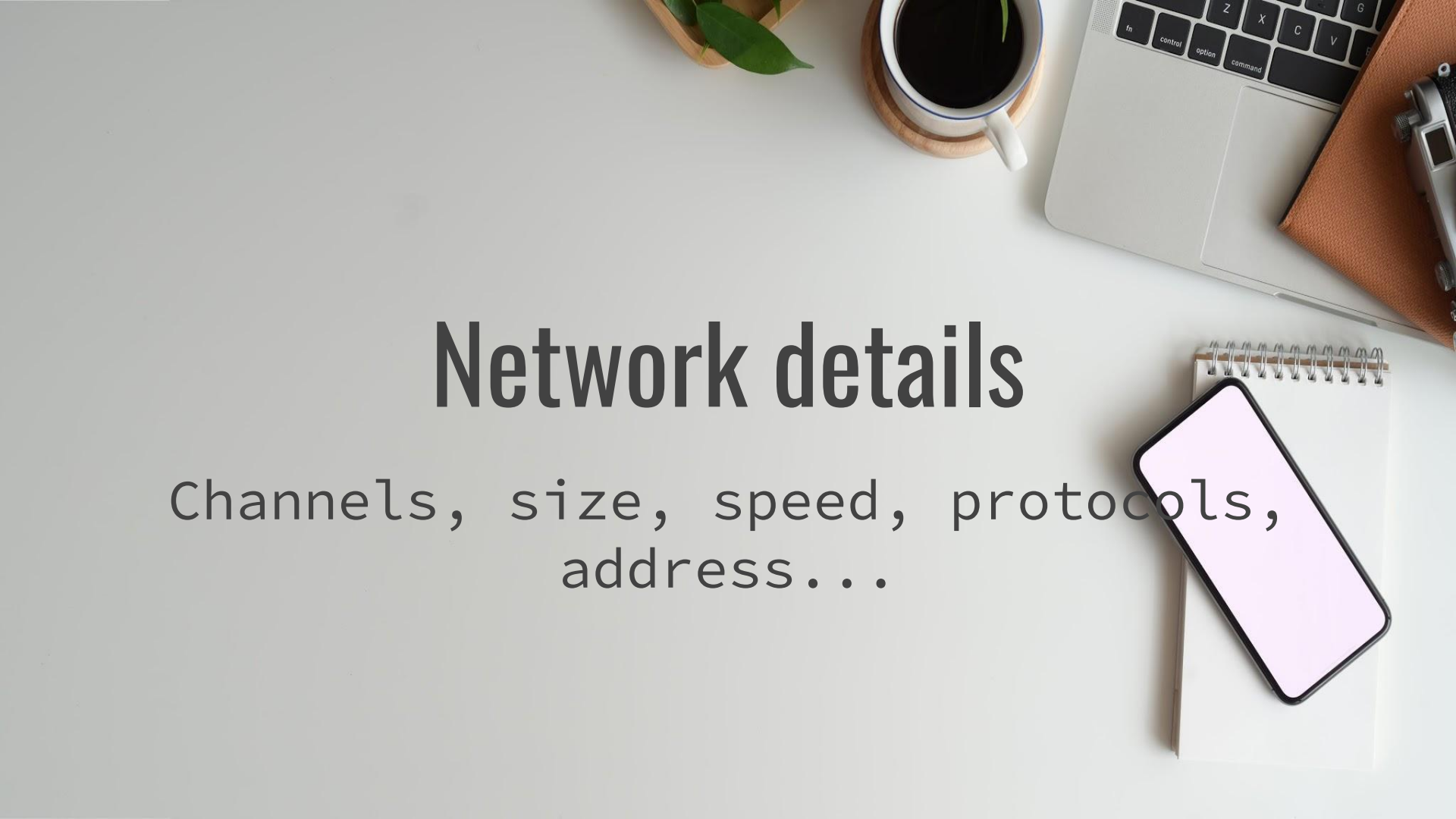


Backbone Ground Fiber Cable



Spine Submarine Fiber Cable

The map illustrates the global network of submarine fiber optic cables. The cables are color-coded by ocean: Pacific (blue), Atlantic (green), Indian (red), and Antarctic (purple). The map shows a high density of connections in Europe, North America, and East Asia, with many cables crossing the Atlantic and Pacific Oceans. The map includes labels for major countries and oceans in Portuguese.

A top-down view of a workspace. In the top right, a silver laptop is partially visible with keys like 'fn', 'control', 'option', 'command', 'Z', 'X', 'C', and 'V'. Next to it is a white cup of black coffee on a wooden coaster. A small green plant in a wooden pot is in the top center. An orange textured folder or notebook is on the right. In the bottom right, a spiral-bound notebook is open with a pink smartphone resting on it.

Network details

Channels, size, speed, protocols,
address...

Signals & Channels

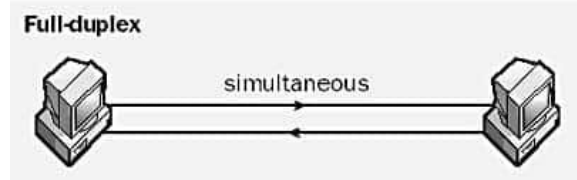
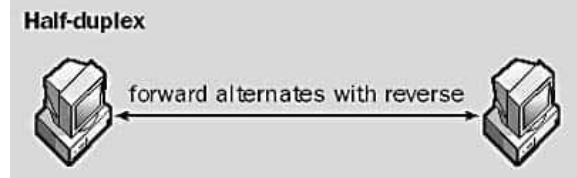
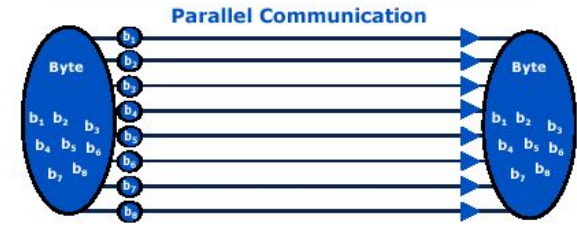
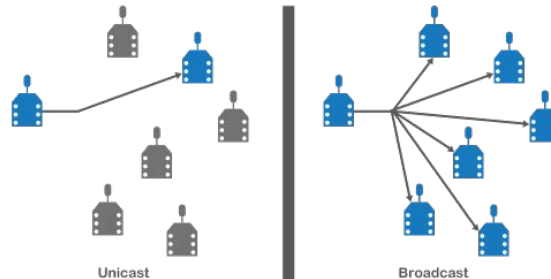
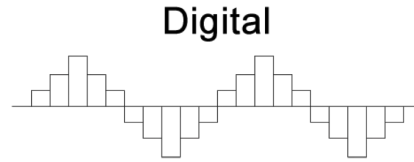
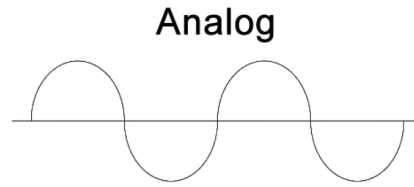
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Analog vs Digital

Unicast vs Broadcast

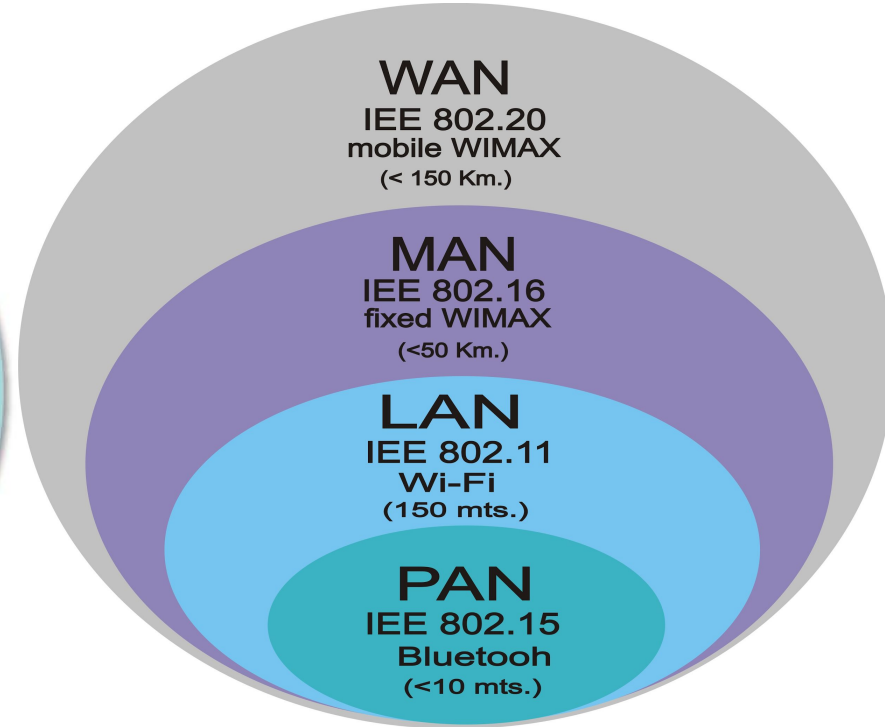
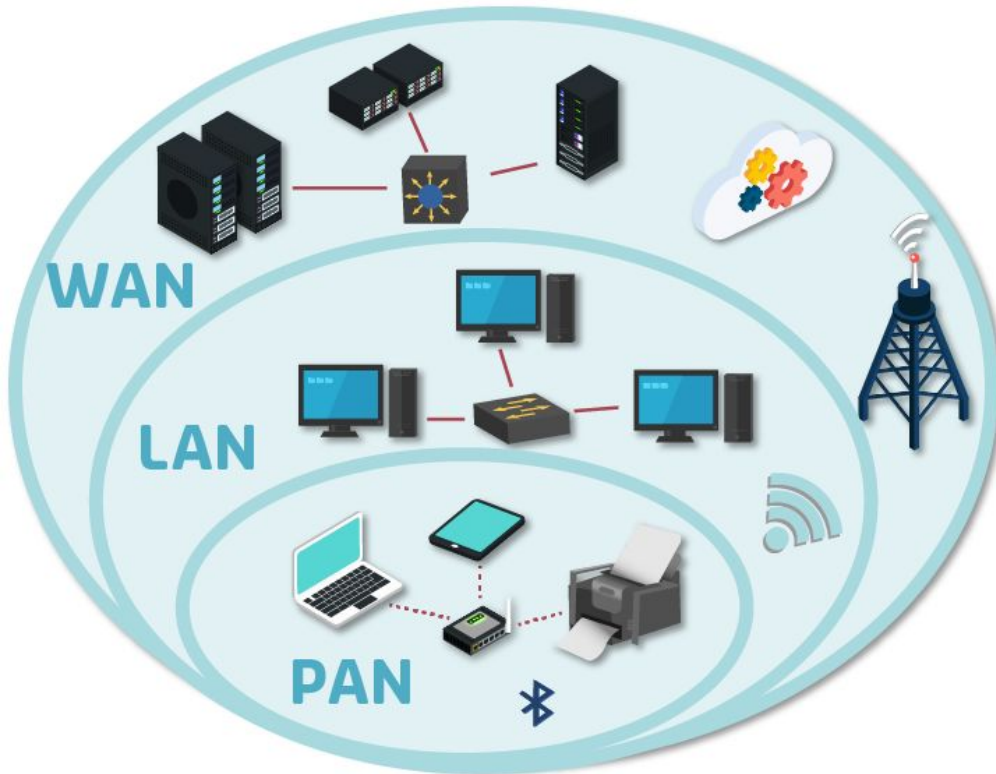
Serial vs Parallel

Wireless vs Wired

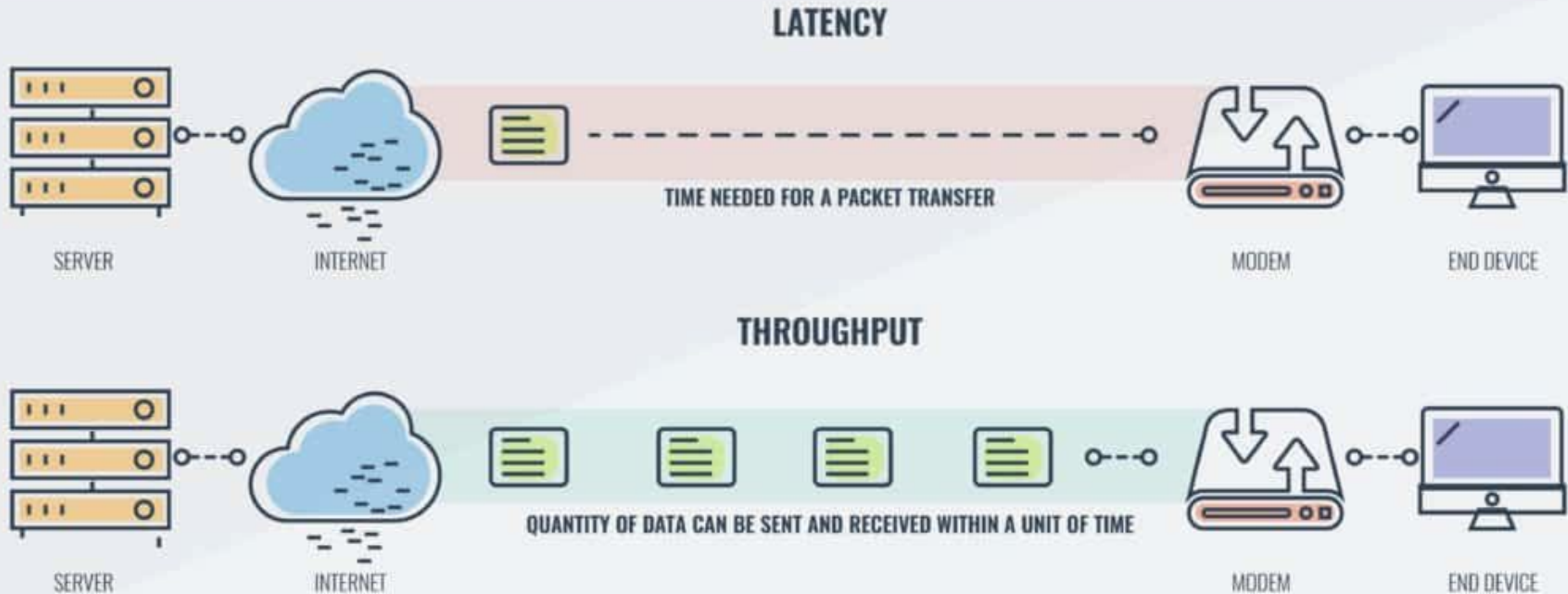
Sinplex, Half-Duplex,
Full-Duplex



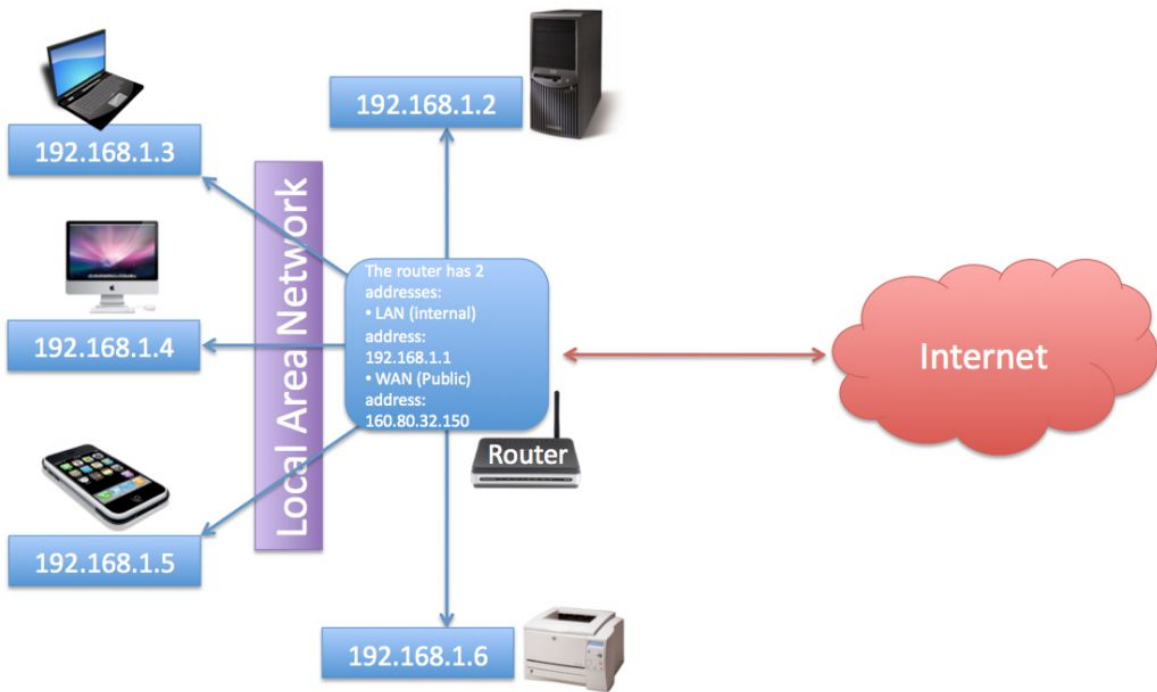
Network size



Latency vs Throughput



IP address



IPv4 address in dotted-decimal notation

172 . 16 . 254 . 1

↓ ↓ ↓ ↓

10101100 .00010000 .11111110 .00000001

8 bits 32 bits (4 bytes)

An IPv6 address (in hexadecimal)

2001:0DB8:AC10:FE01:0000:0000:0000:0000

↓ ↓ ↓ ↓

2001:0DB8:AC10:FE01:: Zeroes can be omitted

0010000000000001:0000110110111000:1010110000010000:1111111000000001:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

IPv6

128 bits each

total range = 340 undecillion
possible addresses

2001:db8::ff00:42:8329

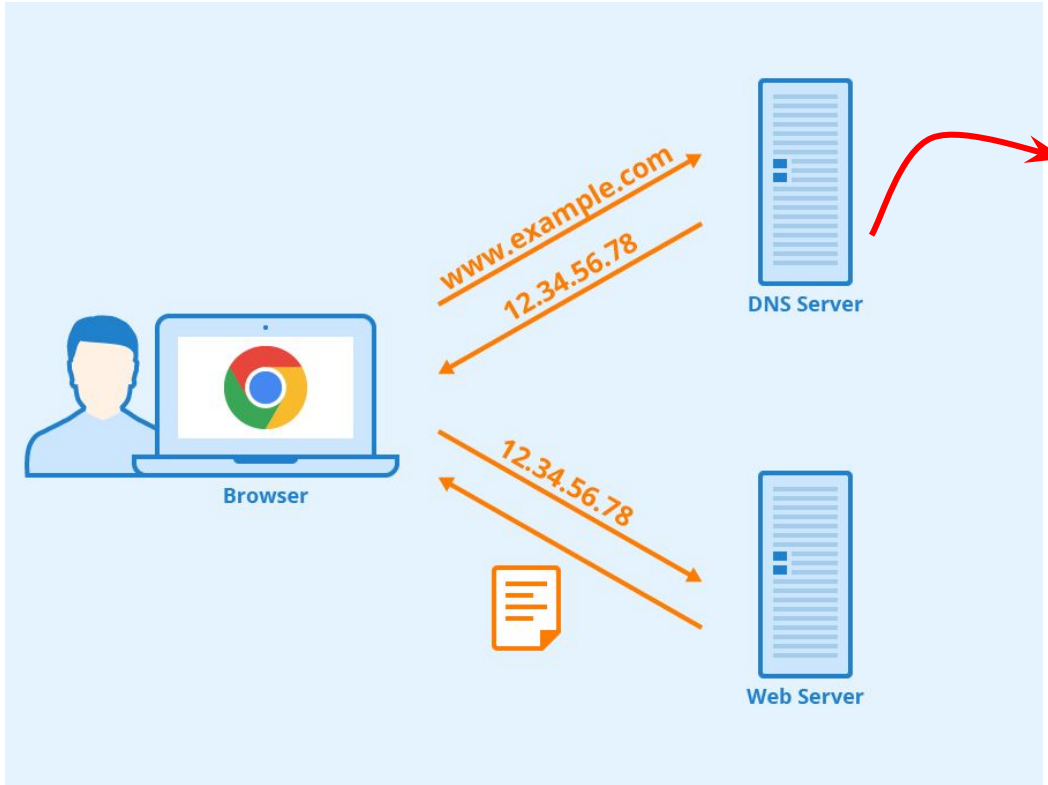
IPv4

4 bytes each

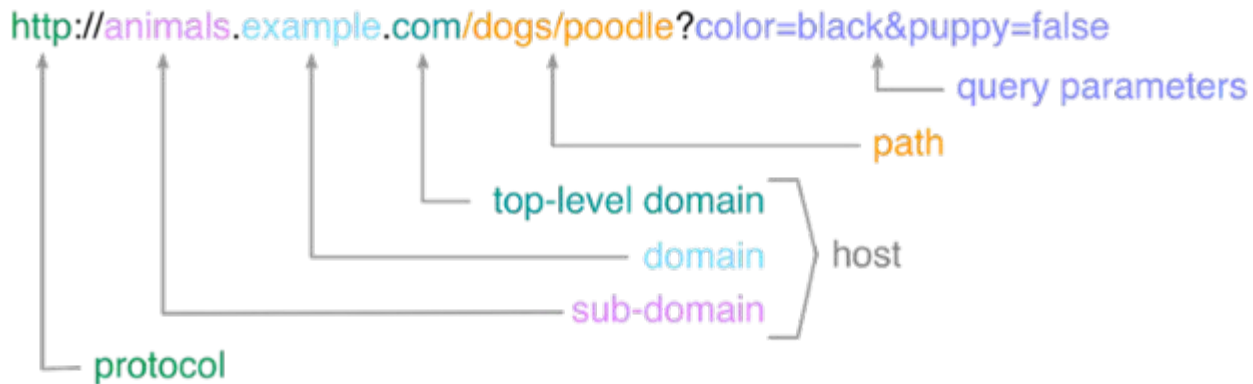
total range = 4.3 billion
possible addresses

123.45.67.89

Domain Name Server



Location of Edge Servers	IP Address	Domain Name
Kolkata	203.197.118.81	www.jaduniv.edu.in
Singapore	137.132.21.27	www.nus.edu.sg
Colombo	192.248.17.88	www.cmb.ac.lk
London	212.113.11.22	www.lse.ac.uk
Chicago	198.101.129.15	www.uchicago.edu
New Delhi	103.27.9.20	www.du.ac.in
Ankara	80.251.40.153	www.ankara.edu.tr
Islamabad	61.5.158.124	www.islamabadairport.com.pk
Santiago	158.170.64.116	www.udesantiago.cl
Mexico	128.123.3.2	www.nmsu.edu
Kingston	130.15.126.136	www.queensu.ca
Buenos Aires	190.224.163.234	www.buenosairesherald.com
Harare	196.201.17.237	www.caaz.co.zw
Cape Town	41.72.141.237	www.capetown.travel
Canberra	137.92.97.88	www.canberra.edu.au



DOMAINS

HOSTING

SERVERS

RESELLERS

MUCH MORE

24/7 SUPPORT

WE ARE DH

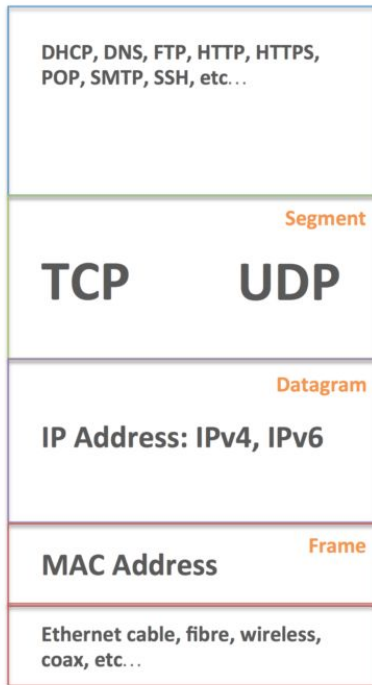
BLOG

LOGIN

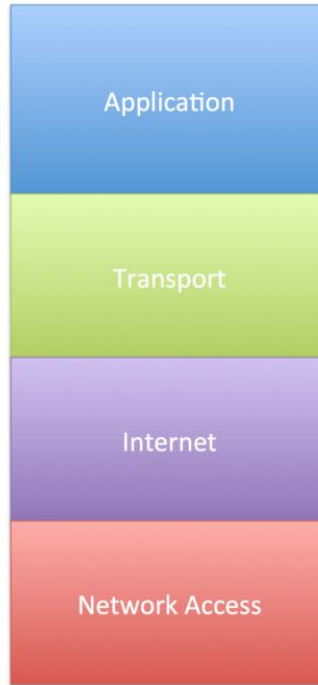
	emlex-informatics.com	8,99€ 14€	Available for registration! :-)	SELECT
	emlex-informatics.es	8,99€ 14€	Available for registration! :-)	SELECT
	emlex-informatics.net	8,99€ 15€	Available for registration! :-)	SELECT
	emlex-informatics.org	8,99€ 16€	Available for registration! :-)	SELECT
	emlex-informatics.online	31,95€	Available for registration! :-)	SELECT
	emlex-informatics.cat	5€ 8,95€	Available for registration! :-)	SELECT
	emlex-informatics.eu	19€	Available for registration! :-)	SELECT

TCP/IP: “envelopes” to send/receive data

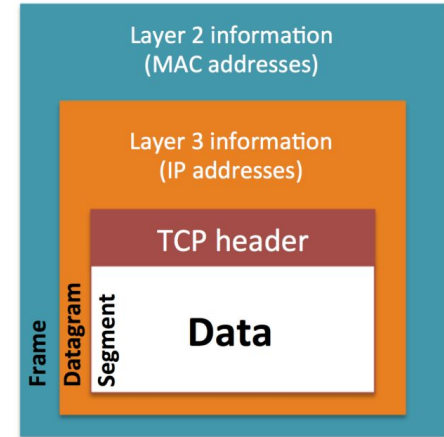
The OSI Model



The TCP/IP Model



The encapsulated structure of TCP/IP packets



This image is part of the Bioinformatics Web Development tutorial at http://www.cellbiol.com/bioinformatics_web_development/

Bibliography

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- http://www.cellbiol.com/bioinformatics_web_development/chapter-1-internet-networks-and-tcp-ip/the-tcpip-family-of-internet-protocols/
- ...

Computers manage
a lot of **information**
connected to **networks**

